



DMS111 Quantitative Methods for Business

Quantitative Methods (University of Nairobi)

LECTURE ONE

SET THEORY

Lecture Outlines

- 1.1 Introduction
- 1.2 Objectives
- 1.3 Definitions of Set Concepts
- 1.4 Set Operations and Set Algebra
- 1.5 Summary

1.1 Introduction

Welcome to the First Lecture in this course unit. In this lecture we are going to learn about set theory.

The study of sets is important and thus popular in the business and economic world for three major reasons:

- Basic understanding of concepts in sets and set algebra provides a form of logical language through which business specialists can communicate important concepts and ideas.
- Set algebra is used in solving counting problems of a logical nature.
- The study of set algebra provides a solid background to understanding of probability and statistics, which are important business decision-making tools.

1.2



Objectives

At the end of this lecture you should be able to:

1. Define a set and set concepts.
2. Use venn diagrams to illustrate sets
3. Perform set operations

1.3 Definitions and Basic Concepts

1. Definition of a set:

A set is a well-defined collection or group of objects. For instance,

- Set of all courses offered at the School of Business, University of Nairobi.
- Set of all European manufactured Mobile Phones in Kenya.
- Set of all female students pursuing a medical degree in Kenyan Universities.

These objects are also referred to as members or elements of a set.

Requirements of a set

- (i) A set must be well defined i.e. it must not leave any room for ambiguities e.g. Set of all students. This raises such questions as which students? Where are these students? What is the time frame, i.e. when?

A well defined set could be a: Set of all female students pursuing a medical degree in Kenyan Universities in the year 2010.

- (ii) The elements of a given set must be distinct i.e. each object will appear once and once only. This means that an element *must appear but only once*.

The following therefore does not qualify to be a set since the element 2 is repeated: $\{1, 2, 4, 2, 7\}$.

The correct set is $\{1, 2, 4, 7\}$

- (iii) The order of presenting the elements of a set is immaterial.

Thus the following four sets are the same: $\{1, 3, 2\} = \{1, 2, 3\} = \{3, 2, 1\} = \{2, 1, 3\}$

2. Specifying or naming of sets

By convention, sets are specified (named) using a capital letter. Further, the elements of a set are designated by either listing all the elements or by using a descriptive characteristic or pattern. The elements of a set are enclosed using curly brackets. For example, consider the set of whole numbers from 0 to 6, inclusive. We can represent them in 3 ways as follows:

- Listing of all elements

$$A = \{0, 1, 2, 3, 4, 5, 6\}$$

- Using a descriptive characteristic

$$A = \{X \text{ such that } X \text{ is a positive integer from } 0 \text{ to } 6, \text{ inclusive}\}$$

- Using a pattern

$$A = \{0, 1, 2, \dots, 6\}$$

3. Set membership

Set membership is expressed by using the Greek letter epsilon; ϵ .

Consider set A above (used to illustrate naming of sets) in which 3 is a member. This is expressed as:

$$3 \in A$$

This can also be used in a plural sense as follows: $0, 5, 1 \in A$ (zero, five and one are members of set A)

4. Finite set

This is a set that consists of a limited or countable number of elements e.g. set A above because it has 7 elements.

An *infinite set* therefore consists of an unlimited or uncountable number of members, e.g. set of all odd numbers.

5. Subset

Any set say S is a subset of set A above if all elements in S are members of A. It is denoted using the symbol; $\subset$

E.g. $S \subset A$ which is read as “Set S is a subset of set A”

If $S = \{1, 5\}$ then $S \subset A$.

Recall $A = \{0, 1, 2, 3, 4, 5, 6\}$

Equally, set A is said to be the superset to set S and is denoted using the symbol; $\supset$

Hence, $A \supset S$

6. Equality of sets

If all elements in set D_1 are also in D_2 and all elements in D_2 are also in D_1 , then;

sets D_1 and D_2 are equal, that is, $D_1 = D_2$

e.g. If $D_1 = \{a, c, f\}$ and $D_2 = \{c, f, a\}$

then $D_1 = D_2$

Further, $D_1 \subset D_2$ and $D_2 \subset D_1$, that is, each set is a subset and superset to itself.

7. Universal set.

It is that set which contains all elements under consideration by the analyst or researcher and is denoted by the symbol; U

For example if $U = \{\text{all University of Nairobi students}\}$, we can have the following subsets:

$$S_1 = \{\text{students at Lower Kabete campus}\}$$

$$S_2 = \{\text{Male students}\}$$

$$S_3 = \{\text{Students studying Engineering}\}$$

8. Null or empty set

This is a set with no elements. It is denoted by the notation; $\{ \}$ or $\emptyset$

where $\emptyset$ is the Greek letter phi.

A good example would be a set of living human beings who are over 200 years old. Since we cannot find such a person, this is said to be an empty set.

9. Complement of a set

If $U \equiv$ Universal set and A is a subset of the universal set, then, the complement of A , denoted A^I or A^C represents all elements in the universal set which are not members of set A .

$$\text{E.g If } A = \{\text{whole numbers from 0 to 6}\}$$

$$\text{and } U = \{\text{whole numbers from 0 to 10}\} \quad \text{then } A^I = \{7, 8, 9, 10\}$$

10. Sets are pictorially represented using Venn diagrams (so named after the 18th Century English logician, John Venn)

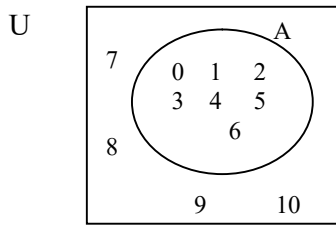
Symbols

Circle:  (is used to represent an Ordinary set ((not a universal set)

Rectangle  is used to represent the Universal set

Example:

Consider the previous universal set and set A



11. Singleton set

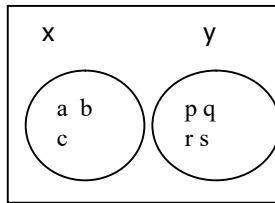
This is a set with only 1 element, e.g. set of current Vice chancellors of University of Nairobi has only 1 member

12. Disjoint sets

These are sets which have nothing in common,

e.g. if $X = \{a, b, c\}$ and $Y = \{p, q, r, s\}$

Then: x & y are disjoint sets



1.4 Set Operations and Set Algebra

These consist of ways or operations whereby sets are combined in order to obtain other sets of interest. This gives rise to set algebra. The operations are union, intersection, difference and symmetric difference.

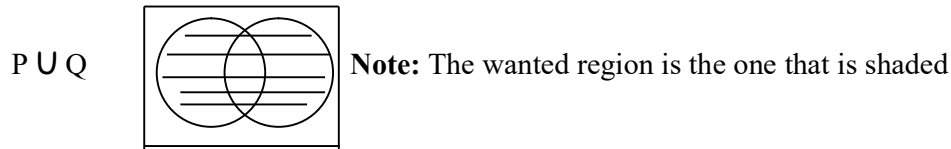
Basic set operations

Let $P = \{1, 3, 2\}$ $Q = \{1, 3, 5, 6\}$

1. Union of sets, denoted; $\cup$

This consists of all elements in P or Q or both P and Q

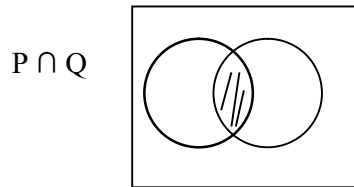
$$P \cup Q = \{1, 3, 2, 5, 6\}$$



2. Intersection of sets, denoted; $\cap$

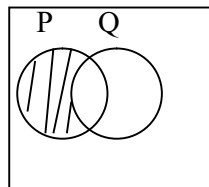
This consists of elements in both sets P and Q (the common elements)

$$P \cap Q = \{1, 3\}$$



3. Set Difference – Also known as set injunction, denoted ($-$)

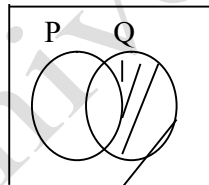
i) $P - Q$ Consists of elements in P but not Q



$$P - Q = \{2\}$$

ii) $Q - P$ Consists of elements in Q but not P.

$$Q - P = \{5, 6\}$$



4. Symmetric difference

The symmetric difference between the two sets, P and Q consists of elements that are in P but not in Q or elements in Q but not in P denoted using the Greek letter delta, Δ .

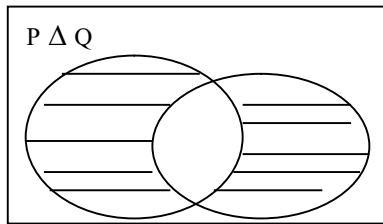
$$\text{Hence } P \Delta Q = \{\text{elements in } P - Q \text{ or } Q - P\}$$

Note:

$$P \Delta Q = (P \cup Q) - (P \cap Q)$$

In the illustration, $P \Delta Q = \{2, 5, 6\}$

Venn diagram for $P \Delta Q$



1.5



Summary

This chapter has introduced you to set concepts and their importance. We have also studied four mathematical operations which are permissible in set theory. In the next lecture, we will continue with the study of sets and in particular, their applications.

LECTURE TWO

APPLICATION OF SET THEORY

Lecture Outlines

- 2.1 Introduction
- 2.2 Objectives
- 2.3 Laws of Set Algebra
- 2.4 Counting Problems of A Logical Nature
- 2.5 Summary

2.1 Introduction

Welcome to our Second Lecture. This is a continuation of the first lecture on sets. In particular, we will learn about the consequences of set operations and their business uses.

Objectives



2.2

At the end of this lecture, you should be able to:

1. Explain the laws of set algebra.
2. Solve logical counting problems.

Laws of Set Algebra

Arising from the set operations we learned in lecture one, we have the following laws of sets:

1. Commutative laws

For any two sets P and Q,

i) $P \cup Q = Q \cup P$ } The order in which sets are combined with union or intersection is irrelevant

ii) $P \cap Q = Q \cap P$

2. Associative laws

For any three sets P, Q and R,

i) $(P \cup Q) \cup R = P \cup (Q \cup R)$ } The selection of 3 or more sets for grouping in a union or intersection is immaterial.

ii) $P \cap (Q \cap R) = (P \cap Q) \cap R$

3. Distributive laws

For any three sets P, Q and R,

i) $P \cup (Q \cap R) = (P \cup Q) \cap (P \cup R)$

ii) $P \cap (Q \cup R) = (P \cap Q) \cup (P \cap R)$

4. Idempotent laws

For any set Q,

i) $Q \cap Q = Q$
ii) $Q \cup Q = Q$

The union or intersection of a set with itself does not change the set.

Other laws

5. $P \cup \emptyset = P$

6. $P \cap \emptyset = \emptyset$

7. $P \cup U = U$

8. $P \cap U = P$

9. $P \cap P' = \emptyset$

10. $P \cup P' = U$

11. De Morgan's laws

For any two sets Q and R,

i) $(Q \cup R)^I = Q^I \cap R^I$

ii) $(Q \cap R)^I = Q^I \cup R^I$

Activity 2.1



1. Rewrite the following:

i) $(A^I \cup B^I)^I$

ii) $(A^I \cap B^I)^I$

2. Simplify the following:

i) $(A \cup B) \cup (A \cap B)$

ii) $P \cup (P^I \cap Q)$

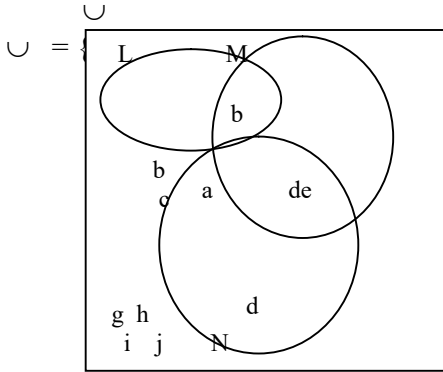
3. The sets L, M and N in a universal set consisting of the first 10 lower-case letters of the alphabet are $L = \{a, b, c\}$ $M = \{b, c, a, e\}$ $N = \{a, d, e, f\}$

Required:

Determine members of the following sets:

- i) $M \cup N$ ii) $L \cup N$ iii) L' iv) $L \cap M \cap N'$ v) $(L \cup M \cup N)'$
vi) $M \cap N$

Solution



- i) $M \cup N = \{b, c, a, e\} \cup \{a, d, e, f\}$
 $= \{a, b, c, d, e, f\}$
- ii) $L \cup N = \{a, b, c\} \cup \{a, d, e, f\}$
 $= \{a, b, c, d, e, f\}$
- iii) $L' = \{d, e, f, g, h, i, j\}$
- iv) $L \cap M \cap N' = \{b, c\}$
- v) $(L \cup M \cup N)' = \{g, h, i, j\}$
- vi) $M \cap N = \{a, e\}$

2.4 Counting Problems of A Logical Nature

Number of elements in a set

Two sets

For any set S , if S contains K elements, we show this as:

$n(S) = K$, e.g. If S has the elements as follows:

$$n = \{10, 0, 17, 2, 12\}$$

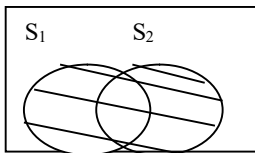
$$n(S) = 5$$

Generally, given any 2 sets S_1 and S_2

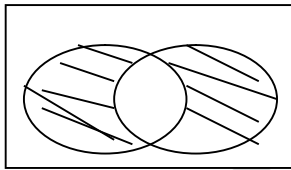
1. $n(S_1 \cup S_2) = n(S_1) + n(S_2) - n(S_1 \cap S_2)$. We have to subtract the number of elements in the intersection to avoid counting them twice. Note, however, that if S_1 and S_2 are disjoint sets, then:

$$n(S_1 \cup S_2) = n(S_1) + n(S_2) \text{ since } n(S_1 \cap S_2) = 0$$

Since $S_1 \cap S_2 = \emptyset$



2. For symmetric difference, $n(S_1 \Delta S_2) = n(S_1 \cup S_2) - n(S_1 \cap S_2)$



Example

In a recent survey of 400 students in a college, 100 were listed as studying typing (T) and 150 were listed as doing accountancy (A).

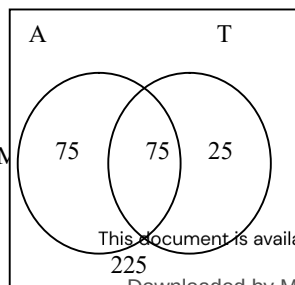
75 were registered for both courses.

Required:

- a) Find the number of students in the college who are not registered for either course.
- b) How many students were registered for typing only?

Solution:

$$n(\cup) = 400$$



$$n(u) = 400$$

$$n(T) = 100$$

$$n(A) = 150$$

$$n(T \cap A) = 75$$

- (a) Number of students not registered for either course are 225.
- (b) Number of students registered for typing only are 25.

Three sets

When solving logical counting problems involving 3 sets, it is conveniently done by use of a Venn diagram.

Assign unknowns to all parts of the Venn diagram and using the known values, we can solve for the values of the unknowns in a series of linear equations.

Example

A survey was conducted on the newspaper readership of 3 dailies; the Mirror, the Citizen and the Times, M, C, T respectively and the following data was obtained:

The number of people who read M, C & T was found to be 55, 45 and 39 respectively.

The number that read M & T = 19

The number that read C & M = 15

The number that read C & T = 14

Those who read all the 3 were found to be 4 people only.

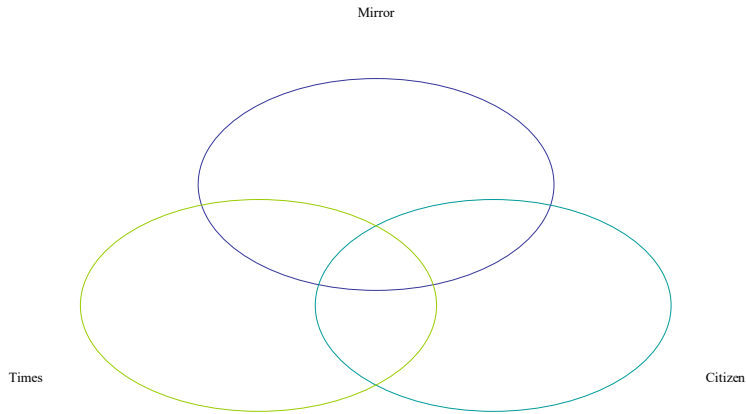
Required:

Determine the number of people who:

- a) Read the Mirror only.
- b) Read Citizen or Times but not the Mirror
- c) The total number of people interviewed if 5 people read none of the papers.

Solution:

We use a Venn diagram to give symbols for the unknowns for all parts as follows:-



From the information given, we find that:

$$n(M) = b + c + d + e = 55 \dots (1)$$

$$n(C) = c + e + h + g = 45 \dots (2)$$

$$n(T) = d + e + f + g = 39 \dots (3)$$

Further,

$$n(M \cap T) = d + e = 19 \dots (4)$$

$$n(C \cap M) = c + e = 15 \dots (5)$$

$$n(C \cap T) = e + g = 14 \dots (6)$$

$$n(M \cap C \cap T) = e = 4$$

Given that we know the value of e , we can solve for the following:

$$d = 19 - 4 = 15 \dots \text{using equation (4)}$$

$$c = 15 - 4 = 11 \dots \text{“ “ (5)}$$

$$g = 14 - 4 = 10 \dots \text{“ “ (6)}$$

Using equation (1), we find that :

$$b = 55 - (11 + 15 + 4)$$

$$= 15$$

Using equation (2):

$$h = 45 - (11 + 4 + 10) = 20$$

Using equation (3):

$$f = 39 - (15 + 4 + 10) \\ = 10$$

If 5 people read none of the three dailies, then $a = 5$

Let us now answer specific questions asked:

- a) $n(M \text{ only}) = b = 5$
- b) $n(C \text{ or } T \text{ but not } M) = 20 + 10 + 10 \\ = 4$
- c) $n(U) = 5 + 25 + 11 + 15 + 4 + 10 + 10 + 20 \\ = 100$

2.5 Summary

This lecture has dealt with the postulates or laws of set theory. This arises from the permissible set operations. Also studied has been the application of set algebra to logical counting problems.

Activity 2.2

Self Test One

- a) A company has a large of computer assistants, each of whom is competent in the use of at least one of 3 utility packages: Word processor (W) Database Management system (D) and a Spreadsheet (S). A survey shows that 30 can use a word processor, 25 can use a Database Management system and 28 are competent in the use of a Spreadsheet. Of the computer facility assistants who can use a Database Management System, 14 can also use a word processor while 6 have no other skill.

6 of the computer assistants can use a word processor and spreadsheet but not a database system while 4 have all three skills.

Required:

Determine the number of computer facility assistants who are members of the following sets:

- i) $W \cup D \cup S'$
 - ii) $(W \cup S)' \cap S$
 - iii) $D \cap S$
 - iv) Universal set
- b) A sample of 100 Young Christian Union voters revealed the following concerning three candidates;
- Ali, Bungei and Chiru, who were running for the Y.C.S Party Chairman, Secretary and Treasurer respectively.
- 14 preferred booth Ali and Bungei
 - 49 preferred Ali or Bungei but not Chiru
 - 21 preferred Bungei but not Chiru or Ali
 - 61 preferred Bungei or Chiru but not Ali
 - 32 preferred Chiru but not Ali or Bungei

7 preferred Ali and Chiru but not Bungei.

Required:

- i) With the aid of a Venn diagram, determine the number of voters that were in favor of all the three candidates. Assume that every member of Y.C.S voted for at least one candidate.

Determine the candidate that went unopposed if a rule of 50% majority were used in such a decision.

LECTURE THREE

FUNCTIONS

Lecture outlines

- 3.1 Introduction
- 3.2 Objectives
- 3.3 Preliminary Concepts and Definitions
 - 3.3.1 Preliminary Concepts
 - 3.3.2 Definitions
 - 3.3.3 Importance of Functions in Business
- 3.4 Types of Functions
- 3.5 Linear functions and Applications
 - 3.4.1 Constant Functions
 - 3.4.2 Polynomial Functions
 - 3.4.3 Multivariate Functions
 - 3.4.4 Logarithmic Functions
 - 3.4.5 Exponential Functions
 - 3.4.6 Continuous Vs Discrete Functions
 - 3.4.7 Step Functions
 - 3.5.1 What is Linear Function
 - 3.5.2 Application of Linear Functions in Business World

3.1 Introduction

Welcome to the Third Lecture of this unit. In this unit, we shall learn about relationships which are known as functions.

Objectives

3.2

At the end of this lecture, you should be able to:

1. Define a function.
2. Describe various types of functions.
3. Describe a linear function and its characteristics.
4. Sketch the graph of a linear function.
5. Fit a linear function given a set of data.
6. Apply linear functions in solving business problems.

3.3. Preliminary Concepts and Definitions

3.3.1 Preliminary Concepts

Before we define a function, there are certain preliminary ideas which we require to understand.

1. A CONSTANT: Is a quantity whose value remains unchanged throughout a particular analysis
e.g. a fixed cost such as salary or rent in a given period.
2. A VARIABLE: Is a quantity, which assumes or takes various values in a particular analysis.

Examples

Suppose an item is sold for Shs.35 per unit and

Let S = Sales revenue

Q = Quantity sold

The price of Sh.35 is a constant but sales revenue (S) and quantity sold (Q) are variables. However, these variables are of two types: independent variable and dependent variable.

Independent and Dependent Variable

An independent variable is the one that determines the quantity or the value of some other variable, which is then termed the dependent variable.

For the preceding illustration, quantity sold (Q) is the independent variable whereas sales revenue (S) is the dependent variable.

Furthermore, since quantity sold “predicts” the sales revenue, which in turn “response” to sales quantity, Q is also called predictor variable and S the response variable.

3.3.2 Definition

A function is a relationship in which values of a dependent variable are determined by the values of one or more independent variable or variables. We use the letter f to express a function. An example is: “Sales is a function of quantity” is expressed as: $\text{Sales} = f(\text{quantity})$.

Types of mappings (Relationships)

Similarly, $\text{Savings} = f(\text{Income})$

Demand=f (Price, income, price of related goods, etc)



Take Note

- Dependent variables is only one
- Independent variables can be one or more

3.3.3 Importance of Functions in Business

Functions are important in establishing relationships among business variables which facilitates their control for achieving organizational objectives, e.g. in making decisions such as:

- Output required
- Prices to charge
- Level of advertising
- Optimal workforce size, etc.

3.4 Types of Functions

Some of the more commonly encountered functions in the business world include the following:

- a) Constant functions
- b) Polynomial functions
- c) Multivariate functions
- d) Logarithmic functions
- e) Exponential, functions
- f) Continuous Versus discrete functions
- g) Step function

We will now have a brief explanation of each of these types of functions.

3.4.1 Constant Functions

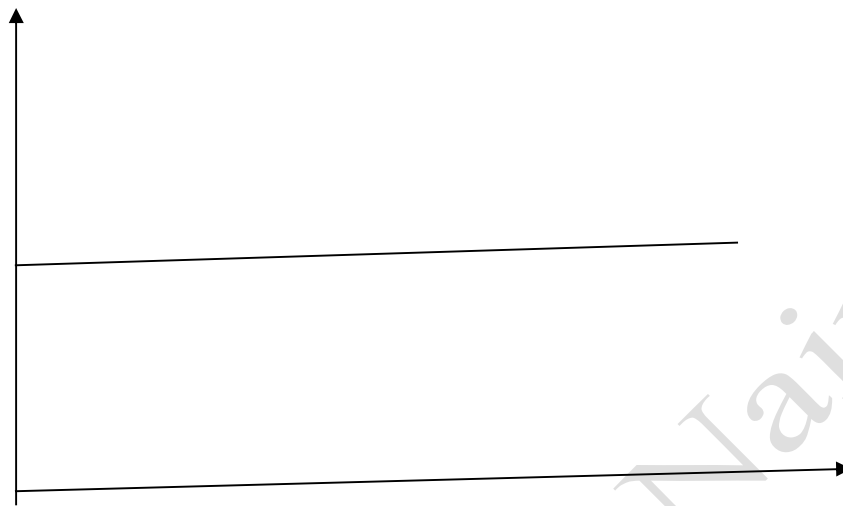
A constant function has the same value of the dependent variable irrespective of the values of the independent variable.

Example is the function, $Y = 4$ where x = independent variable and y = dependent variable.

An example of a constant function in business is fixed cost (such as rent and salaries) which do not change in the short run.

Explanation: “Short run” as used in business usually means a period within one year.

Sketch of a Constant Function



3.4.2 Polynomial Functions

Any function of the form:

$$y = a + b_1 x + b_2 x^2 + \dots + b_n x^n$$

where $y \equiv$ dependent variable

$x \equiv$ independent variable

$a, b_1, b_2, b_n =$ constants

is known as a polynomial function. N , which is the highest power of x , is known as the degree of the polynomial.

Examples of polynomials

1st degree polynomial (linear function) $\rightarrow n = 1$

$$y = a + bx$$

- (1) $y = 10 - 3x$
- (2) $y = 20 + x$
- (3) $y = 2x$

2nd degree polynomial (Quadratic functions) $\rightarrow n = 2$ and $b_2 \neq 0$.

$$y = a + bx + b_2x^2$$

e.g. (1) $y = 25 + 2x - 3x^2$

(2) $y = 20 + 0.3x^2$

(3) $y = x^2$

3rd degree polynomial (Cubic functions) $\Rightarrow n = 3$ and $b_3 \neq 0$

$$y = a + bx + b_2x^2 + b_3x^3$$

e.g (1) $y = 15 - x + 2x^2 + 10x^3$

(2) $y = x^3$

(3) $y = x^2 - 2x^3$

For most business applications, we do not usually go beyond cubic functions and so we will only study the first three types, i.e. linear, quadratic and cubic functions.

3.4.3 Multivariate functions:

These are functions with more than one independent variable.

NB: - One independent variables function is known as univariate function.

- Two independent variables function is known as bivariate function

- Three or more independent variable function is known as Multivariate function.

3.4.4 Logarithmic Functions:

These are functions whose at least one of the terms is in logarithmic form. This term could be in the independent or dependent variable or both.

e.g. (1) $3 \log y = 7x$

(2) $y = 7 \log 2x$

(3) $5 \log 3y = 6 \log x$

Example:

(1) $Y = 3x - 4Z^2$ - Bivariate function (x and z are independent variables)

$Y = 2x - 12z^3 + 2w^2$ - Multivariate function (x, z and w are independent variables).

3.4.5 Exponential Functions:

These are functions whereby the predictor variable is at least part of an exponent or power

Examples

1) $Y = 10^{2x}$

2) $\text{Log } y = 15 - 2x^2 + 12$

3) $3 \log y = 7^x + 2w - 3z^2$

Note that a function need not be classified in only one way, e.g. number 3 apart from being an exponential function, it is also a multivariate and a logarithmic function.

3.4.6 Continuous Vs Discrete Functions

A continuous function is one in which the values of a dependent variable are defined for all values of the independent variables.

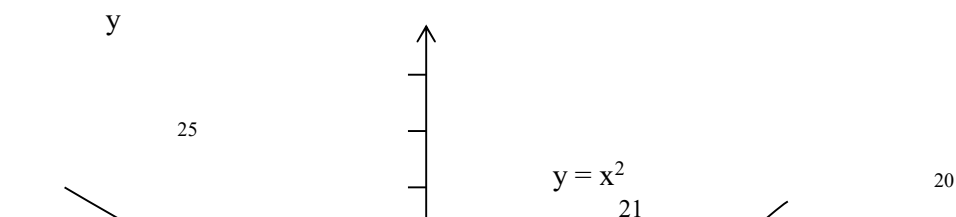
On the other hand, for a discrete function the dependent variable is defined only for particular values of the independent variables i.e. there is existence of gaps.

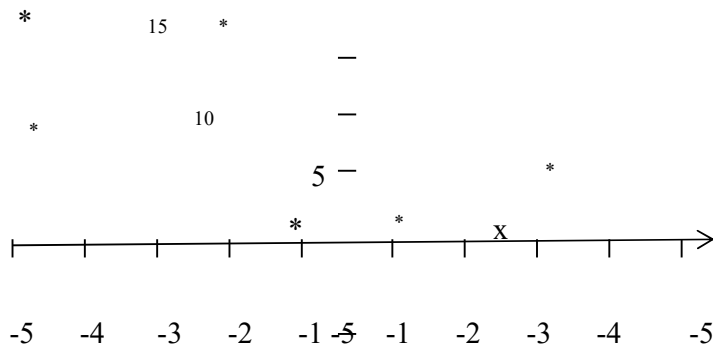
Example of continuous functions

Exercise:

Plot the graph of the function $y = x^2$

y =	x^2	-5	-3	-1	1	3	5
y		25	9	1	1	9	25





From the graph, note that for $y=x^2$, there is no gap. Examples of discrete functions are

- 1) $y = x$ where x = number of children in a household
- 2) $y = x$ where x = number of typographical errors in a textbook.

3.4.7 Step Functions:

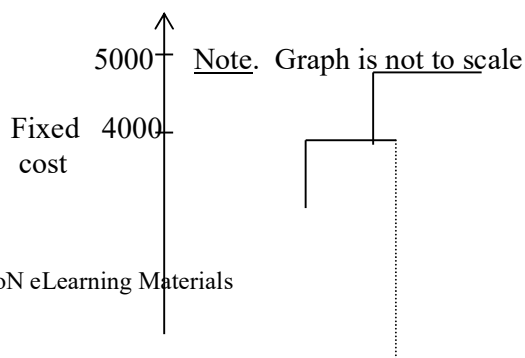
These are functions which take a constant value (for a given range), then the constant changes e.g. fixed cost for production increasing capacity.

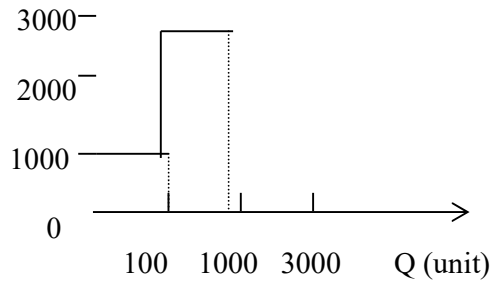
e.g. The following function can be plotted graphically as follows:

Production level (Units) Fixed cost (Sh)

1 - 100	1000
101 - 1000	3000
1001 - 3000	4000
Beyond 3000	5000

Graphically:





3.5 Linear Functions and Applications

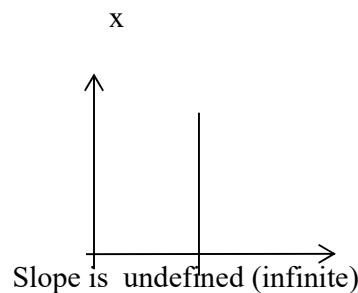
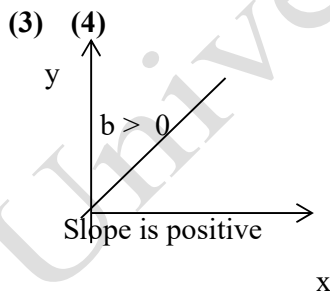
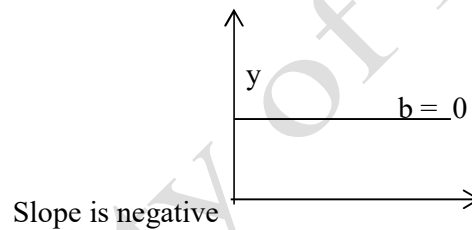
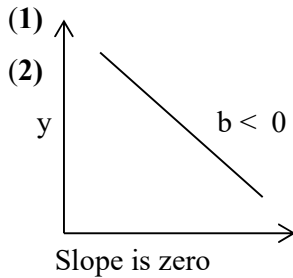
3.5.1 What is Linear Function

The linear function is a polynomial of the form $y = a + b x$

Where $a \equiv$ y intercept i.e. value of y when $x = 0$

$b \equiv$ The slope or gradient i.e. change in y per unit change in x.

Sketches of the linear function



Characteristics of the Straight-line

These are the features or properties of the linear function and there are three main ones:

- i) It has only one solution (root) i.e. it can cross the x axis a maximum of once.

Consider the general linear function:

$$y = a + bx$$

when $y = 0$

$$a + bx = 0$$

$$x = -\frac{a}{b}$$

- ii) It has no turning (stationary/critical) point
- iii) It is completely defined once either;
 - Two points on the line are specified or
 - one point and the slope are given.

Exercise

1. Determine the linear function that goes through the following points when

$$x = 2, y = 5$$

$$x = 5, y = 17$$

Solution

Substitute in the general form $y = a + b x$ as follows:

$$1) \quad 5 = a + 2b$$

$$17 = a + 5b$$

$$-12 = 0 - 3b$$

$$b = \frac{-12}{-3}$$

$$b = 4$$

Substituting in equation (i)

$$5 = a + 2b$$

$$5 = a + 2(4)$$

$$5 = a + 8$$

$$a = 8 - 5$$

$$a = -3$$

Equation: $y = -3 + 4x$

2. What is the equation of the linear function which goes through the points?
 - i) $(x, y) = (-3, 5)$ and
 - ii) $(x, y) = (-2, 2)$

Solution

Substituting:

$$1) \quad 5 = a - 3b$$

$$2) \quad 2 = a - 2b$$

$$3) \quad = (1) - (2):$$

$$3 = -b \quad b = -3$$

Using equation (1):

$$4 = a - 3(-3)$$

$$5 = a + 9 \Rightarrow a = 5 - 9$$

$$a = -4$$

Hence, equation is $y = -4 - 3x$

3. What is the straight line which has slope $b = -0.5$ and goes through $(x, y) = (10, 18)$?

Solution:

$$Y = a + bx$$

$$18 = a - 0.5(10)$$

$$18 = a - 5 \quad a = 18 + 5; \quad a = 23$$

$$\text{Equation } y = 23 - 0.5x$$

3.5.2 Application of Linear Functions in the Business World

Introduction

This section will give four different areas where functions are applied in business. These are in the form of numerical examples. This is to highlight how pervasive use of functions is in solving business problems.

1. Computation of Commissions

A salesman's daily wages is composed of a fixed amount and a variable component, which is dependent on the number of office cream units sold. He finds that when he sells 10 units on a given day, he earns Sh 600 whereas when he doubles his sales his earnings increase by only Sh 100.

Determine:

- i) Fixed daily earnings;
- ii) Level of commission per unit sold and hence;

- iii) What are the salesman's earnings if he sells 30 units?
- iv) On a given day the salesman is determined to earn Sh 3500. Suppose on the previous day he had guaranteed orders of 20 units, how many more must he sell in order to achieve his target earnings?

Let

E = total daily earnings

Q = Total quantity of ice cream sold in a day

$E = a + bQ$

Hence, to get the relationship between E and Q , we need to solve for the values of a and b .

When $d = 10$, $E = 600$ and when $Q = 20$, $e = 700$.

Equation:

$$(2) \quad 600 = a + 10b$$

$$(2) \quad 700 = a + 20b$$

$$600 = a + 10(10)$$

$$600 = a + 100$$

$$a = 500$$

and substituting when $a = 500$; we have $b = 10$

Equation: $E = 500 + 10Q$

Hence,

i) Fixed daily earning = $a = \text{Sh.}500$

i) Level of commission per unit sold

$b = \text{Sh.}10$ this is the slope

iii) What are the salesman's earnings if he sells 30 units?

$$Q = 30$$

$$E = 500 + 10(30)$$

$$E = 500 + 300$$

$$E = \text{Sh.}800$$

$$(iv) E = 3500$$

Let no. of ice cream units required be x

$$E = 5$$

$$3500 = 500 + 10(20 + x)$$

$$3000 = 200 + 10x$$

$$3000 - 200 = 10x$$

$$2800 = 10x \quad x = 2800/10$$

$$x = 280 \text{ units}$$

2. Demand and Supply Functions and Market Equilibrium

Suppose a certain commodity has linear demand and supply functions going through the following points

(i) When $P = \text{Sh } 7500$, $q = 1000$ units

$P = \text{Sh } 4625$, $q = 750$ units

(ii) When $P = \text{Sh } 2525$, $q = 100$ units

$P = \text{Sh } 1525$, $q = 200$ units

(a) Obtain the linear functions that go through the points given in (i) and (ii) above and clearly explain which the supply is and which the demand function is. Assume this is a normal commodity.

(b) Explain what is meant by market equilibrium and obtain the same for the above. Indicate your results on a graphical sketch.

Solution

(1) $p = a + bq$ where $p = \text{price}$ $Q = \text{quantity}$

$$7500 = a + 1000b \text{ -----(1)}$$

$$4625 = a + 750b \text{ -----(2)}$$

Solving for (1) and (2), we have;

$$7500 = a + 1000(11.5)$$

$$2875 = 250b$$

$$b = 2875/250 = 11.5$$

$$7500 = a + 11500$$

$$a = -4000$$

The General equation is

$$P = -4000 + 11.5Q$$

Comment:

This is a supply function since it has a positive slope i.e. $b = 11.5$

(2) **Substituting:**

$$p = a + bQ \quad 2525 = a + 100b$$

$$(1) \quad 2525 = a + 100b \quad 2525 = a + 100(-10)$$

$$(2) \quad 1525 = a + 200b \quad 2525 = a - 1000$$

$$(1) - (2)$$

$$1000 = -100b \quad a = 2525 + 1000$$

$$b = -1000/100 \quad a = 3525$$

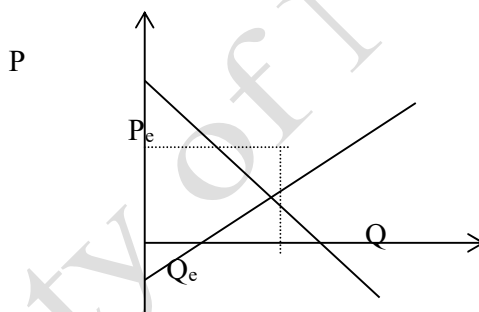
$$b = -10 \text{ (demand function)}$$

Equation $P = 3525 - 10Q$. It is a demand function since slope is negative

b) Market equilibrium is the point where demand equal supply for a commodity

$$Q_d = 3525 - 10Q$$

$$Q_s = -4000 + 11.5Q$$



P_e = equilibrium price

Q_e = equilibrium quantity

At market equilibrium: Demand function = supply function

$$Q_d = d_s$$

$$3525 - 10Q = -4000 + 11.5Q$$

$$\text{Since prices are equal } 3525 + 4000 = 11.5Q + 10Q$$

$$7525 = 21.5Q$$

$$Q = \frac{7525}{21.5}$$

$$21.5$$

$$Q = 350 \text{ units}$$

$$\text{Price at equilibrium } P_e = 3525 - 10Q$$

$$P_e = 3525 - 10(350)$$

$$P_e = 3525 - 3500$$

$$P_e = \text{Sh.}25$$

3. Fixed Assets Accounting, Straight Line Method of Depreciation

Tombe Transporters Ltd depreciates its fleet of trucks using a straight line method. The current accounting year is coming to an end and external auditors are examining the books of accounts. However, they cannot get complete records concerning a truck which was acquired 3 years ago. Its current book value is Sh 1,800,000 while its purchase cost was Sh 4,200,000. This type of truck is usually disposed off after 5 years.

- Determine the linear function $v = a + bt$ which relates the book value v and time in years t . Interpret a and b .
- What is the book value at the end of the 2nd year of the truck?
- Determine the disposal value of the truck stating any assumptions you may make.

Solution

Equation 1: $4,200,000 = a + b(0)$ $a = \text{Sh.}4200000$ purchase price

Substituting

$$1800000 = 4200000 - 3b$$

$$1800000 - 4200000 = -3b$$

$$-2400000 = -3b$$

$$b = \frac{2400000}{3}$$

$$b = -800,000$$

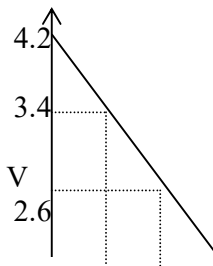
$$b = -800,000$$

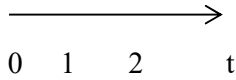
$$b = -\text{Sh.}800,000$$

Equation

$$V = 4200000 - 800000t$$

$$b = \text{Sh. } 800,000 = \text{annual depreciation}$$





b) When $t = 2$

$$v = 4.2 - 0.8(2)$$

$$v = 4.2 - 1.6$$

$$v = \text{Sh.}2.6 \text{ million}$$

c) When $t = 5$ $v = 4.2 - 0.8(5)$

$$= 4.2 - 4$$

$$v = 0.2 \quad v = \text{Sh.}200,000$$

Assumption:

The disposal value is the same as the book value.

4. Cost-Volume Profit (C-V-P) Analysis

This is also known as profit planning and breakeven analysis.

Profit is a function of prices, costs, volume and other factors.

Problem:

How does management manipulate factors which determine profits in order to achieve the objective of maximizing profits?

For linear c-v-p analysis, we make the following assumptions.

- i) Linearity. We assume that the revenue, cost and hence profit functions are linear with respect to the level of activity i.e. (quantity produced and sold)
 - For revenue function to be linear, price per unit must be constant e.g. there should be no quantity discounts.
 - For cost function to be linear.
 - unit variable costs are constant e.g. there are no changes in direct material or direct labour costs.
 - Fixed costs remain so
- ii) All costs can be classified as either fixed or variable. In particular, there are no semi-variable costs.
- iii) All units produced are sold i.e. inter-period inventory changes are negligible.
- iv) The only factor which influences revenues, cost and hence profits is level of activity.

- v) There is no demand restrictions i.e. there are no constraints.
- vi) All factors which influence profits are known with certainty in advance.

Equation approach

Let R = total revenue in monetary terms

P = unit price

V = unit variable costs

X = sales quantity (physical units)

f = fixed cost

C = total cost

Π = profit

V = total variable costs

a) Sales in physical units (x)

1. Profit equation

By definition:

$$\pi = R - C$$

$$R = px \quad p - v = \text{Contribution margin}$$

$$C = vx + f \quad \text{per unit (CM)}$$

$$\pi = px - (vx + f)$$

$$\pi = px - vx - f$$

$$\pi = (p - v)x - f$$

$$\pi = \text{cm} \cdot x - f$$

2. Unit sales X for target profit, T (Sh)

From equation above, $\pi = T$

$$\text{Hence } T = \text{CM} \cdot x - f$$

Making x the subject

$$x = \frac{f + T}{\text{CM}}$$

CM

3. Breakeven sales units, X b/e (b/e = breakeven)

At B.E.P, $R = C$ so that $\Pi = 0$

From equation 2 above $T = 0$ at B.E.P

$$\therefore x \text{ b/e} = \frac{f}{\text{CM}}$$

b) Sales in monetary units (R = sales revenue)

1) Profit function

$$\pi = (p-v)x - f$$

$$R = px = x = \frac{R}{p} (p-v) \equiv \text{Contribution margin}$$

$\frac{p-v}{p}$ Ratio (CMR)

$$\pi = (p-v) \frac{R}{p} - f$$

P

$$\pi = \frac{(p-v)}{p} R - f$$

P

$$\therefore \Pi = \text{CMR} * (R - f)$$

2) Sales revenue required for target profit, T Sh

From equation above $\pi = T$

$$\therefore T = \text{CMR} \times R - f$$

$$R = \frac{f+T}{\text{CMR}}$$

CMR

3) Sales revenue required for breakeven sales revenue, R b/e

$$\therefore R \text{ b/e} = \frac{f}{\text{CMR}}$$

At B.E.P, $\Pi = 0$

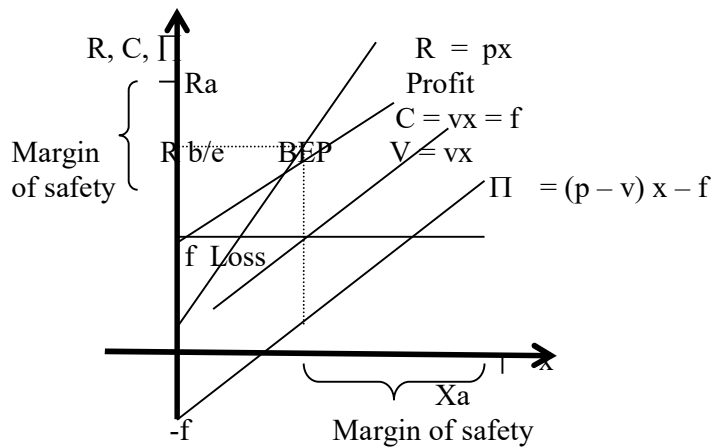
$$\text{Note: CMR} = \frac{p-v}{p} \quad \text{CMR} = \frac{p}{p} - \frac{v}{p}$$

$$\frac{v}{p} = \text{Variable cost ratio (VCR)}$$

$$\text{CMR} = 1 - \text{VCR} \quad \therefore \text{CMR} + \text{VCR} = 1$$

$$\text{Also } \frac{v}{p} = \frac{\text{Total variable costs}}{\text{Total sales}} = \frac{v \times x}{p \times x}$$

Graphical Representation



Let actual sales (beyond breakeven sales) be x_a

$\therefore X_a - X_{b/c}$ the extent by which actual sales exceeds breakeven sales is known as Margin of Safety (MoS).

This is also the extent by which sales would have to fall before the firm begins to make losses.

Example 1 – Sales in Physical Units

A product has selling price as sh. 200 whereas unit variable cost is sh. 140. the annual fixed cost is sh. 720,000. You are required to determine the following (B.E.P.)

- breakeven sales units
- profit to be made if 20000 units are sold
- Sales required for a profit of Sh. 2,000,000

Solution:

a) B.E.P. is given by the formula:

$$b) E(\text{ in units}) = \frac{\text{fixed cost}}{\text{CM}} \quad \text{cm} = p - v = 200 - 140 = \text{Sh } 60$$

$$= \frac{720,000}{60} = 12000 \text{ units}$$

The profit function is:

$$\Pi = 60 \times 20000 - 720000 = \text{sh. } 480,000.$$

- c) If a target profit is Sh T, then units, x required to make this profit is given as:

$$X = \frac{f + T}{CM}$$

CM

$$X = \frac{720000}{60} + \frac{2000000}{60} x = 2720000$$

$$x = 45333 \text{ units (to the nearest whole number).}$$

Example 2 – Sales in Monetary Units

A firm sells a product whose data in two periods is as follows:

Period	Sales (Shs)	Variable cost (Shs)	Profit (Shs)
I	100000	60000	20000
II	150000	90000	40000

Assume the price, unit variable cost, and fixed costs are the same in the two periods.

Required:

- Determine the fixed cost
- Determine the breakeven sales revenue
- What is the profit when sales are sh. 600000
- What is the sales required for a profit of sh.110000
- Determine the profit if variable cost incurred is Sh 300000

Solution:

- When sales are in monetary terms, profit function is given as:

$$\Pi = CMR \times R - f$$

We substitute the two given points to get the following equations:

$$(1) \quad 20000 = CMR \times 100000 - f$$

$$(2) \quad 40000 = CMR \times 150000 - f$$

To eliminate f, we subtract thus: (2) – (1):

$$(3) \quad - (1): 20000 = 50000 \text{ CMR}$$

$$\therefore CMR = \frac{20000}{50000}$$

$$50000 \text{ CMR} = 0.4$$

Using equation (1):

$$20000 = 0.4 \times 100000 - f$$

$$\therefore 20000 = 0.4 \times 100000 - f$$

$$\text{OR } f = -20000 + 0.4 \times 100000$$

$$= f = \text{sh. } \underline{20000}$$

$$\text{b) B.E.P.,} = \frac{f}{\text{CMR}}$$

$$\therefore \text{B.E.P.} = \frac{20000}{0.4} = \text{sh. } \underline{50000}$$

$$\text{c) For } R = \text{sh. } 600000$$

$$\Pi = 0.4 \times 600000 - 20000$$

$$= \text{sh. } \underline{220000}$$

d) For a target profit T, sales revenue required:

$$R = \frac{f + T}{\text{CMR}}$$

$$\text{CMR}$$

$$\therefore R = \frac{20000 + 110000}{0.4}$$

$$R = \frac{130000}{0.4}$$

$$R = \text{sh. } \underline{325000}$$

e) To answer this part, we need to know the sales revenue which is consistent with a variable cost of sh. 300000. We use the equation:

$$\text{CMR} + \text{VCR} = 1 \text{ and since } \text{CMR} = 0.6, \text{ it follows that } \text{VCR} = 0.6$$

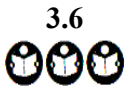
$$300000 = 0.6/R$$

$$\Rightarrow R = \frac{300000}{0.6}$$

$$R = \text{Sh } \underline{500000}$$

$$\text{Hence profit, } \Pi = 0.4 \times 500000 - 20000$$

$$= \text{Sh } \underline{180000}$$



3.6

Summary

This lecture has introduced you to functions, that is relationships between variables. An overview of the more commonly used functions in business was given. We have studied the linear function in greater detail in terms of its properties and uses. In the next lecture, you will study non-linear functions.

Activity 3.1

Aldox Limited produces two items, Alon (x) and Balon (y). Each unit of x requires in its production 30 units of raw material A and 15 units of raw material B. Each unit of y requires 45 units of raw material A and 75 units of raw material B. There is a limited supply of only 1800 units of raw material A and 1425 units of raw material B.

Required:

An equation, which expresses the number of units of x and y that can be produced using 1800 units of raw material A.

- i) An equation which expresses the number of units of x and y that can be produced using exactly 1425 units of raw material of B.
- ii) By solving equations obtained in parts (i) and (ii) simultaneously, determine the number of units of x and y which can be produced by using exactly 1800 units of raw material A and exactly 1425 units of raw material B.

iii) Suppose the unit prices of x and y are Sh. 10,000 and Sh. 7,000 respectively whereas their respective variable production costs per unit are Sh. 5,000 and Sh. 4,900. Further, their combined fixed costs are Sh. 50,000.

Required:

Determine the total profit that Aldox Industries will make given production levels obtained in part (iii) above.

LECTURE FOUR

QUADRATIC AND CUBIC FUNCTIONS

Lecture Outlines

- 4.1 Introduction
- 4.2 Objectives
- 4.3 Quadratic Functions and Applications
- 4.4 Cubic Functions and Applications
- 4.5 Summary

4.1 Introduction

Welcome to our fourth lecture. In lecture three, we learned the properties and applications of linear functions. This lecture is in a sense a continuation of the same concepts to quadratic and cubic functions.

4.2 	Objectives At the end of this lecture, you should be able to: 1. Describe quadratic and cubic functions and their properties 2. Draw the graphs of quadratic and cubic functions 3. Fit quadratic and cubic functions to given sets of data 4. Solve quadratic functions using the “formula” 5. Apply quadratic and cubic functions in solving business problems
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Quadratic Functions and Applications

The quadratic function is a polynomial of the form:

$$y = a + b_1X + b_2X^2$$

Where: y = dependent variable

x = independent variable

$a_1, b_1, b_2 \equiv$ Constants and $b_2 \neq 0$.

Properties:

1. Number of solutions, (roots) = 2 ----- **Note:** This is the maximum number of times the function can cross the x axis.

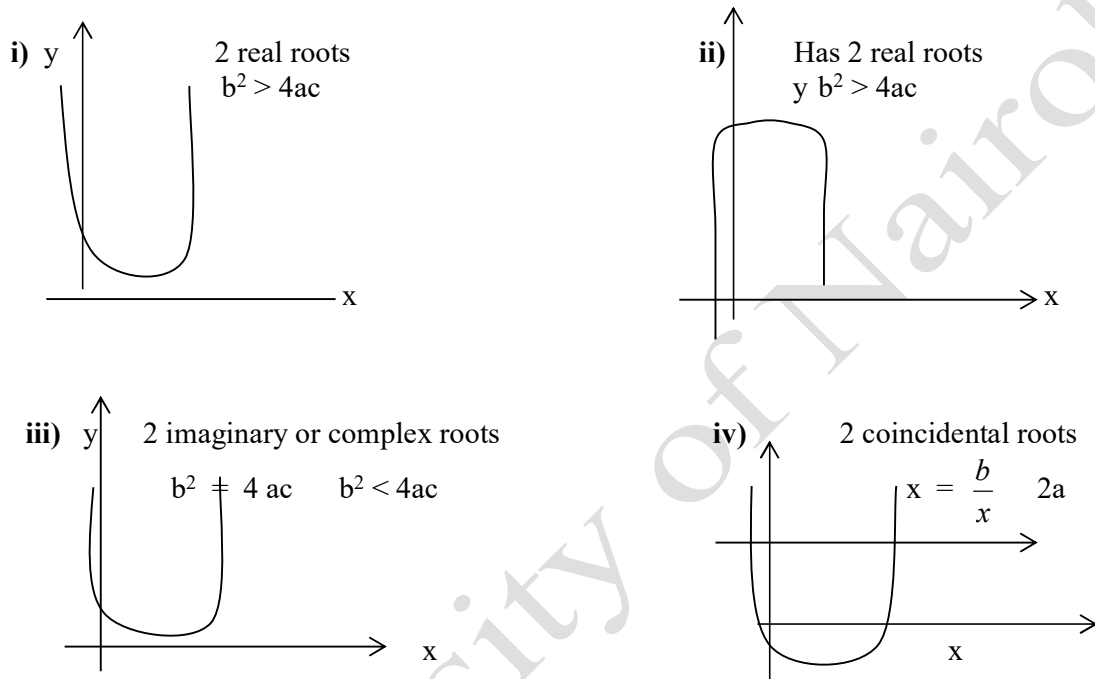
Recall

If $ax^2 + bx + c = 0$

$$\text{Then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. A quadratic function has a single turning point.
3. It is completely specified once three points which lie on the curve are given.

Quadratic sketches:



Equations:

The supply function of a commodity is quadratic and passes through the points shown below:

P	30	40	50
Q	500	3600	6300

as $q = f(p)$, i.e. in the form:

Solution

$$(1) \quad 500 = a + 30b_1 + 900b_2$$

$$(2) \quad 3600 = a + 40b_1 + 1600b_2$$

$$(3) \quad 6300 = a + 50b_1 + 2500b_2$$

$$\text{Equation 2 - Equation 1: } 3100 = 10b_1 + 700b_2 \dots (4)$$

$$\text{Equation 3 - Equation 2: } 2700 = 10b_1 + 900b_2 \dots (5)$$

$$(4) - (5) \quad 400 = -200 b_2$$

$$\therefore b_2 = \frac{400}{-200} = -2$$

Substituting:

$$3100 = 10b_1 + 700(-2)$$

$$3100 = 10b_1 - 1400$$

$$\frac{310000 + 1400}{10} b_1 = 450$$

Using equation 1:

$$500 = a + 30(450) + 900(-2)$$

$$500 = a + 13500 - 1800$$

$$\therefore a = 500 - 13500 + 1800$$

$$a = -11,200$$

Thus the Supply function is

$$q = -11200 + 450p - 2p^2$$

Example;

The demand function of a certain commodity is quadratic and passes through the points (p,q) = (5,1600); (10,900); (20,100);

Determine the function in form; $q = a + b_1 p + b_2 p^2$

Solution:

$$q = a + b_1 p + b_2 p^2$$

Equations

$$(1) \quad 1600 = a + 5b_1 + 25b_2$$

$$(2) \quad 900 = a + 10b_1 + 100b_2$$

$$(3) \quad 100 = a + 20b_1 + 400b_2$$

$$(4) = (1) - (2): \quad 700 = -5b_1 - 75b_2$$

$$(5) = (2) - (3): \quad 800 = -10b_1 - 300b_2$$

$$2: \quad 400 = -5b_1 - 150b_2$$

$$2: \quad = (4) - (6): \quad 300 = 75b_2$$

$$b_2 = 300/75 = 4$$

Substituting in equation (4):

$$700 = -5b_1 - 75(4) \quad 700 = -5b_1 - 300$$

$$\therefore b_1 = \frac{700 + 300}{-5} = \underline{-200}; b = -5$$

Using equation 1

$$1600 = a + 5(-200) + 25(4)$$

$$1600 = a - 1000 + 100$$

$$\therefore a = 1600 + 1000 - 100$$

$$a = 2500$$

Thus, the Demand function is:

$$q = 2500 - 200p + 4p^2$$

Example:

A revenue function is quadratic in nature. When $x=5$, $R=50$ whereas when $x=4$, $R=48$. Determine the revenue function.

Solution:

Let the revenue function R be:

$$R = a + b_1x + b_2x^2$$

When $x = 0$, revenue = 0

$$\therefore a = 0$$

hence $R = b_1x + b_2x^2$

Equations

$$(1) \quad 50 = 5b_1 + 25b_2$$

$$(2) \quad 48 = 4b_1 + 16b_2$$

$$(3) \quad = (2) \times 1.25 \quad 60 = 5b_1 + 20b_2$$

$$(4) \quad = (1) - (3) \quad -10 = 5b_2$$

$$b_2 = \frac{-10}{5}$$

$$b_2 = -2.$$

Using equation 1

$$50 = 5b_1 + 25(-2)$$

$$\therefore b_1 = \frac{50 + 25(2)}{5}$$

$$B = 20$$

$$\therefore \text{Revenue function is: } R = 20x - 2x^2$$

Extension:

Determine the demand function and hence price when quantity is $x = 5$ units

$$\text{Demand function: } P = R/X = \frac{20 + 2x^2}{x} \quad P$$

$$= \underline{20 - 2x}$$

$$\text{When } x = 5, p = 20 - 2(5) = \text{Sh } 10$$

Cubic Functions and Applications

These are 3rd degree polynomials and their general form is given as:

$$Y = a + b_1 x + b_2 x^2 + b_3 x^3$$

Where y = dependent variable

x = independent variable

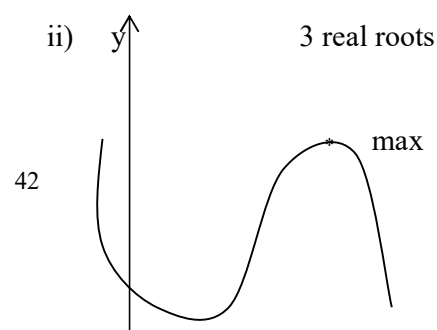
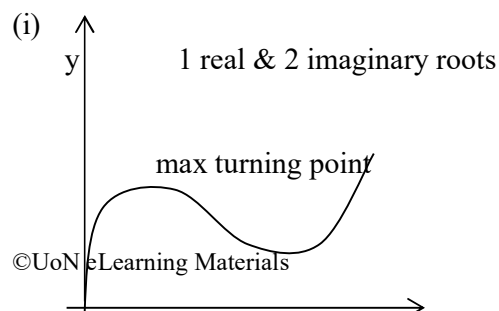
a, b_1, b_2, b_3 = Constants

Note that $b_3 \neq 0$

Properties:

1. It will have at most 3 real roots (i.e. the maximum number of times it can cross the x – axis is three times)
2. It will have either 2 turning points (one a maximum and the other a minimum) or one point of inflexion.
3. It is completely defined once 4 points which lie on the curve are specified.

Cubic sketches



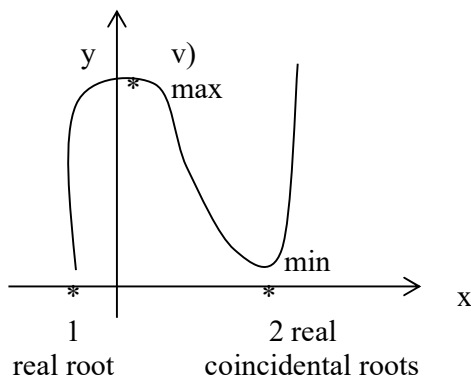
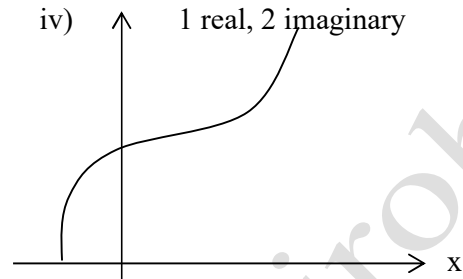
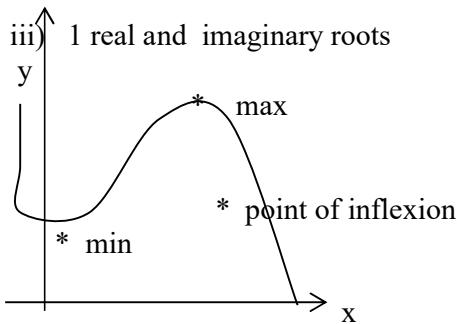
*

* minimum turning point

x

→ x

* min



A management accountant is studying the relationship between the number of units of output in one year and the total cost incurred from a given product. From the records of the firm, the following data was extracted:

Output, Q	Total Cost, C
0	120
1	124
3	120
5	140

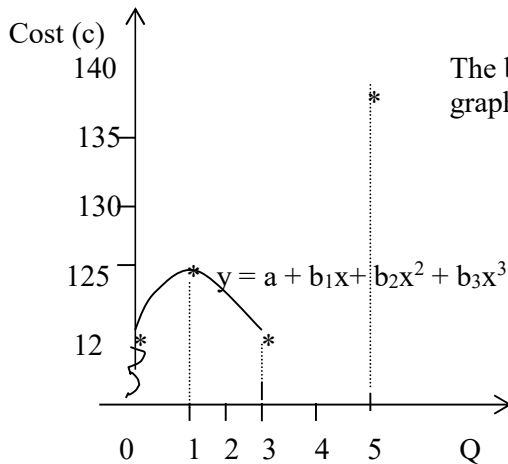
- Determine the firm's fixed cost.
- Plot the above data on a graph and hence recommend the best functional form within the given range.
- Without prejudice to your answer in b) above, fit a 3rd degree polynomial to the data above, i.e. of the form: $C = a + b_1Q + b_2Q^2 + b_3Q^3$ and hence estimate the total cost if level of output equals eleven (11) units.

Solution

a) Fixed cost is the cost when $Q = 0$

Hence, fixed cost = Sh.120

b) **Graph**



The best functional form is cubic since the graph has two turning points.

c) General form: $C = a + b_1 Q + b_2 Q^2 + b_3 Q^3$

Equations:

When $Q = 0$ $C = 120 \rightarrow a = 120$

$$1) 124 = 120 + b_1 + b_2 + b_3 \leftrightarrow b_1 + b_2 + b_3 = 4$$

$$2) 120 = 120 + 3b_1 + 9b_2 + 27b_3 \leftrightarrow 3b_1 + 9b_2 + 27b_3 = 0$$

$$b_1 + 3b_2 + 9b_3 = 0 \text{ (division by 3)}$$

$$3) 140 = 120 + 5b_1 + 25b_2 + 125b_3 \leftrightarrow 5b_1 + 25b_2 + 125b_3 = 20$$

$$b_1 + 5b_2 + 25b_3 = 4 \text{ (division by 5)}$$

Equation 1 - Equation 2

$$(4) = (1) - (2): -2b_2 - 8b_3 = 4$$

$$(5) = (2) - (3): -2b_2 - 16b_3 = -4$$

$$(6) = (4) - (5): 8b_3 = 8$$

$$\therefore b_3 = \frac{8}{8} \quad b_3 = 1$$

Substituting in equation 4 Using equation 1:

$$\begin{array}{l|l} -2b_2 - 8(1) = 4 & b_1 + (-6) + 1 = 4 \\ -2b_2 = 8 + 4 & \therefore b_1 = 4 + 6 - 1 = \underline{9} \end{array}$$

$$-2b_2 = 12$$

$$b_2 = \frac{12}{-2} = -6$$

Hence the total cost function is

$$C = 120 + 9Q - 6Q^2 + Q^3$$

When $Q = 11$,

$$C = 120 + 9(11) - 6(11)^2 + 11^3$$

$$= \underline{\text{Sh.824}}$$

4.5

Summary



In this lecture, we have looked at quadratic and cubic functions. We have studied their properties, how to get their equations and how to sketch their graphs. We have also looked at the importance and application of these functions in solving business problems. In the next lecture, you will study multivariate, exponential and logarithmic functions.

Activity 4.1



Sosina Ltd. specializes in renting out a certain type of equipment. In their planning Endeavour, they have invited you to analyze the relationship between profit and the number of units of the equipment rented out. You obtain the following data from their records:

Number of units rented out per day	Total Daily profit (\$)
20	60
30	110
40	140

Further, the firm does not rent out more than 100 units daily and within this relevant range (i.e. 0 – 100) , you think that a quadratic model is the most suitable.

Required:

- (a) Determine the function relating daily profit to the number of units rented out.
- (b) From the function above, determine the daily level of fixed cost and the average daily profit per unit of equipment.
- (c) At what level of equipment rental are total daily profit equal to zero (i.e. the break even point)?
- (d) Determine the level of equipment to be rented out per day that maximizes daily profit and this profit.

LECTURE FIVE

MULTIVARIATE, EXPONENTIAL AND LOGARITHMIC FUNCTIONS

Lecture Outlines

- 5.1 Introduction
- 5.2 Objectives
- 5.3 Multivariate Functions
- 5.4 Exponential Functions
- 5.5 Modified Exponential Functions
- 5.6 Logarithmic Functions
 - 5.6.1 Properties of Logarithms
 - 5.6.2 Solving Logarithmic Exponential Equations
 - 5.6.3 Application of Exponential & Logarithmic Functions
- 5.7 Summary

5.1 Introduction

Welcome to lecture five. We are still studying functions. In this lecture, we shall study functions which have more than one independent variables (multivariate functions) and functions which are quite useful in describing growth and decay phenomena (exponential and logarithmic functions).

5.2	Objectives At the end of this lecture, you should be able to: 1. Describe and apply multivariate functions. 2. Identify and sketch general exponential and logarithmic functions. 3. Solve simple exponential and logarithmic equations. 4. Use the exponential functions to model growth and decay processes.
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5.3 Multivariate Functions

In business, there are many dependent variables of interest which are determined by two or more independent variables, for example:

- Savings can be determined by income, household size and interest rates;
- Sales level can be determined by level of advertisement and prices of related goods and so on.

These are known as multivariate functions.

.Example:

Two products which are substitutes, X and Y are sold by Kilifi Traders. They have recently become interested in their profit function and a consultant has found out that fixed costs amount to Sh 100.

Required:

If the function of profit is given by the following form:

$$\Pi = ax + by - cx^2y^2 - f$$

Where a, b and c are the constants (coefficients) of the first three terms respectively, and f is fixed cost, establish the profit function.

Hence determine the profit when $x = y = 5$ units.

Solution:

We substitute the 3 given points to the general structure of the function to obtain the following equations:

$$(i) \quad 249 = a + b - c - 100$$

$$(ii) \quad 966 = 4a + 7b - 4^2 + 7^2 c - 100$$

$$\text{Therefore: } 966 = 4a + 7b - 784c - 100$$

$$(iii) \quad 96 = 8a + 6b - 8^2 * 6^2 c - 100$$

$$\text{Which gives: } 96 = 8a + 6b - 2304c - 100$$

Summary of the simplified equations:

$$(i) \quad 249 = a + b - c - 100 \text{ which gives: } a + b - c = 349$$

$$(ii) \quad 966 = 4a + 7b - 784c - 100 \text{ which gives: } 4a + 7b - 784c = 1066$$

$$(iii) \quad 96 = 8a + 6b - 2304c - 100 \text{ which gives: } 8a + 6b - 2304c = 196$$

Exercise

Complete the solution to the above equation to obtain $a = 200$, $b = 150$, $c = 1$ so that the multivariate profit function is:

$$\Pi = 200x + 150y - x^2 - y^2 - 100$$

Thus, when $x = y = 5$ units,

$$\text{Profit, } \Pi = 200(5) + 150(5) - 5^2 - 5^2 - 100$$

$$= \underline{\text{Sh. 1600}}$$

5.4 Exponential Functions

These are functions whose at least one term has the independent variable as part of an exponent or power e.g.

(1) $y = 15^{2x}$ (2) $y = 72 - 23^x$ (3) $y = 2x + 21^{0.1x}$

An important class of exponential functions is those of the form

$$y = ab^{mx}$$

where a , b and m are constant and also b has to be greater than zero but b cannot be 1.

i.e. $b > 0$

$b \neq 1$

Consider the functions:

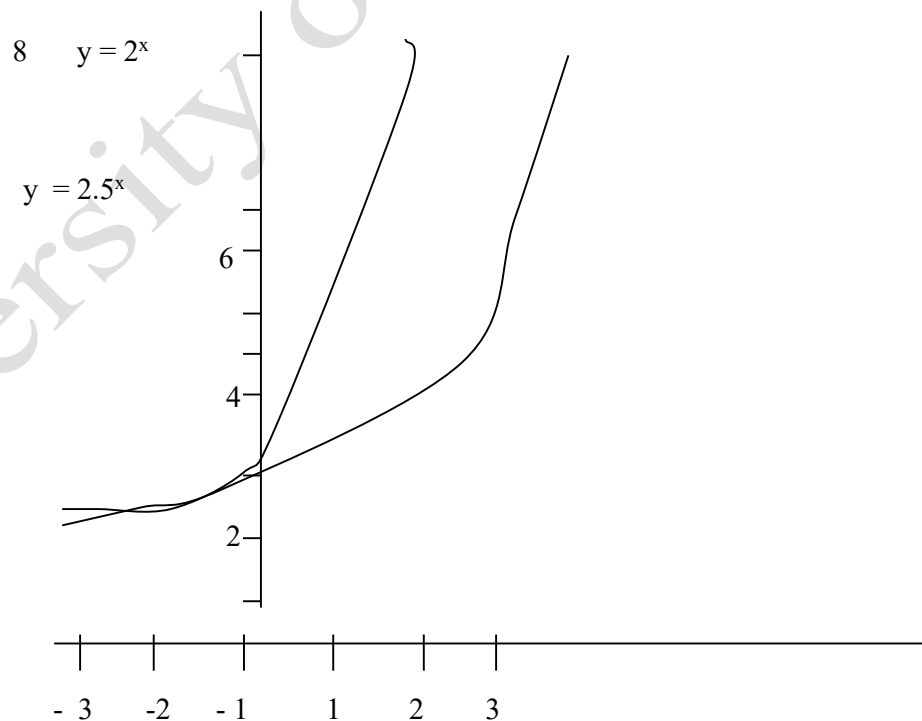
(a) $y = 2^x$ and ;

(b) $y = 2.5^x$

Sketch the graph

a) $y = 2^x$

X	-3	-2	-1	0	1	2	3
Y	0.125	0.25	0.5	1	2	4	8



b) $y = 2.5^x$

X	-3	-2	-1	0	1	2	3
y	0.064	0.16	0.4	1	2.5	6.25	15.625

Characteristics of exponential functions

$$y = b^x$$

where $b > 0$ but $b \neq 1$

- 1) The graph of the function lies entirely above the x – axis i.e. y is always positive.
- 2) The graph y is asymptotic to the x – axis i.e. the value of y approaches but never reaches the x-axis as x approaches $-\infty$
- 3) The y intercept occurs at (0,1) i.e. when x is 0, $y = 1$
- 4) Y is an increasing function of x.
- 5) The larger the value of the base b, the greater the rate of increase in y as x increases.

This class of functions is useful in modeling growth processes.

Consider exponential functions of the form:

$$y = b^x \text{ where } 0 < b < 1.$$

In particular sketch the graphs of the functions:

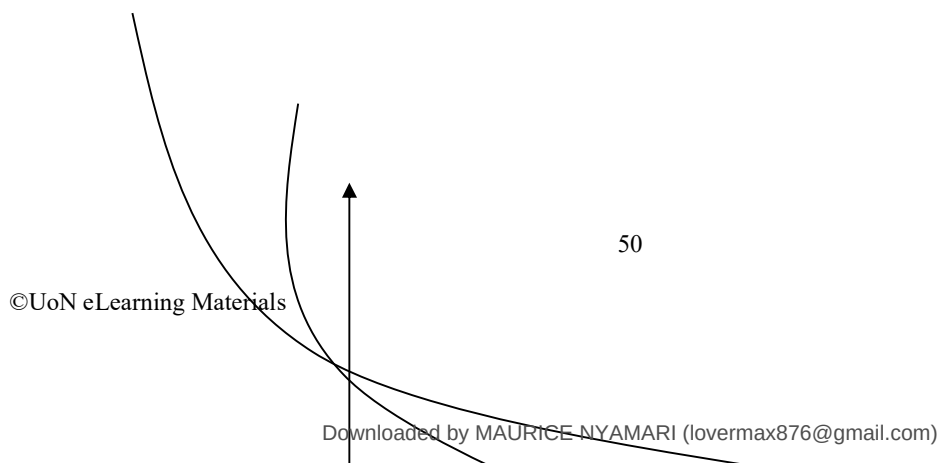
- (a) $y = 0.4^x$ and (b) $y = 0.8^x$

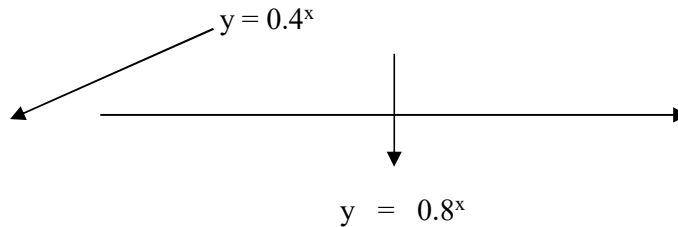
$$y = 0.4^x$$

X	-3	-2	-1	0	1	2	3
Y	15.625	6.25	2.5	1	0.4	0.16	0.064

$$Y = 0.8^x$$

X	-3	-2	-1	0	1	2	3
Y	1.953	15.625	1.25	1	0.8	0.64	0.512





Characteristics of exponential functions $y = b^x$ where $0 < b < 1$

1. The graph of the function lies entirely above the x – axis, i.e. y is always positive.
2. The graph is asymptotic to the x -axis; i.e. the value of y approaches but never reaches the x axis as x approaches to ∞
3. The y intercept occurs at $(x, y) = (0, 1)$
4. Y is a decreasing function of x .
5. The larger the value of the base, b , the greater the rate of decrease in y as x increases.

This class of functions is useful in modeling decay processes.

Base Exponential Functions

Base e exponential function is a special class of exponential functions taking the form: $y = ae^{mx}$ where e = Euler's constant.

This is associated with natural growth or natural decay. Euler's constant, e , occurs in nature just like 3.14 or the speed of light. It becomes more accurate as we make n larger in the expression $e = (1 + 1/n)^n$

e.g. when $n = 1$

$$e = 1 + \left(\frac{1}{1}\right)^1 = 2$$

when $n = 2$

$$e = \left(1 + \frac{1}{2}\right)^2 = 2.25$$

when $n = 10$

$$e = \left(1 + \frac{1}{10}\right)^2 = 2.5937$$

when $n = 50$

$$e = \left(1 + \frac{1}{50}\right)^{10} = 2.691$$

when $n = 100$

$$e = \left(1 + \frac{1}{100}\right)^{100} = 2.7048$$

when $n = 1000$

$$e = \left(1 + \frac{1}{1000}\right)^{1000} = 2.7169$$

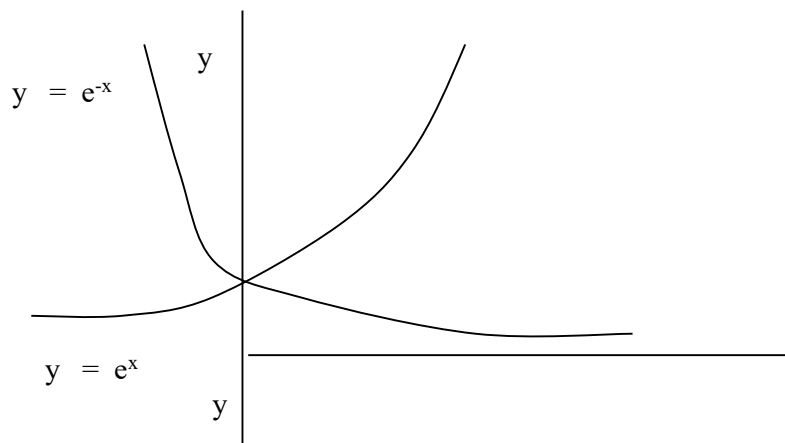
From a calculator, the value of e is obtained by raising e to power 1 in the function e^x , to four decimal places, $e^1 = e = 2.7183$

Activity 5.1

Draw the graph

$$1) \quad \begin{array}{c|ccccccc} y = e^x & x & -3 & -2 & -1 & 0 & 1 & 2 & 3 \\ \hline y & & 0.050 & 0.135 & 0.367 & 1 & 2.718 & 7.389 & 20.085 \end{array}$$

$$2) \quad \begin{array}{c|ccccccc} y = e^{-x} & x & -3 & -2 & -1 & 0 & 1 & 2 & 3 \\ \hline y & & 20.08 & 7.389 & 2.718 & 1 & 0.367 & 0.135 & 0.050 \end{array}$$



5.5 Modified Exponential Functions

Certain applications of exponential functions involve functions of the form:

$$y = k - ae^{mx}$$

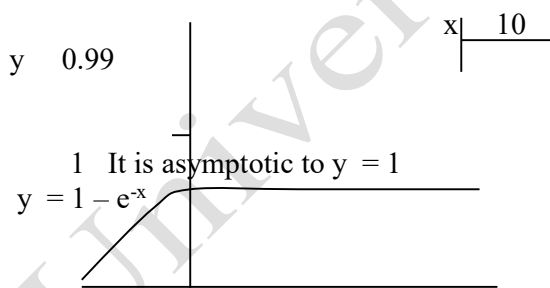
where k, a, m = constants

Example

Draw the graph

3)	$y = 1 - e^{-x}$	x	-3	-2	-1	0	1	2	3
	y		-19.086	-6.89	1.718	0	0.632	0.8647	0.9502

Graph



As $x \rightarrow \infty$ The symbol $\rightarrow$ is used to mean approaches “or” “tends to”

$$y = 1 - e^{-x} \rightarrow 1$$

$$\text{Since } e^{-x} \rightarrow 0$$

Applications of Exponential functions

Areas of Use

- Growth processes in business increase;
- Population growth;
- Appreciation in the value of assets,
- Inflation growth;
- Rate at which particular resources are used (e.g. demand for energy), growth in GNP (Gross Net Product, etc.

Decay processes

Examples of decay processes are: Declining value of assets e.g. motor vehicles (depreciation), decline in the incidences of certain diseases as medical research advances, decrease in the purchasing power of the shilling etc.

When a growth process is characterized by a constant percentage increase in value it is referred to as exponential growth process and when it is characterized by a constant percent decrease. It is referred to as an **exponential decay process**.

Although exponential decay and growth functions are usually stated as functions of time, the independent variable may represent something other than time.

Regardless of the nature of the independent variable, the effect is that equal increases in the independent variable result in constant percent changes (increases or decreases) in the value of dependent variable

Illustration

1. Population growth

The population of a country was 100 million in 1990. It has grown since that time at 4% p.a and the growth is exponential.

Required

- (a) What is the function which describes population growth through time?

$$\text{i.e. } P = f(t)?$$

- b) What is the population in year 2010?

Solution

(a) The function is:

$$y = ae^{kt}$$

where $a \equiv y$ when $t = 0$

$k \equiv$ decay or growth rate

1990

$$t = 0$$

$$k = 0.04$$

$$P = 100e^{0.04t}$$

(b) In 2010: $t = 2010 - 1990$ $t = 20$ years.

$$P = 100e^{0.04 \times 20}$$

$$p = 100e^{0.8}$$

$$= 222.5541 \text{ million}$$

1. Asset valuation

The depreciation function, V , of a certain piece of industrial equipment has been found to be described by the function:

$$V = 100e^{-0.1t}$$

Where $t \equiv$ time in years

$V =$ book value at time t

Required:

- a) What is the annual depreciation rate?
- b) What was the purchase cost of the asset?
- c) What is the book value after
 - i) 6 years?
 - ii) 11 years?

Solution

- a) Annual depreciation rate is 10%
- b) $V = 100e^0 = 100,000$
- c) i) $V = 100e^{-0.1 \times 6} = 100e^{-0.6} = \text{Sh } 54,881$
ii) $V = 100e^{-0.1 \times 12}$
 $V = 30,120$

2. Debtors managements

For a certain good, the collection percentage of credit issued in any month is an exponential function of the time since credit was issued. Specifically the function which approximates this relationship is

$$P = 0.95 (1 - e^{-0.7t}) \quad t \geq 0$$

P = percentage of debtors (in shillings) collected t months after the credit is granted

Required

- a) Calculate the percentage of debtors recovered after
 - i) 3 months
 - ii) 7 months
- b) Given this function, what is the percentage of debtors which should be set aside as bad debt?

Solutions

a) $P = 0.95 (1 - e^{-0.7(3)}) = 0.834$

$$P = 0.95 (1 - e^{-2.1}) = 83.4\%$$

b) $P = 0.95 (1 - e^{-0.7(7)}) = 0.943$

$$P = 0.95 (1 - e^{-4.9}) = 94.2\%$$

c) As $t \rightarrow \infty$ $e^{-0.7t} \rightarrow 0$

$$P \rightarrow 95\%; \text{ Bad debts} = 5\%$$

3. Compound interest

A certain amount of money P is deposited in an account which earns compound interest at a rate of

i % p.a. the amount after n years the total sum, S (principal + interest) is given as:

$$S = P (1 + i)^n$$

e.g. let $P = 1000$

$$i = 8\%$$

What is the sum, S after 25 years if interest is paid annually?

$$S = 1000 (1.08)^{25} = 6848.475$$

Let us investigate the effect of more frequent compounding within the year.

Semiannual compounding

$$S = 1000 \left(\frac{1 + 0.08}{2} \right)^{25 \times 2}$$
$$= 1000 (1.02)^{50} = 7106.68$$

Quarterly compounding

$$S = 1000 \left(\frac{1 + 0.08}{4} \right)^{25 \times 4}$$
$$= 1000 (1.04)^{100} = 7244.65$$

Let the number of compounding periods in the year be m and number of years be t

$$\therefore S = \left(\frac{P(1+i)}{m} \right)^{mt}$$

What is the formula for S if interest is compounded continuously?

Solution

$$S = P \left(\frac{1+i}{m/i} \right)^{mt}$$

This can be rewritten as

$$S = P \left(1 + \frac{i}{m/i} \right)^n \quad \text{since } m/i \cdot t = mt$$

Recall

$$e = \text{Limit} \left(1 + \frac{1}{n} \right)^n \quad \text{as } n \rightarrow \infty$$

If $m/i = n$ then it follows that $s = e^{it}$

Therefore, in the proceeding illustration for continuous compounding the sum S after 25 years at 8% interest rate p.a. is given as:

$$S = 1000 e^{0.08 \times 25}$$
$$S = 7389.06$$

Logarithmic Functions

A logarithm is a power to which a base must be raised in order to give a certain number i.e. a logarithm is an exponent

Consider the following equation

$$2^3 = 8$$

Figure 3 is the logarithm to base 2 of the number 8 written as

$$= \text{Log}_2 8$$

Generally, if $y = b^x$, then

$$x = \log_b y$$

Equivalent logarithmic and Exponential forms

Exponential equation Logarithmic equation

$$10^4 = 10,000 \quad 4 = \text{Log}_{10} 10,000$$

$$2^4 = 16 \quad 4 = \log_2 16$$

$$5^2 = 25 \quad 2 = \text{Log}_5 25$$

$$10^2 = 100 \quad 2 = \text{Log}_{10} 100$$

Note

Although logarithms can be taken to any base, the most commonly used bases are base 10 and base e. Base 10 (denoted log) are also known as briggsian logarithms (after Henry Briggs who first used them).

Base e (denoted ln) are also known as natural logarithms or naperian logarithms (after John Napier).

5.6.1 Properties of Logarithms

(a) The log of UV = Log U + Log V

$$\begin{aligned} \text{e.g. } \text{Log}_{10}(1000) &= \log_{10} 10 + \text{Log}_{10} 100 \\ &= 1 + 3 = 4 \end{aligned}$$

(b) Log U/V = Log U - Log V

$$\begin{aligned} \text{e.g. } \text{Log}_{10} 1000 - \text{Log}_{10} 100 &= 1 \\ \text{Log}_{10} 1000 - \text{Log}_{10} 100 \\ &= 3 - 2 = 1 \end{aligned}$$

(c) $\text{Log}_b u^n = n \log_b u = \log_{10} 10000 = 4$

$$\text{e.g. } \text{Log}_{10} 100^2 = \text{Log}_{10} 100 \times 2 = 4$$

(d) $\text{Log}_b b = 1$

e.g. $\text{Log}_{10} 10 = 1$

since $10^1 = 10$

(e) $\text{Log}_b 1 = 0$

Since $b^0 = 1$ provided b is not zero

(f) $b^{\log_b x} = x$

e.g. $100^{\log_{10} 100} = 100$

(g) $\text{Log}_b b^x = X \log_b b$

$= x$ since $\log_b b = 1$

e.g. $\text{Log} 2^5 = \log_2 32 = 5$

5.6.2 Solving Logarithmic and Exponential Equations

a) Sol $b_3 = 1$ ve for x in the following equation

Solution

$$\ln x^2 + \ln x = 9$$

$$2 \ln x + \ln x = 9$$

$$3 \ln x = 9$$

$$\ln x = 9/3 = 3$$

$$e^3 = x$$

$$x = 20.0855$$

b) $\ln (x^2 + 2) - (\ln x^2) = 2$

$$\ln = \frac{2(x^2 + 2) = 2}{x^2}$$

$$\log_e \frac{x^2 + 2}{x^2} = 2$$

$$e^2 = \frac{x^2 + 2}{x^2}$$

$$\frac{x^2 + 2}{x^2} = 7.3891 \times 2x^2$$

$$x^2 + 2 = 7.3891 x^2$$

$$2 = 7.3891 x^2 - x^2$$

$$2 = 6.3891x^2$$

$$x^2 = \frac{2}{6.3891}$$

$$x = \pm \sqrt{\frac{2}{6.3891}}$$

$$x = \pm 0.5595$$

c) $e^{2x} = 5$

$$\ln e^{2x} = \ln 5 \quad \text{Note } \ln e = \log_e e = 1$$

$$2x \ln e = \ln 5$$

$$2x = \ln 5$$

$$x = \frac{\ln 5}{2}$$

$$x = 0.8047$$

5.6.3 Application of Exponential & Logarithmic Functions

(a) Cost Analysis

The cost C of processing an application for a certain task is given as

$$C = 0.001x^2 - 5 \ln x + 60$$

where x = number of analysts

Required:

Calculate the cost of

(i) $x = 40$

(ii) $= 50$

Solution

(i) For $x = 40$, $\text{Cost} = 0.001(40)^2 - 5 \ln 40 + 60$
 $= 43.16$

(ii) For $x = 50$, $\text{cost} = 0.001(50)^2 \ln 50 + 60$
 $= 42.94$

(b) Learning Curves

The number of items, y , produced each day by an assembly line worker, x days after an initial training period, is modeled by:

$Y = 120 - 80e^{-0.30x}$ where y = number of units completed per day and x = number of days of experience of employees.

Required:

- a) Calculate the number of units produced per day after:
 - (i) one day after training
 - (ii) 10 days after training
- b) What is the daily rate after 10 days of experience?
- c) After how many hours will production rate be 90 units?
- d) What is the production rate after many days of experience i.e. steady state?

Solution

(a) (i) $Y = 120 - 80e^{-0.3 \times 1}$
 $= 120 - 80e^{-0.3}$
 $= 61 \text{ units}$

(ii) $y = 120 - 80e^{-0.3 \times 10}$
 $y = 120 - 80e^{-3}$
 $= 116 \text{ units}$

(b) Daily rate after 10 days of training is therefore 116 units

(c) $90 = 120 - 80e^{-0.3x}$
 $80e^{-0.3x} = 120 - 90$

$$\frac{80e^{-0.3x}}{80} = \frac{30}{80}$$

$$e^{-0.3x} = 0.375$$

$$\text{Lne}^{-0.3x} = \text{Ln } 0.375$$

$$-0.3x \text{ Lne} = \text{Ln } 0.375$$

$$\text{but Lne} = 1$$

$$\therefore -0.3x = \text{Ln } 0.375$$

$$x = \frac{\text{Ln}0.375}{-0.3}$$

$$x = 3.27 \text{ hours}$$

$$\text{when } x \rightarrow \infty$$

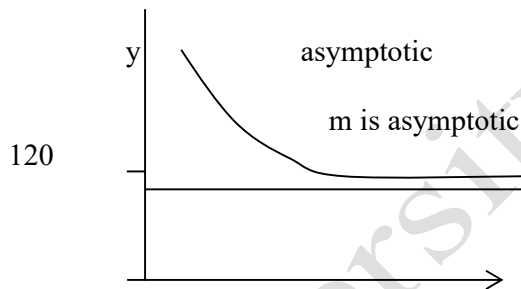
$$80 e^{-0.3x} \rightarrow 0$$

$$\therefore y \rightarrow 120$$

(c) After many days of experience, or steady state.

This can be represented as below:

$$y = 120 \text{ units per hour}$$



5.7

Summary



In this lecture, we have looked at multivariate functions, which are useful in describing situations of many predictor variables. We also studied exponential and logarithmic functions, which are important in modeling growth and decay processes. This has marked the end of the study of functions for this course. In the next lecture, we begin the study of the mathematical analysis of change or calculus.

Activity 5.2



Question One

The monthly profit of Sweet Chew Chocolate Ltd. is represented as a function of monthly sales S , units of the small size bar x , and the big size bar y , sold each month. Specifically the function is of the form

$$S = ax^2 + by^2 + cxy$$

Where a , b , and c are constants. Observations in recent past indicate that monthly sales were K£50,950 when 500 and 1,000 units of small and large bars respectively were sold; K£ 377,500 when 1500 and 2,500 respectively were sold; and K£ 603,300 when 2,000 and 3,000 units respectively were sold.

Required:

Determine the:

- (i) Sales function;
- (ii) Sales when 1,000 and 2,000 units of small and large bars respectively are sold.

Question Two

Kwale Traders sells all its products on credit. Data gathered over time indicate that the collection percentage for credit sales (debtors) issued in any month is an exponential function of the time since the sales were made. Specifically, the function approximating this relationship is:

$$P = 0.94(1 - e^{-0.085t}), t \geq 0$$

Where P equals the percentage of debtors (in shillings) collected t months after the sales are made.

(a) Required:

- (i) What percentage is expected to be collected after 1 month?
- (ii) What percentage is expected after 3 months?
- (iii) What value does P approach as t increases without limit and hence what is the expected percentage of bad debts?
- (iv) Provide a graphical sketch of the above function.

(b) Twaomin Ltd is hiring persons to work in its plant. For the kind of job available, it has been estimated by efficiency experts that the average cost, AC of performing the task is a function of the number of persons hired, x . Specifically,

$$AC = 0.00005x^2 - \ln 2x + 6$$

Required:

- (i) Determine the number of persons who should be hired to minimize the average cost.
Ensure it is minimum
- (ii) What is the minimum average cost?
- (iii) What is the minimum total cost?

Question Three

An advertising company is interested in the retention rate of a person exposed to an advert t hours after the subject views the advert. In one such advert, subjects were asked to look at a picture that contained many different objects.

At different intervals after this, they would be asked to recall as many objects as they could. Based on the experiment, the following function was developed:

$$R = 90 - 20 \ln t$$

Where R equals the average percentage recall and t equals time in hours since studying the picture.

- (i) Determine the recall percentage after 2 hours and after 10 hours
- (ii) Determine the time when the recall percentage is 50%
- (iii) At what time is the recall percentage estimated to be zero?
- (iv) Provide a graphical sketch of the above function

LECTURE SIX

CALCULUS

Lecture Outlines

- 6.1 Introduction
- 6.2 Objectives
- 6.3 Importance of Calculus
- 6.4 Linear functions and slopes
- 6.5 Summary

6.1 Introduction

Welcome to Lecture Six of this course unit. In Lectures three to five, we studied relationships between business variables which we called functions. In this lecture, we are now concerned about how changes in independent variables affect changes in the dependent variable.

This is important since it enables decision makers to control these changes to achieve desired objectives. For example, how does change in output affect change in profit?

The mathematical analysis of change or movement is known as calculus.

Background

Calculus is concerned with mathematical analysis of change or movement. There are 2 basic operations in calculus:

- Differentiation
- Integration

The two basic operations are inverse to one another as addition and subtraction or multiplication and division.

6.2



Objectives

At the end of this lecture, you should be able to:

1. Discuss the importance of calculus in solving business problems.
2. Distinguish between differentiation and integration.
3. Calculate and interpret the slope of a straight line.

6.3 Importance of Calculus in the Business and Economic World

There are two main uses of calculus in business:

1. Often we may be involved in optimisation of business variables, e.g.

Maximize: sales, productivity, profits etc or

- Minimize: costs, waste, losses, fatigue etc

This is also called classical optimisation

2. Calculus is used in marginal analysis eg. obtaining a total cost function from a marginal cost function, total productivity from marginal productivity etc.

Differentiation

Differentiation is concerned with the rates of change. Examples of rates of change of interest in business are:

- Profits with respect to output
- Savings with respect to income levels
- Revenues with respect to advertising levels
- Production with respect to hours worked
- Profits with respect to output and advertising level etc.

Rates of change and gradients (slopes)

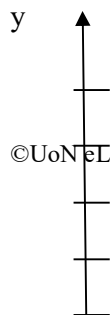
6.4 Linear functions and slopes

Consider the linear functions:

$y = 4 + 3x$	x-	1	0	1	2	3	4
Y		14	7	10	3	1	6

$y = -1 + 3x$	x-	1	0	1	2	3	4
Y-4-		1	2	5	8	1	1

y



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$$12 \quad y = 4 + 3x$$

10

$$8 \quad y = -1 + 3x$$

6

4

2

-1 0 1 2 3 4

-2

-4

Observation:

The two lines are parallel i.e. the rate of change of y with respect to x are the same or they have the same slope/gradient.

In both cases, when x changes by one unit, y changes by 3 units.

Therefore

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{3}{1} = 3 \quad \Delta = \text{Delta, sign for change}$$

Example: Consider the change from point (0,4) to (3,13) in equation 1.

$$\text{i.e. } \frac{\Delta y}{\Delta x} = \frac{13-4}{3-0} = \frac{9}{3} = 3$$

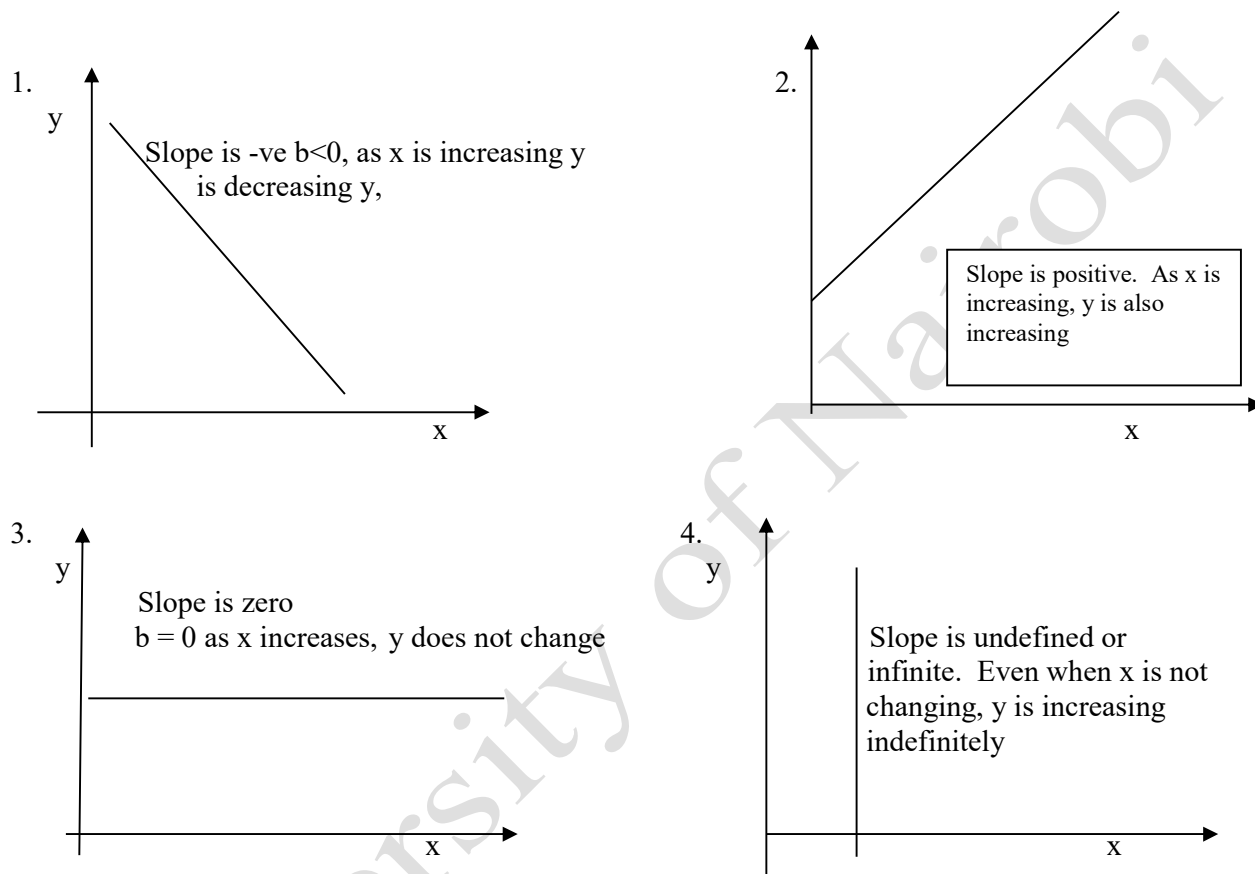
Thus, for any linear function, $y = a + bx$

$$\frac{\Delta y}{\Delta x} = b$$

..... The slope of a linear function is constant.

Forms of slopes of linear functions:

A linear function will have one of four types of slopes as shown by the following graphs:



Exercises

- i. What is the slope of line joining the points (2,5) and (4,13)

Solution

$$\frac{\Delta y}{\Delta x} = \frac{13-5}{4-2} = \frac{8}{2} = 4$$

- ii. What is the equation of the line joining the points (-3, -3) and has a slope = 2?

Solution

The general linear function is $y = a + bx$. But the slope is given, i.e. $b = 2$.

So, $y = a + 2x$.

To obtain the value of a , we substitute the known point as follows:

$$-3 = a + 2(-3)$$

$$-3 = a + 6$$

Therefore, $a = -3 - 6$

$$a = -9$$

Hence the linear function $y = -9 - 2x$.

6.5 Summary

This lecture has introduced you to calculus or the mathematical analysis of change. We have studied the difference between differentiation and integration and their significance in business. We have also learned how to calculate the slope of a straight line. In the next lecture, we shall explore how to determine slopes of non linear functions.

Activity 6.1



The demand function for a product is linear and it is of the form $p = a + bq$ where p = price and q = quantity. The values a and b are constants. When the price is Sh. 2, quantity demanded is 9 units. For every increase in price by Sh. 2, quantity demanded falls by 1 unit and vice versa.

Required:

- Determine the demand function and hence determine the price that leads to a demand of 4 units.
- What price results in zero demand?
- Determine the rate of change of price with unit with respect to quantity and interpret it.

LECTURE SEVEN

GENERAL PRINCIPLES OF DIFFERENTIATION

Lecture Outlines

- 7.1 Introduction
- 7.2 Objectives
- 7.3 Nonlinear Functions and Slopes
- 7.4 Concept of Limits and Continuity
- 7.5 Differentiation from 1st Principles
- 7.6 Rules and Techniques of Differentiation
- 7.7 Higher order Derivatives
- 7.8 Summary

7.1 Introduction

Welcome to Lecture Seven of this course unit. In Lecture Six, we discussed slopes of linear functions. Although linear functions represent many business and economic problems, there are also many other problems which are described by non-linear functions. In this lecture, we are concerned with slopes, or rates of change of non-linear functions.

7.2 Objectives



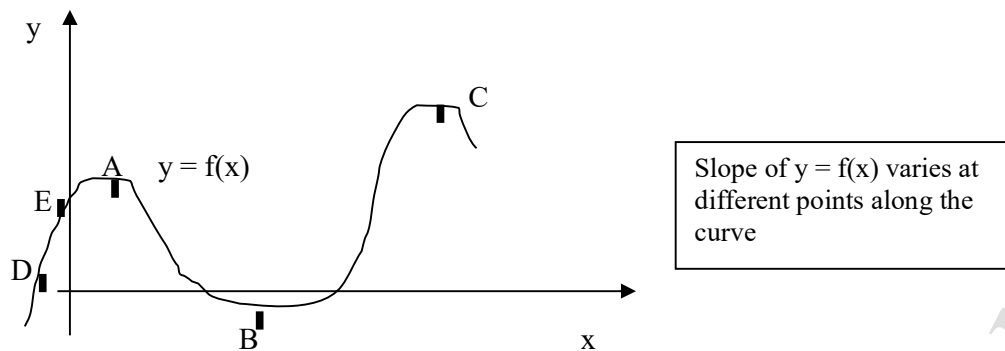
At the end of this lecture, you should be able to:

1. Distinguish between slope of a linear and a non-linear function.
2. Explain the concept of continuity.
3. Solve problems relating to limits.
4. Define the derivative of a function.
5. Use the rules of differentiation to find the derivatives of power functions.
6. Obtain higher order derivatives.
7. Obtain higher order derivatives.

Non-linear functions and Slopes

The slope of a straight line is constant. However, slopes of other (non-linear) functions are different at different points or sections of the function.

Consider the following general functions:



Observations

- To the left of point A, the slope is positive
- Between A and B, the slope is negative
- Between points B and C, slope is positive
- Slope at D is greater than slope at E, i.e. it is more steep at point D than at point E.

The problem then is to calculate the slope of non-linear functions at any points of interest given the function.

Concept of Limits and Continuity

An idea toward understanding differentiation is the concept of the limit. To understand this concept, we shall use simple examples.

Examples:

1. Find the limit of the function $y = x + 2$ as $x \rightarrow 0$ (= approaches/tends to)

as $x \rightarrow 0$, $y \rightarrow 2$ As x increasingly gets closer to 0, y gets closer to 2.

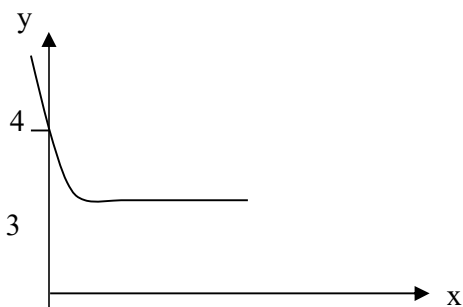
2. What is the limit of $y = 3 + (\frac{1}{2})^x$ as x increases without limit?

We can answer this question by constructing the following table:

x	1	1	2	4	1	0
y	3.5	3.2	5	3	.06253	.00097653

Note that as x is increasing, y is getting closer and closer to 3, written as follows:

As $x \rightarrow \infty$, $y \rightarrow 3$ since $(1/2)^x \rightarrow 0$. Below is the sketch of this information.



3. What is the limit of $y = \frac{1}{x^4}$ i) as $x \rightarrow \infty$?

ii) As $x \rightarrow 0$?

Solution:

As $x \rightarrow 100$

i)	x	1	2	3	4	5	10	15		y	$0 \rightarrow$
	y	1	0.625	0.1230	0.0039	0.0016	0.0001	0.0000198			$\rightarrow$

ii)

x	1	0.1	0.01	0	As $x \rightarrow 0$	$\rightarrow$
y	1	10,000	100,000,000	∞	$y \rightarrow \infty$	$\rightarrow$

4. Evaluate the following limit $y = \frac{x^2 - 4}{x - 2}$ as $x \rightarrow 2$

Solution

The limit can be approached from below or from above as given in the following tables:

(i) From below:

x	1.9	1.99	1.999
y	3.9	3.99	3.999

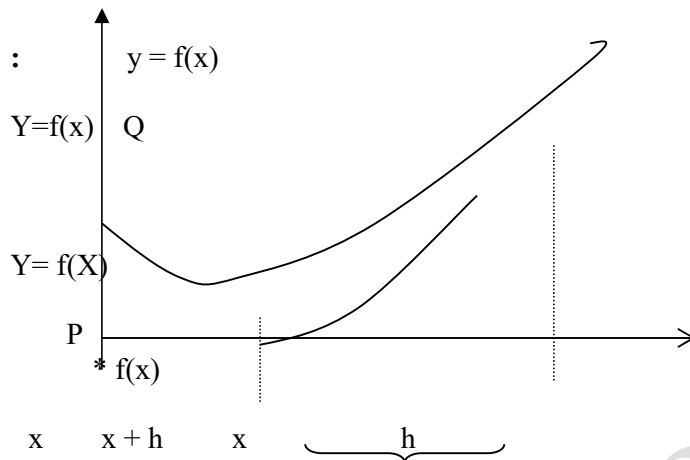
as $x \rightarrow 2$, $y \rightarrow 4$

(ii) From above

x	2.1	2.01	2.001	as x → 2
y	4.1	4.01	4.001	y → 4

7.5 Differentiation From First Principles

Consider two points P and Q in the general function $y = f(x)$



We are interested in obtaining the slope of the function at a particular point, say point P. We can approximate it using the slope of line PQ. At point P, the independent variable has a value of x and the corresponding value of the dependent variable can be determined by evaluating $f(x)$. At point Q, the independent variable has changed in value to $x+h$, and the corresponding value of the dependent variable can be determined by evaluating $f(x+h)$. In moving from point P to point Q, the change in the value of x is $(x+h) - x$ or h . The associated change in the value of y is:

$f(x+h) - f(x)$. Thus, the slope of line PQ is:

$$\frac{\text{Change in } y}{\text{Change in } x} = \frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{h}$$

If we want the slope at exactly point P, we must make h to be smaller and smaller, i.e. we make it tend towards zero, i.e. ($h \rightarrow 0$)

As h tends to 0, $\frac{\Delta y}{\Delta x}$

$\frac{\partial y}{\partial x}$ where $\frac{\partial y}{\partial x}$ is the rate of change at a point and it is also referred to as the instantaneous rate of change. It is also called the derived function or simply the derivative.

Definition: The derivative

Given a function of the form $y = f(x)$, the derivative of the function is:

$$\frac{\partial y}{\partial x} = \frac{f(x+h) - f(x)}{h} \text{ as } h \rightarrow 0$$

Activity 7.1

1. Determine the slope (instantaneous rate of change) for $y = x^2$ when $x = 2$

Solution:
$$\frac{\Delta y}{\Delta x} = \frac{(x+h)^2}{h} - x^2 = \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$\frac{\partial y}{\partial x} = 2x + h$$

$$\text{As } h \text{ tends to } 0, \frac{\partial y}{\partial x} \rightarrow \frac{\partial y}{\partial x} = 2x$$

This is the equation of the slope and enables the slope to be calculated at any point on the curve, e.g.

$$\text{when } x = 2, \text{ slope } \frac{\partial y}{\partial x} = 2 \times 2 = 4$$

$\therefore \frac{\partial y}{\partial x} = 2x$ is said to be the derived function (or simply the derivative) of the function $y = x^2$.

2. Obtain the derivative of the function

$y = 4x^2 - x + 3$ and hence calculate the slope when $x = 11$

$$\frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{h} = \frac{\{4(x+h)^2 - (x+h) + 3\} - \{4x^2 - x + 3\}}{h}$$

$$4(x^2 + 2xh + h^2)$$

$$\frac{\Delta y}{\Delta x} = \frac{4(x^2 + 8xh^2 - x - h + 3 - 4x^2 + x - 3)}{h}$$

$$\frac{\partial y}{\partial x} = 8x + 4h - 1$$

$$\text{As } h \rightarrow 0, \quad \frac{\partial y}{\partial x} \rightarrow \frac{\partial y}{\partial x} = 8x - 1$$

$$\text{Hence: Slope at } x = 11, \quad \frac{\partial y}{\partial x} = 8(11) - 1 = \underline{87}$$

7.6 Rules and Techniques of Differentiation: (Power Functions)

Differentiation from first principles can be tedious, although it greatly assists in the understanding of what the slope is. However, there are certain techniques for the more commonly encountered functions which, when followed eases the task of differentiation. These are what we study next.

1. The derivative of a constant is zero.

$$\text{e.g. if } y = 40, \text{ then } \frac{\partial y}{\partial x} = 0$$

2. The derivative of the function $y = ax^n$

where a, n = constants is:

$$\frac{\partial y}{\partial x} = nax^{n-1}$$

Examples:

$$i) \quad y = x^2, \quad \frac{\partial y}{\partial x} = 2x$$

$$y = 4x^5, \quad \frac{\partial y}{\partial x} = 20x^4$$

$$y = 3x, \quad \frac{\partial y}{\partial x} = 3$$

$$ii) \quad y = 4x^{1/2}, \quad \frac{\partial y}{\partial x} = \frac{1x^{1/2-1}}{2} = 2x^{-1/2}$$

$$iii) \quad y = \frac{2}{x^5} \quad \therefore y = 2x^{-5} \quad \frac{\partial y}{\partial x} = -10x^{-6} \quad \text{or} \quad \frac{-10}{x^6}$$

$$iv) \quad y = \frac{1}{3x^{4/3}} \quad \therefore y = x^{-1/3}, \quad \frac{\partial y}{\partial x} = \frac{-1/3x^{-4/3}}{1} \quad \text{or} \quad \frac{-1}{3x^{4/3}}$$

2. The derivative of a sum or difference

When we have more than one term in a function, we aggregate the derivatives of the different terms as follows:

if $y = u + v$, where u and v are functions of x .

$$\text{Then } \frac{\partial y}{\partial x} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}$$

Examples:

$$i) \quad y = 3x - 1 \quad \frac{\partial y}{\partial x} = 3$$

$$ii) \quad y = 4x^2 - x + 3, \quad \frac{\partial y}{\partial x} = 8x - 1$$

$$iii) \quad y = 2x - \frac{4}{x^{1/6}} + x^2 - 2 \text{ is same as } y = 2x - 4x^{-1/6} + x^2 - 2$$

$$\frac{\partial y}{\partial x} = 2 + 4x \cdot \frac{1}{6} x^{-1/6-6/6} + 2x$$

$$\frac{\partial y}{\partial x} = 2 + \frac{2}{3} x^{-7/6} + 2x$$

3. The derivative of a product

If $y = uv$ (where u, v = functions of x)

Then: $\frac{\partial y}{\partial x} = u \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial x}$ ----- Product rule.

Examples

i) $y = (2x^2 - 1)(x^3 + 3x)$

$$\begin{aligned} \text{Let } u &= 2x^2 - 1 & \frac{\partial u}{\partial x} &= 4x \\ v &= x^3 + 3x & \frac{\partial v}{\partial x} &= 3x^2 + 3 \end{aligned}$$

$$\therefore \frac{\partial y}{\partial x} = (2x^2 - 1)(3x^2 + 3) + (x^3 + 3x)4x$$

ii) $y = (3x + 2x^2)(4x - 1)(x^3 + 15x)$

$$\underbrace{(3x + 2x^2)}_u \underbrace{(4x - 1)}_v (x^3 + 15x)$$

$$y = (3x + 2x^2)(4x - 1)$$

$$\text{Let } u = (3x + 2x^2)(4x - 1) \text{ Let } m = (3x + 2x^2)$$

$$\text{and } n = (4x - 1)$$

$$\frac{\partial u}{\partial x} = \frac{m \partial n}{\partial x} + n \frac{\partial m}{\partial x} \rightarrow \frac{\partial m}{\partial x} = (3x + 2x^2)4 + (4x - 1)(3 + 4x)$$

$$\text{and } v = x^3 + 15x \quad \frac{\partial v}{\partial x} = 3x + 15$$

$$\frac{\partial y}{\partial x} = (3x + 2x^2)(4x - 1)(3x^2 + 15) + (x^3 + 15x)\{(3x + 2x^2)4 + (4x - 1)(3 + 4x)\}$$

4. The derivative of a quotient

Let $y = \frac{u}{v}$ where $u, v \equiv$ functions of x

$$\frac{\partial y}{\partial x} = - \frac{\frac{u \partial v}{\partial x} + \frac{v \partial u}{\partial x}}{v^2} \quad \text{----- Quotient rule}$$

Examples

$$\text{i) } y = \frac{7x - 2u}{4x^2 + 4} = 7x - 2 \frac{\partial u}{\partial x} = 7$$

$$v = 4x^2 + 4 \quad \frac{\partial v}{\partial x} = 8x$$

$$\therefore \frac{\partial y}{\partial x} = \frac{(7x - 2)8x + (4x^2 + 4)7}{(4x^2 + 4)^2}$$

$$\text{ii) } y = \frac{10x^3 - 4x + 1}{x^3 + 4}$$

$$u = 10x^3 - 4x$$

$$= \frac{\partial u}{\partial v} = 30x^2 - 4$$

$$v = x^3 + 4 \quad \frac{dv}{dx} = 3x^2$$

$$\therefore \frac{\partial y}{\partial x} = \frac{(10x^3 - 4x + 1)3x^2 + (x^3 + 4)(30x^2 - 4)}{(x^3 + 4)^2}$$

5. Derivative of function of a function

If $y = f(u)$ where $u = f(x)$

Then: $\frac{\partial y}{\partial x} = \frac{dy}{du} * \frac{du}{dx}$ ----- Chain Rule

Example

i) $y = (3x^2 - 7x + 4)^6$

Let $u = 3x^2 - 7x + 4$ $\frac{\partial u}{\partial x} = 6x - 7$

$$y = u^6 \quad \frac{\partial y}{\partial u} = 6u^5$$

$$\therefore \frac{\partial y}{\partial u} = 6u^5 * (6x - 7)$$

$$\frac{\partial y}{\partial x} = 6(3x^2 - 7x + 4)^5 (6x - 7)$$

ii) $y = 3(2x^4 + 1 - 5x^3)^{10}$

Let $u = 2x^4 + 1 - 5x^3$ $\frac{\partial u}{\partial x} = 8x^3 - 15x^2$ $y = 3u^{10}$ $\frac{\partial y}{\partial u} = 30u^9$

$$\therefore \frac{\partial y}{\partial u} = 30(2x^4 + 1 - 5x^3)^9 (8x^3 - 15x^2)$$

$$\frac{\partial y}{\partial x} = 30u (2x^4 + 1 - 5x^3)^9 (8x^3 - 15x^2)$$

Alternative method

Consider the function: $y = (3x^2 - 7x + 4)^6$ differentiated earlier. Let what is in the bracket be big X.

Therefore, $y = X^6$ which when differentiated yields $\frac{\partial y}{\partial x} = 6X^5$

To obtain the eventual derivative, multiply this result by the derivative of the bracket as follows;

$$\frac{\partial y}{\partial x} = 6X^5 (6x - 7)$$

But we know that $X = 3x^2 - 7x + 4$ so that:

$$\frac{\partial y}{\partial x} = 6(3x^2 - 7x + 4)^5 (6x - 7)$$

Other examples

i) $y = 3(2x^4 + 1 - 5x^3)^{10} \quad 3x^{10} = 30x^9$

$$\frac{\partial y}{\partial x} = 30(2x^4 + 1 - 5x^3)^9 (8x^3 - 15x^2)$$

ii) $y = 4(17x^5 + 12x^2 + x - 7)^{3/2} \quad 4x^{3/2} = 6x^{1/2}$

$$\frac{\partial y}{\partial x} = 6(17x^5 + 12x^2 + x - 7)^{1/2} (85x^4 + 24x + 1)$$

7.7 Higher order Derivatives

Sometimes its necessary to differentiate a function more than once and the results of successive differentiations are known as higher order derivatives.

e.g. Consider a moving body e.g. a car.

Distance = $f(\text{time})$ Initial (parent) function.

d (distance) = speed/velocity ----- 1st order derivative
d(time)

d(speed) = Acceleration ----- 2nd order derivative

d(time)

d = (Acceleration)= ?

d(time)

Activity 7.2



Determine the successive differentiation for the following functions

$$\text{i) } y = x^2 \quad \frac{\partial y}{\partial x} = 2x \quad \frac{\partial y}{\partial x^2} = 2 \quad \frac{\partial^2 y}{\partial x^3} = 0$$

$$\text{ii) } y = 3 - x^4 + 8x$$

$$\frac{\partial y}{\partial x} = -4x^3 + 8 \quad \frac{\partial^2 y}{\partial x^2} = -12x^2 \quad \frac{\partial^3 y}{\partial x^3} = -24x$$

$$\frac{\partial^4 y}{\partial x^4} = -24 \quad \frac{\partial^5 y}{\partial x^5} = 0$$

$$\text{iii) } \frac{y}{x} = y = 2x^{-1} \quad \frac{\partial y}{\partial x} = -2x^{-2}$$

$$\frac{\partial^2 y}{\partial x^2} = 4x^{-3} \quad \frac{\partial^3 y}{\partial x^3} = -12x^{-4} \quad \frac{\partial^4 y}{\partial x^4} = 48x^{-5}$$

7.8



Summary

In this lecture, we have extended the ideas of differentiation to non-linear functions. We have seen the concept of the limit and its importance in differentiation. We also dealt with differentiation using the definition of the derivate, that is, differentiation using first principles. We then looked at the basic techniques or rules of differentiation which greatly simplify the differentiation process. Finally, we looked at successive differentiation, also known as higher order derivatives.

LECTURES EIGHT

APPLICATIONS OF DIFFERENTIATION

Lecture Outlines

- 8.1 Introduction
- 8.2 Objectives
- 8.3 Turning Points
- 8.4 Revenue, Cost and Profit Functions
- 8.5 Elasticity
- 8.6 Summary

8.1 Introduction

Welcome to Lecture Eight. In Lecture Seven, we learned the techniques of differentiation. We are now ready to use these tools in solving business and economic problems.

8.2 Objectives

At the end of this lecture, you should be able to:

1. Identify stationary points.
2. Distinguish between a maximum and a minimum turning point.
3. Apply differentiation techniques to optimize business variables.
4. Determine price elasticity of demand.

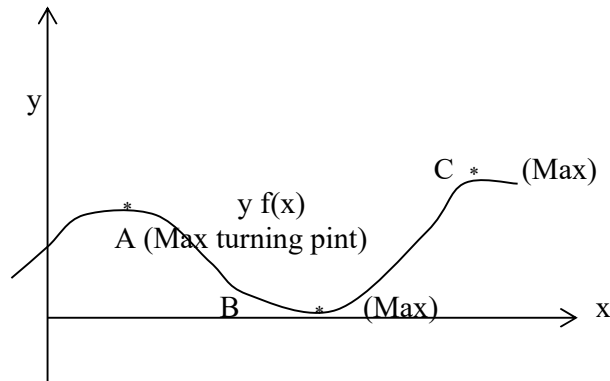
8.3 Turning points

Optimisation, which is the process of either maximising or minimising some commercial activity or quantity is at the core of most business problems e.g. maximise revenue, output, productivity etc.

Minimisation has to do with minimising costs, waste, loss etc.

The key to optimisation is to identify turning points (also known as critical or stationary points) for the relevant functions.

Consider the general function, $y = f(x)$ represented by the following graph:



Observations

- To the left of A slope is positive.
 - Between points A and B the slope is negative.
 - Between points B and C the slope is positive.
 - After C, slope becomes negative etc.
- ∴ Slope changes sign at points A, B and C. They are referred to as

Stationary/turning/critical points.

Conditions for turning points:

1. First Order Conditions (FOC)

Whether a turning point is maximum or minimum, its slope is zero i.e. $\frac{\partial y}{\partial x} = 0$ at any turning point. (This is a necessary condition for a turning point but is not sufficient) Second Order Conditions (SOC)

- a) For a maximum turning points, slope changes signs from positive to negative i.e. slope

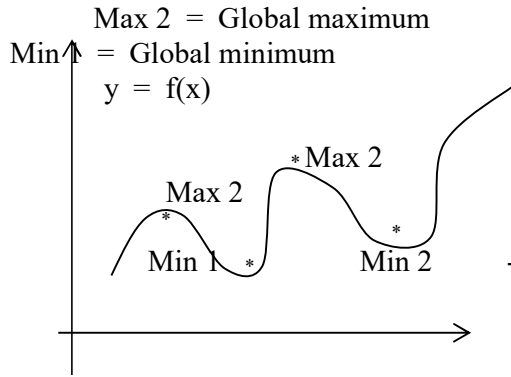
decreases so that $\frac{\partial^2 y}{\partial x^2} < 0$ ---- (2nd derivative is negative)

- b) For a minimum turning points, slope changes signs from negative to positive i.e. slope

increases so that $\frac{\partial^2 y}{\partial x^2} > 0$ ---- (2nd derivative is positive)

Since: SOC helps to distinguish between maximum and minimum, they are also called

sufficient conditions.



- However each max or min is so within its locality i.e. it is a local max or min.

8.4 Revenue Cost and Profit Analysis

Background

- a) Given a demand function, $P = f(Q)$
 P = price Q = quantity

The revenue function $R = PQ$

e.g. If $P = 20 - 3Q$, then $R = P \times Q = \{(20 - 3Q) Q\}$

$$R = 20Q - 3Q^2$$

- b) Average revenue cost or profit is usually on the basis of quantity produced or sold in a given period of time.

E.g. average revenue, $AR = \frac{R}{Q}$ = price demand curve below average revenue type

Average cost (AC) = $\frac{C}{Q}$ = where C = Total cost

- c) Marginal revenue, cost or profit is the first derivative of the relevant function with respect to quantity.

$$\text{i.e. } MR = \frac{\partial R}{\partial Q}, \quad MC = \frac{\partial C}{\partial Q}$$

e.g. $R = 20Q - 3Q^2$

$$MR = \frac{\partial R}{\partial Q} = 20 - 6Q \quad AR \text{ (price)} = \frac{R}{Q} = 20 - 3Q$$

Exercises

A firm's demand function is given as $P = 24 - 3x$

Where P = price and x quantity produced and sold.

Determine the output for maximum revenue and show it is a maximum. What is the price at maximum revenue.

Solution:

$$P = 24 - 3x$$

$$P = PX = (24 - 3x)x \quad R = 24x - 3x^2$$

$$\text{FOC: For maximum or minimum} \quad \frac{\partial R}{\partial X} = 0$$

$$\frac{\partial R}{\partial X} = 24 - 6x = 0 \quad X = \frac{24}{6} = \underline{4 \text{ units}}$$

$$\text{SOC: } \frac{\partial^2 R}{\partial x^2} = -6 < 0 \quad \rightarrow \quad \text{Max}$$

$$\therefore \text{Maximum revenue, } R_{\text{maximum}} = 24(4) - 3(4)^2 = \underline{\text{Sh.48}}$$

$$\text{Hence price when revenue is maximum} = P = 24 - 3(4) = \underline{\text{Sh.12}}$$

Problem

The revenue function of a product is $R = 28q - q^2$ and the unit variable cost $v = q - 8$ while fixed cost is Sh.64.

Determine the following:

- Total cost function
- Profit function
- Output and price for maximum profit.
- Show that the output for maximum profit is not necessarily the same as the output for maximum revenue.
- Represent the above functions on a graphical sketch.

Solution:

$$\text{a) Total cost } C = \text{Total variable cost} + \text{Fixed cost}$$

$$C = (q - 8)q + 64 = 64 + q^2 - 8q$$

$$C = \underline{q^2 - 8q + 64}$$

b) Profit function, $\pi = R - C$

$$\pi = 28q - q^2 - (q^2 - 8q + 64)$$

$$\pi = \underline{36q - 2q^2 - 64}$$

b) Output and price for maximum profit:

$$\text{FOC: For maximum or minimum } \frac{\partial \pi}{\partial q} = 36 - 4q = 0$$

$$q = \frac{36}{4} = \underline{9 \text{ units}}$$

$$\text{SOC: } \frac{\partial^2 \pi}{\partial q^2} = -4 < 0 \rightarrow \text{Maximum turning point}$$

$$\text{Maximum profit, } \pi \text{ maximum} = 36(9) - 2(9)^2 - 64 = \underline{\text{Sh.98}}$$

Price to be charged to obtain maximum profits (π maximum)

$$\frac{P}{Q} = \frac{R}{q} = \underline{28q - q^2} = 28 - q$$

$$\text{When } q = 9 \text{ units, } P = 28 - 9 = \underline{\text{Sh.19}}$$

d) Output which maximizes profit, $q = 9$ units.

Output for maximum revenue.

$$R = 28q - q^2$$

$$\text{FOC For maximum or minimum } \frac{\partial R}{\partial q} = 28 - 2q = 0$$

$$q = \frac{28}{2} = \underline{14 \text{ units}}$$

$$\text{SOC: } \frac{\partial^2 R}{\partial q^2} = -2 < 0 = \text{Maximum}$$

Hence, whereas profit is maximum when $q = 9$ units, revenue is maximum when $Q = 14$ units; thus the two are not maximum at the same point.

e) Graphical sketch

The functions:

$$R = 28q - q^2$$

$$C = q^2 - 8q + 64$$

$$\pi = 36q - 2q^2 - 64$$

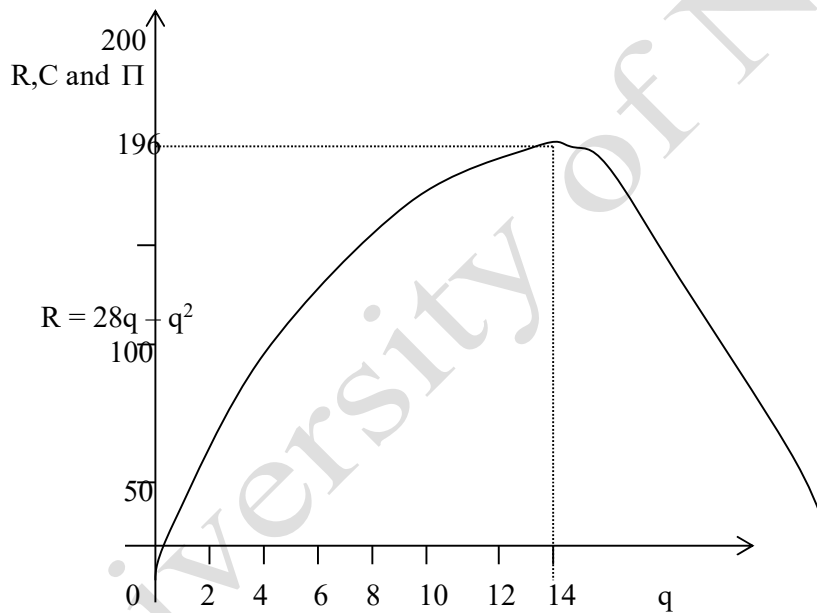
R Function

When $q = 0$, $R = 0$

From d above, R is maximum $q = 14$ units.

$$R \text{ maximum} = 28(14) - 14^2 = \underline{196}$$

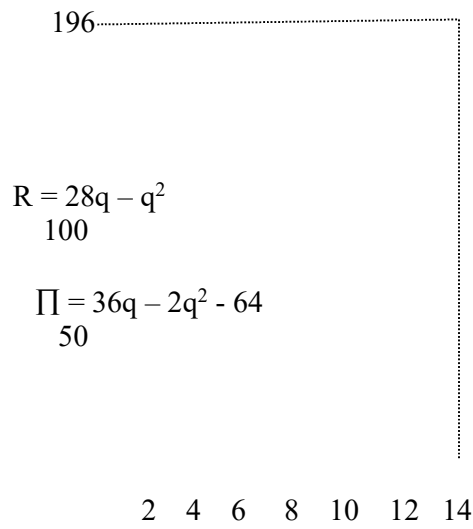
Graph 1



Graph 2



200
R, C and Π



Cost function

When $q = 0$, $C = 64$

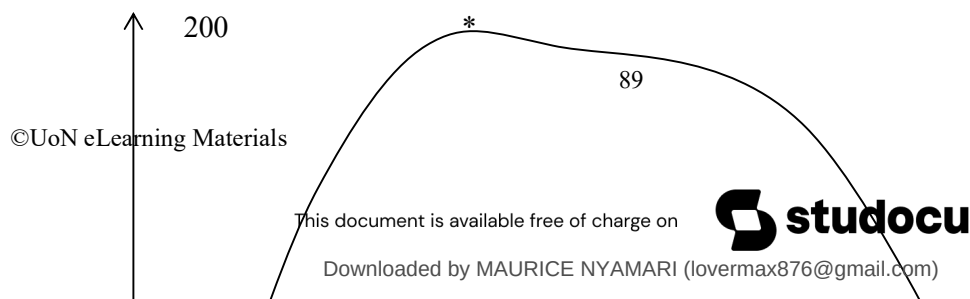
FOC: $\frac{\partial c}{\partial q} = 2q - 8 = 0$

$$\frac{2q}{2p} = \frac{8}{2} = 4 \text{ units}$$

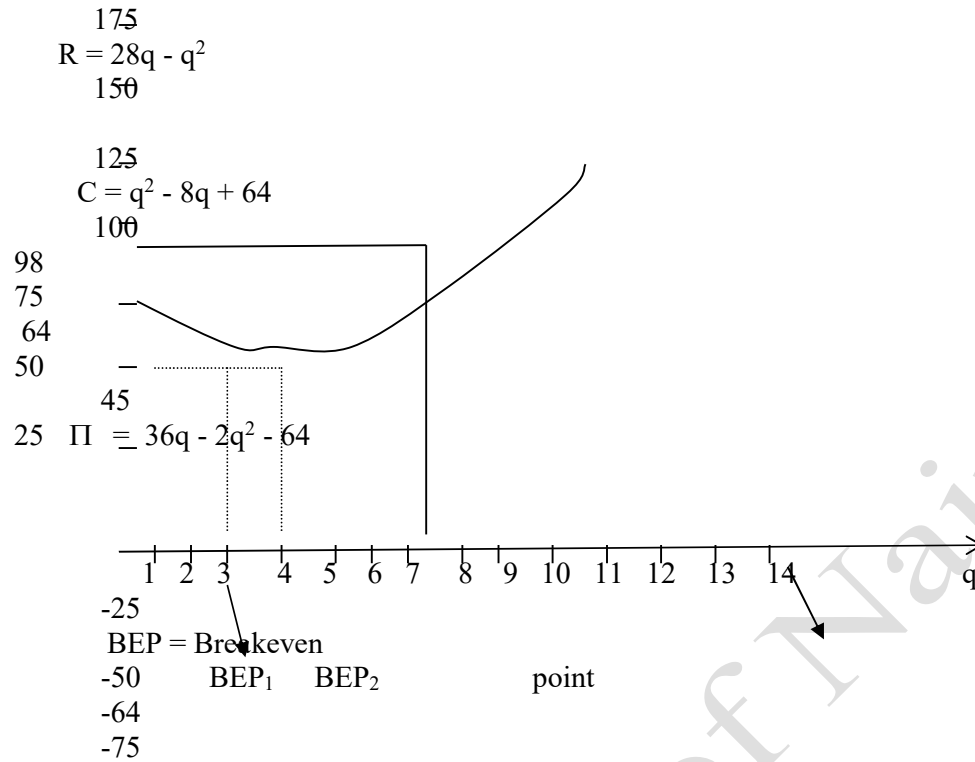
SOC: $\frac{\partial^2 c}{\partial q^2} = 2 > 0$ Min

$$C_{\min} = 4^2 - 8(4) + 64 = \text{Sh.}48$$

Graph 3



R, C and Π



Problem:

Determine BEP1 and BEP2 and explain the correct BEP from a business standpoint.

At BEP, $R = C$ so that $\pi = 0$

$$\therefore 36q - 2q^2 - 64 = 0$$

$$-2q^2 + 36q - 64 = 0 \text{ OR Recall: If } ax^2 + bx + C = 0$$

$$\text{then: } \frac{x = -b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$q = \frac{-36}{2(-2)} \pm \frac{\sqrt{36^2 - 4(-2)(-64)}}{2(-2)}$$

$$q = \frac{-36 - 28}{-4q_1} = \frac{-36 - 28}{-4} = 2$$

$$q_2 = \frac{-36 - 28}{-4} = 16$$

The correct break even point is = 2 units because thereafter the firm begins to make profit.

8.5 Elasticity

8.6 Summary

This lecture has dealt with applications of differentiation. The concept of turning point and its importance in optimization was explained.

We have also discussed the application of differentiation to revenue, cost and profit analysis and the calculation of elasticity. In the next lecture, we extend these ideas on differentiation to multivariate functions.

Activity

a) Explain and graphically illustrate what is meant by the breakeven point in business.

b) The average total cost of producing a commodity is given by

$ATC = 1000/q + 100 - 5q + q^2$ where p is the price in shillings and q is the quantity in kilogrammes. The expected sales of commodity are 390 units at the selling price of Sh. 10 and the sales would fall to 380 units if the price is doubled to Sh.20

Required:

- i) By using an example, explain what $1000/q$ in the ATC equation represents economically.
- ii) Determine the linear demand function in the form $p = a + bq$ where p = price and q = quantity.
- iii) Determine the revenue function.
- iv) Determine the profit function, output and price for maximum profit and the maximum profit. Show that it is a maximum point.
- v) Given that the point elasticity of demand is defined

$$\text{as: } \eta = \frac{p \partial q}{q \partial p}$$

Find η at the point of maximum profit and interpret it.

University of Nairobi

LECTURE NINE

FURTHER DIFFERENTIATION: MULTIVARIATE FUNCTIONS

Lecture Outlines

- 9.1 Introduction
- 9.2 Objectives
- 9.3 Partial Differentiation
- 9.4 Classical Optimisation: Presence of an Equality Constraint
- 9.5 The Total Differential
- 9.6 Summary

9.1 Introduction

Welcome to this lecture. In Lecture eight, we learned the application of differentiation to functions of a single independent variable. For many business problems, dependent variables are determined by two or more independent variables. This Lecture is concerned with differentiation of such functions and their applications.

9.2 Objectives



At the end of this lecture, you should be able to:

1. Determine first order partial derivatives.
2. Determine second order partial derivatives.
3. Apply partial derivatives in optimizing business variables.

9.3 Partial Differentiation

So far, we have done differentiation of functions of one independent variable e.g.

- i) Price = $f(\text{output})$
- ii) Revenue = $f(\text{output})$

However there are many situations where the dependent variable is determined by two or more independent variables e.g.

Sales = $f(\text{price, advertising budget, output etc})$

Problem:

Optimisation of the dependent variable when we have two or more independent variables.

Rule:

If a function has more than one independent variable, then the function may be differentiated with respect to any one of the independent variables if the other independent variables are taken to be constant. The result is then referred to as the partial derivative of the function with respect to the particular independent variable.

Example:

$$\text{i) } y = x^2 + 3xz + 4z^3 = \frac{\partial y}{\partial x} = 2X + 3Z \text{ ---- } -z \text{ is constant}$$

$$\frac{\partial y}{\partial z} = 3x + 12Z^2 \text{ ---- } x \text{ is constant}$$

$$\text{ii) } y = x^4 - 4z^{1/2} + 3x^{1/2} Z$$

$$\frac{\partial y}{\partial x} = 4x^3 + 1.5x^{-1/2} z \text{ } Z \text{ is constant}$$

$$\frac{\partial y}{\partial z} = 2Z^{-1/2} + 3x^{1/2} \text{ } x \text{ is constant}$$

$$\text{iii) } y = 3x^2 - 4xzw^3 + x^2 w - 5z^2 w + z^5$$

$$\frac{\partial y}{\partial z} = 6x - 4zw^3 + 2xw$$

$$\frac{\partial y}{\partial z} = -4xw^3 - 10zw + 5z^4$$

$$\frac{\partial y}{\partial w} = -12xzw^2 + x^2 - 5z^2$$

Higher Order Partial Derivatives:**Example:**

$$1. y = x^2 + 3xz + 4z^3$$

$$\frac{\partial y}{\partial x} = \underline{2x + 3z} \quad \frac{\partial^2 y}{\partial x^2} = \underline{2} \quad \frac{\partial^2 y}{\partial x \partial z} = \underline{3}$$

$$\frac{\partial y}{\partial x} = 3x + 12Z^2 \frac{\partial^2 y}{\partial z^2} = \underline{24z} \frac{\partial^2 y}{\partial z \partial x} = \underline{3}$$

Note:

$$\frac{\partial^2 y}{\partial z \partial t} \quad \frac{\partial^2 y}{\partial z \partial x} \text{ ----- Young's theorem}$$

2. $y = 3xw - w^3 + 2x^2 w$

iii) $\frac{\partial y}{\partial x} = 3x + 4xw$

iv) $\frac{\partial y}{\partial w} = 3x - 3w^2 + 2x^2$

iii) Show that $\frac{\partial^2 y}{\partial x \partial w} = \frac{\partial^2 y}{\partial w \partial x}$

$$\left. -\frac{\partial^2 y}{\partial x \partial w} = 3 + 4x \right\}$$

Equal

$$-\frac{\partial^2 y}{\partial w \partial x} = 3 + 4x$$

Maxima and Minima for Two Independent Variables

Consider the function

$$Y = f(x, z)$$

To obtain the critical/stationary points.

FOC: $= \frac{\partial y}{\partial x} = \frac{\partial y}{\partial z}$

SOC: For both maximum and minimum turning points the determinant of Hessian Matrix, H must be positive i.e. $\frac{\partial^2 y}{\partial w^2}$

$$H = \begin{bmatrix} \frac{\partial^2 y}{\partial x^2} & \frac{\partial^2 y}{\partial x \partial z} \\ \frac{\partial^2 y}{\partial z \partial x} & \frac{\partial^2 y}{\partial z^2} \end{bmatrix}$$

NOTE: If $|H| > 0$ then we have maximum or minimum

If $|H| < 0$ then the results are inconclusive (indeterminate) $\rightarrow$ Implying we perform further tests.

If $|H| = 0$, then we have a saddle point i.e. a point which is maximum with respect to one variable and minimum with the other variables.

Distinctions between Maximum and Minimum:

- For a maximum turning point

$$\frac{\partial^2 y}{\partial x^2}, \frac{\partial^2 y}{\partial z^2} < 0$$

- For a minimum turning point

$$\frac{\partial^2 y}{\partial x^2}, \frac{\partial^2 y}{\partial z^2} > 0$$

Illustration 1

A firm sells 2 products which cost £10 and £15 to produce. The profit function is considered to be

$$\Pi = 60x + 150y - x^2 - 3y^2$$

Determine the values of x and y which will maximize profit

Solution:

$$\Pi = 60x + 150y - x^2 - 3y^2$$

FOC: For a maximum or minimum $\frac{\partial \pi}{\partial x} = \frac{\partial \pi}{\partial y} = 0$

$$\frac{\partial \pi}{\partial x} = 60 - 2x = 0 \quad x = \frac{60}{2} = 30 \text{ units}$$

$$\frac{\partial \pi}{\partial y} = 150 - 6y = 0 \Rightarrow y = \frac{150}{6} = 25 \text{ units}$$

$$\left. \begin{array}{l} SOC: \frac{\partial^2 \pi}{\partial^2 y} = -2 \\ Both < 0 \frac{\partial^2 \pi}{\partial y^2} = -6 \end{array} \right\} \text{Maximum provided } |H| > 0$$

$$H = \begin{pmatrix} \frac{d^2 \pi}{dx^2} & \frac{d^2 \pi}{dxdy} \\ \frac{d^2 \pi}{dydx} & \frac{d^2 \pi}{dy^2} \end{pmatrix} = \begin{pmatrix} \frac{d^2 \pi}{dx^2} & \frac{d^2 \pi}{dxdy} \\ \frac{d^2 \pi}{dydx} & \frac{d^2 \pi}{dy^2} \end{pmatrix}$$

$$|H| = (-2x - 6) - (0)12 > 0 \rightarrow \text{Maximum}$$

$$\pi_{\max} (\text{Max profit}) = 60(30) + 150(25) - 30^2 - 3(25)^2 = \underline{\underline{\text{£}2775}}$$

Illustration 2

A car selling firm has determined through regression analysis that its sales (S) are a function of the amount of advertising (measured in units) in two different media. This is given by the following relationship:

$$S(x,y) = 200x + 100y - 10x^2 - 20y^2 + 20xy$$

where x = newspapers, and

y = magazines

Required:

Determine the firm's optimal level of sales.

Solution:

$$S(x,y) = 200x + 100y - 10x^2 - 20y^2 + 20xy$$

$$FOC: \frac{\partial S}{\partial x} \text{ for max or min } \frac{\partial S}{\partial x} = \frac{\partial S}{\partial x} = 0$$

To determine the firm's optimal level of sales;

$$\text{FOC: } \frac{\partial s}{\partial x} = 200 - 20x + 20y = 0$$

$$\frac{\partial s}{\partial y} = 100 - 40y + 20x = 0$$

$$200 - 20x + 20y = 0 +$$

$$100 - 40x + 20x = 0 \quad \leftarrow$$

$$300 - 20y = 0$$

$$-20y = -300 = y - 300 = \underline{\underline{15}}$$

Substituting:

$$200 - 20x + 20(15) = 0$$

$$\frac{x = -20(15) - 200x}{-20} = \underline{\underline{25}}$$

$$\text{SOC: } \left. \begin{array}{l} \frac{\partial^2 s}{\partial x^2} = -20 \\ \frac{\partial^2 s}{\partial y^2} = -40 \end{array} \right\} \text{Max. If /H/ is positive}$$

$$\left(\frac{\partial^2 s}{\partial y^2} \quad \frac{\partial^2 s}{\partial x \partial y} \right) = \left(\begin{array}{cc} -20 & -20 \\ 20 & -40 \end{array} \right)$$

$$\frac{\partial^2 s}{\partial y \partial x} = \frac{\partial^2 s}{\partial x \partial y} = 20$$

$$\therefore |H| (-20 \times -40) - (20)^2 = 400 > 0 \text{ Maximum}$$

$$\text{Maximum sales, } S_{\max} = 200(25) + 100(15) - 10(25)^2 - 20(15)^2 + 20(25)(15) = \underline{\underline{3250 \text{ units}}}$$

$$\text{a) } \text{B.E.P (units), } X_{b/e} = \frac{\text{fixed cost}}{\text{Contribution margin CM}} = \frac{\text{fixed cost}}{P - v}$$

P = price, v = unit variable cost.

$$f = \text{Sh.}1800,000 \text{ P} = \text{annual revenue per car} = 21000 \times 8 \times 2 \text{ session} \times 4 \text{ students} = \underline{\text{Sh.}1,344,000}$$

$$\sqrt{} = \text{Variable cost per car per year } 180,000 + 660,000 = \underline{\text{Sh.}840,000}$$

Hence B.E.P (No. of cars)

$$X \text{ b/e} = \frac{1800,000}{1344000 - 840000} = \underline{4 \text{ cars.}}$$

With 5 cars the firm will break even since the B.E.P is 4 cars. It will actually make a profit.

- b) i) Sh.100 is the fixed cost e.g. rent, salaries, interest
 ii) Where revenue is maximised is not necessarily, the same as the point where profit is maximised. This is because whereas revenue represents sales, profit is revenue less costs.

$$\text{Revenue, } R = pq = 600q - 2q^2$$

$$\text{Note: } P = 600 - 2q$$

$$\text{FOC: } \frac{\partial R}{\partial q} = 600 - 4q = 0 \Rightarrow q = \frac{600}{4} = \underline{150 \text{ units}}$$

$$\frac{\partial^2 R}{\partial q^2} = -4 < 0 \rightarrow \text{Maximum}$$

$$\text{Profit, } \pi = R - C \quad \pi = 600q - 2q^2 - (100 + 216q)$$

$$\pi = 384q - 2q^2 - 100$$

$$\text{FOC: } \frac{\partial \pi}{\partial q} = 384 - 4q = 0 \Rightarrow q = \frac{384}{4} = \underline{96 \text{ units}}$$

$$\text{SOC: } \frac{\partial^2 \pi}{\partial q^2} = -4 < 0 \rightarrow \text{Maximum}$$

Hence, profit is maximised where $q = 96$ units whereas revenue is maximised where $q = 150$ units.

iii) Point elasticity of demand $E = \frac{\% \Delta q_d (\text{quantity demanded})}{\% \Delta \text{ in price}}$

$$= \frac{\Delta q / q}{\Delta p / p} = \frac{p}{q} = \frac{\Delta q}{\Delta p}$$

$$E = \frac{p}{q} \frac{\partial p}{\partial q} \quad p = 600 - 2q$$

$$\frac{\partial p}{\partial q} = -2 \quad \therefore \frac{\partial p}{\partial q} = \frac{-1}{-2}$$

Revenue is maximised where $q = 150$ units.

$$p = 600 - 2(150)$$

$$= \underline{\text{Sh.300}}$$

$$\text{Point elasticity of demand } E = \frac{300}{150} \times \frac{1E}{-2} = \underline{\underline{-1}}$$

Interpretation:

When $E = -1$ (unitary elasticity) a given % change in price results in the same % change in quantity demanded. (negative implies that this is a normal good such that as the price increases demand falls).

Illustration 3

Two competing soap bars x and y are manufactured by a monopolist

Their profiles follow:

Product x: Variable cost per unit = Sh.9

$$\text{Price} = P_x$$

$$\text{Demand function: } x = 2(P_y - P_x) + 4$$

Product y: Variable cost per unit = Sh.12

$$\text{Price} = P_y$$

$$\text{Demand function } Y = \frac{P_x}{4} - \frac{5P_y}{4} + 52$$

Solution:

$$R = f(\text{price}), C = f(\text{prices}), \pi = f(\text{prices})$$

Product x

$$\begin{aligned} \text{Revenue, } RX &= XP_x = 2(py - p_x) + 4 \\ &= (2py - 2p_x + 4) p_x \quad RX = 2P_x P_y - 2P_x^2 + 4P_x \end{aligned}$$

Product y

$$\begin{aligned} R_y &= y p_y = 0.25x - 2.25P_y + 52) p_y \\ R_y &= -0.25p_x p_y - 2.5p_y^2 + 52 P_y \\ \text{Total cost function, } C &= c_x + c_y \quad C = 9x + 12 y \\ C &= v_x \{ (2py - 2px) + 4 \} + v_y (0.25Px - 2.5Py + 52) \end{aligned}$$

$$C = 9 \{ (2py - 2px) + 4 \} + 12 \quad 0.25 px - 2.5py + 52$$

$$C = 18py - 18px + 36 + 3px - 30py + 624$$

$$C = -12py - 15px + 660$$

$$\text{Profit, } \pi = R - C$$

$$\pi = 2pxpy - 2px^2 + 4px + 0.25pxpy - 2.5 py^2 + 52py - (-12py - 15px + 660)$$

$$\pi = 2.25pxpy - 2px^2 + 19px - 2.5py^2 + 64 py - 660$$

$$\text{FOC:} = 2.25PY - 4PX + 19 = 0$$

$$\begin{aligned} \frac{\partial \pi}{\partial py} &= 2.25px - 5py + 64 = 0 \times 0.45 + \quad \swarrow \\ &= 1.0125px - 2.25py + 28.8 = 0 \\ &= -2.9875 px + 47.8 = 0 \\ &= -2.9875px = -4px = \frac{-47.8}{-2.9875} = \underline{\underline{\text{Sh}16}} \end{aligned}$$

Substituting:

$$2.25px - 4(16) + 19 = 0$$

$$\therefore py - 19 + 4(16) = \text{Sh.20}$$

$$2.25$$

$$\text{SOC: } \left. \begin{aligned} \frac{\partial^2 \pi}{\partial py^2} &= -4 < 0 \\ \frac{\partial^2 \pi}{\partial py^2} &= -5 < 0 \end{aligned} \right\} \text{Maximum provided } |H| > 0$$

$$|H| = \begin{pmatrix} \frac{\partial^2 \pi}{\partial px^2} & \frac{\partial^2 \pi}{\partial px \partial py} \\ \frac{\partial^2 \pi}{\partial py^2} & \frac{\partial^2 \pi}{\partial py^2} \end{pmatrix} = \begin{bmatrix} 4 & 2.25 \\ 2.25 & -5 \end{bmatrix}$$

$$\frac{\partial^2 \pi}{\partial py \partial px} = 2.25$$

$$|H| = (-4 \times -5) - (2.25)^2 = 14.9375 > 0 \text{ Maximum hence confirmed}$$

Π Maximum

$$\Pi = 2.25(16)(20) - 2(16)^2 + 19(16) - 2.5(20)^2 + 64(20) - 660 = \text{Sh.132}$$

$$x = 2(20 - 16) + 4 = \underline{12 \text{ units}}$$

$$y = \frac{16}{2} - \frac{5}{20}(20) + 52 = \underline{6 \text{ units}}$$

9.4 CLASSICAL OPTIMISATION: PRESENCE OF AN EQUALITY CONSTRAINT.

(Lagrange Multiplier Approach)

Frequently, we may want to maximise or minimise a function which is non-linear subject to an equality constraint.

The general form: 2 decision variables x and y

Suppose $f(x,y)$ is to be minimised or maximised subject to the constraint $g(x, y) = 0$

STEPS

- i) Form the objective function $L = f(x,y) + \lambda g(x,y)$ where quantity λ (lambda) is called the Lagrangian multiplier and the function on L is known as the Lagrange multiplier.
- ii) Perform first order conditions i.e. find values of x , y and λ such that

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial \lambda} = 0$$

- iii) Confirm the nature of the turning points by performing second order conditions i.e.

For both maximum and minimum, $|H| > 0$

$$H = \begin{pmatrix} \frac{\partial^2 L}{\partial x^2} & \frac{\partial^2 L}{\partial x \partial y} \\ \frac{\partial^2 L}{\partial y \partial x} & \frac{\partial^2 L}{\partial y^2} \end{pmatrix}$$

To distinguish between maximum and minimum points if

$$\frac{\partial^2 L}{\partial x^2} < 0 \rightarrow \text{Max.}$$

$$\frac{\partial^2 L}{\partial y^2} > 0 \rightarrow \text{Min.}$$

Examples:

1. A factory manufactures two types of heavy duty machines in quantities x and y . The joint cost function is $C = x^2 + 2y^2 - xy$. How many machines of each type should be produced if they have to be a total of 8?

Solution:

$$\text{i) Min } C = x^2 + 2y^2 - xy \text{ Subject to (st) } x + y = 8 \leftrightarrow x + y - 8 = 0$$

$$\text{Lagrange function, } L = x^2 + 2y^2 - xy + \lambda (x + y - 8)$$

$$\text{ii) FOC: } \frac{\partial L}{\partial x} = 2x - y + \lambda = 0 \quad (1)$$

$$\frac{\partial L}{\partial y} = 4y - x + \lambda = 0 \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = x + y - 8 = 0 \quad (3) \times 5$$

$$3x - 5y = 0 \quad (4)$$

Equation: $3 \times 5 =$

$$4 + \text{Equation 5} \quad \left[\begin{array}{r} 5x + 5y = 40 \\ 3x - 5y = 0 \\ \hline 8x - 40 = 0 \end{array} \right]$$

$$= \frac{40}{8} = 5 \quad y = 3$$

Using equation 1:

$$2(5) - 3 + \lambda = 0 \quad \lambda = -10 + 3$$

$$\lambda = \underline{-7}$$

$$\text{iii)SOC: } \left. \begin{array}{l} \frac{\partial^2 L}{\partial x^2} = 2 \\ \frac{\partial^2 L}{\partial y^2} = 4 \end{array} \right\} > 0 \rightarrow \text{Maximum provided } |H| > 0$$

$$\frac{\partial^2 L}{\partial x \partial y} = \frac{\partial^2 L}{\partial y \partial x} = \begin{pmatrix} -1 & 2 \\ 1 & 4 \end{pmatrix}$$

$$|H| = 7 > 0 \text{ minimum confirmed.}$$

$$C \min = 5^2 + 2(3)^2 - 5 \times 3 = \underline{\text{Sh.28}}$$

Interpretation of (Lambda)

λ represents the sensitivity of the objective function given a unit change in the constraint such that if the constraint changes by 1 unit, the objective function value will change by λ .

λ is also called the shadow price or opportunity cost of the related resource (constraint).

In the illustration above, if number of machines increase by 1 unit i.e. to 9 machines, then total cost will increase by Sh.7 to $28 + 7 = \underline{\text{Sh.35}}$

Example 2

2. Jaribu Ltd. manufactures two types of milling machines, the Ace and the champion (y) which cost £10 and £15 per unit to produce respectively. There is a budget constraint of £850 and it has been established that the profit function is

$$\Pi = 60x + 150y - x^2 - 3y^2$$

Required:

Determine the number of each machine type which should be produced and sold in order to maximise the profits subject to the budget constraint. the number of machines produced should just exhaust the budget.

Solutions: Objective function

i) $\max \Pi = 60x + 10y - x^2 - 3y^2$

$$\text{st } = 10x + 15y = 850$$

$$\Rightarrow 10x + 15y - 850 = 0$$

$$\text{Lagrange function } L = 60x + 150y - x^2 - 3y^2 + \lambda (10x + 15y - 850)$$

ii) FOC:

$$\frac{\partial L}{\partial x} = 60 - 2x + 10\lambda = 0 \quad (1) \times 1.5 = \text{Equation 4)}$$

$$\frac{\partial L}{\partial y} = 150 - 6y + 15\lambda = 0 \quad (2)$$

$$\frac{\partial L}{\partial x} = 10x + 15y - 850 = 0 \quad (3)$$

$$(\text{Equation 1 } \lambda \times 1.5) + 20 - 3x + 15\lambda = 0 \quad (4)$$

$$(\text{Equation 2} - \text{Equ. 4}) \quad 60 - 6y - 3x = 0 \quad (5) \times 2.5$$

$$= \text{Equ. (6)} = \text{Equ. (5)} \times 2.5$$

$$(5) \times 2.5 \quad 150 - 15y + 7.5x = 0 \quad (6)$$

$$(3) + (6)$$

$$= -850 - 15y + 10x$$

$$\frac{+150 + 15y}{-700 + 17.5x} = 0$$

$$x = \frac{700}{175} = \underline{40 \text{ units}}$$

Using Equation 3:

$$10(40) + 15y - 850 = 0$$

$$y = \frac{850 - 10(40)}{15} = \underline{30 \text{ units}}$$

Using Equation 1

$$60 - 2(40) + 10\lambda = 0 \quad \}$$

$$\lambda = \frac{-60 + 2(40)}{10} = \underline{\pounds 2}$$

iii) **FOC:**

$$< 0 \quad \text{maximum if } |H| > 0$$

$$\frac{\partial^2 L}{\partial y^2} = -6$$

$$\frac{\partial^2 L}{\partial y \partial x} = \frac{\partial^2 L}{\partial x \partial y} = 0$$

$$H = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix} \quad |H| = 12 > 0 \quad \text{Maximum confirmed}$$

$$\therefore \Pi \text{ maximum} = 60(40) + 150(30) - 40^2 - 3(30)^2 = \pounds 2600$$

Interpretation of λ :

If the budget changed by £1 profit changes by £2

Generalisation

Lagrange multiplier approach can be used in optimisation of any number of decision variables and constraints provided all the constraints are of equality type.

Objective function: $F = f(x_1, x_2 \dots \dots \dots)$

Subject to (1) $g_1(x_1, x_2 \dots \dots) = 0$

(2) $g_2(x_1, x_2 \dots \dots) = 0$

,

,

,

Form the lagrange function L, such that:

$$L = f(x_1, x_2 \dots \dots) + \lambda_1 g_1 + \lambda_2 g_2 + \dots \dots$$

Optimizing L in the usual manner gives the appropriate values of x_1, x_2 ----- λ_1, λ_2 -----

Question 2 c June 2003

Objective function: Max. $R = q^2 + 3qa + a^2$

$$\text{St } q + a = 100 \quad \leftrightarrow \quad q + a - 100 = 0$$

Lagrangian function: $L = q^2 + 3qa + a^2 + \lambda (q + a - 100)$

Constrained optimisation {Lagrange multiplier approach}

Frequently, we may want to maximise or minimise a function which is non-linear subject to EQUALITY constraints.

GENERAL FORM: 2 Decision variables:

Suppose $f(x,y)$ is to be maximised or minimised subject to the constraint $d(x,y) = 0$

STEPS

1. Form the $L = f(x,y) + \lambda g(x,y)$ where the quantity is formed the Lagrange multiplier. The function L is termed the Lagrangian function

2. Form F.O.C i.e. find the values of x, y and such $\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial \lambda} = 0$

3. Perform S.O.C to determine the nature of the critical point

$|H| > 0 \Rightarrow$ maximum if 2nd derivative writ x, y $\Rightarrow$ minimum if 2nd derivative writ x, y

$|H| > 0 \Rightarrow$ the test fails i.e. results are inclusive; investigate the behaviour of the function in the neighbourhood of the critical point.

H = Hessian matrix constructed thus:

$$\left[\frac{\partial^2 L}{\partial x^2} \quad \frac{\partial^2 L}{\partial x \partial y} \right]$$

$$H = \begin{bmatrix} \frac{\partial^2 L}{\partial y \partial L} & \frac{\partial^2 L}{\partial y^2} \end{bmatrix}$$

Note:

For n variable:

Maxima If $H_2 > 0, H_3 < 0, H_4 > 0$

Minima If $H_2 < 0, H_3 < 0, H_4 < 0$

Example 1

Find the maxima and the minima, if any of

$$F = 5x^2 + 6y^2 - xy$$

$$\text{Subject to: } x + 2y = 24$$

i) Obtain $g(x,y) = 0$

$$\longleftrightarrow x + 2y = 24$$

Solution:

1. Lagrangian function

$$L = 5x^2 + 6y^2 - xy + \lambda (x + 2y - 24)$$

2. F.O.C for max or min

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial \lambda} = 0$$

$$\frac{\partial L}{\partial x} = 10x - y + \lambda = 0$$

$$\frac{\partial L}{\partial y} = 12y - x + 2\lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = x + 2y - 24 = 0$$

Solving for x, y, λ

$$20x - 2y + 2\lambda = 0$$

$$-x + 12 + 2\lambda = 0$$

$$21x - 14y = 0 \quad x + 2y = 24$$

$$7x + 2y = 24 \quad 2y = 18$$

$$21x - 14y = 0 \quad y = 9$$

$$\begin{array}{r} + 7x + 14y = 168 \\ \hline 28x = 168 \end{array}$$

$$x = \frac{168}{28} = 6 \quad \lambda = -51$$

S.O.C

$$\left. \begin{array}{l} \frac{\partial^2 L}{\partial x^2} = 10 \\ > 0 \end{array} \right\} \text{critical point is minimum provide } H > 0 \quad |$$

$$\frac{\partial^2 L}{\partial x^2} = 12$$

$$\frac{\partial^2 L}{\partial x \partial y} = -1 = \frac{\partial^2 L}{\partial x \partial y} \text{ Young's theorem } \}$$

$$\left[\frac{\partial^2 L}{\partial x^2} \quad \frac{\partial^2 L}{\partial x \partial y} \right]$$

$$H = \frac{\partial^2 L}{\partial x \partial x} \quad \frac{\partial^2 L}{\partial y^2}$$

$$H = \begin{bmatrix} 10 & -1 \\ 1 & 12 \end{bmatrix} = 119 > 0 \therefore \text{Critical point is minimum}$$

$$\begin{aligned} F_{\min} &= 5(6)^2 + 6(9)^2 - 6 \times 9 \\ &= \underline{612} \end{aligned}$$

Example 2

A factory manufactures 2 types of heavy duty machines in quantities x and y. The joint cost function is given by

$C = x^2 + 2y^2 - xy$. How many machines of each type should be produced if they have to be a total of exactly 8?

Solution:

$$C = x^2 + 2y^2 - xy$$

$$\text{S.t. } x + y = 8$$

$$L = x^2 + 2y^2 - xy + \lambda (x + y - 8)$$

$$\text{F.O.C for max or min } \frac{\partial L}{\partial x} \quad \frac{\partial L}{\partial y} \quad \frac{\partial L}{\partial \lambda} = 0$$

$$\frac{\partial L}{\partial x} = 2x - y + \lambda = 0$$

$$\frac{\partial L}{\partial y} = 4y - x + \lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = x + y - 8 = 0$$

Solving:

$$2x - y + \gamma = 0$$

$$\begin{array}{rcl} -x & = & 4y + \lambda = 0 \\ 3x - 5y & = & 0 \end{array} \quad \begin{array}{l} 3x - 5(3) = 0 \\ 3x = 15 \end{array}$$

$$\begin{array}{rcl} 3(x + y) & = & 8 \\ 3x - 5y & = & 0 \end{array} \quad x = 5$$

$$\begin{array}{rcl} -3x + 3y & = & 2 \\ -8y & = & -24 \end{array} \quad \begin{array}{l} 10 - 3 = -\lambda \\ 7 = -\lambda \end{array}$$

$$y = 3 \quad \lambda = -7$$

S.O.C

$$\frac{\partial^2 L}{\partial x^2} = 2$$

$$\left. \begin{array}{l} \frac{\partial^2 L}{\partial y^2} = 4 \\ \frac{\partial^2 L}{\partial y \partial x} = -1 \end{array} \right\} > 0 \text{ critical point is minimum provided } |H| > 0$$

$$\frac{\partial^2 L}{\partial y \partial x} = \frac{\partial^2 L}{\partial x \partial y} = -1$$

$$|H| = \begin{vmatrix} 2 & -1 \\ 1 & 4 \end{vmatrix} = 7 > 0 \text{ critical point is maximum}$$

$$C_{\min} = 5^2 + 2(3)^2 - 5 \times 3$$

$$= \text{Sh.28}$$

Interpretation of λ

Lambda, λ , represents the sensitivity of the Objective Function to a unit change in the constraint such that if that constraint changes by 1 unit, the O.F value will change by approximately λ . For instance in illustration2, the number of machines increase cost by about Sh.7.

λ is also called the shadow price or opportunity cost.

Generalisation

The Lagrange multiplier approach can be used in optimisation of any number of decision variables and constraints provided all the constraints are of equality type.

$$\text{O.F max /min} \quad F = f(x_1, x_2), \dots\dots\dots)$$

$$\text{St (1) } g_1(x_1, x_2, \dots) = 0$$

$$(2) \ g_2(x_1, x_2, \dots) = 0$$

$$\text{Lagrangian function } L = f(x_1, x_2, \dots) + \lambda_1(g_1) + \lambda_2(g_2)$$

Thus optimisation of L in the usual manner yields the appropriate values of $x_1, x_2, \dots$ and λ_1, λ_2

Question 2 June 1996

a) Lagrangian function

$$L = 100 - \frac{x^2}{20000} - \frac{9y^2}{20000} + \lambda(x + y - 400)$$

$$\text{F.O.C for max or min} = \frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial \lambda} = 0$$

$$\frac{\partial L}{\partial x} = \frac{-2x}{20000} + \lambda = 0$$

$$\frac{\partial L}{\partial y} = \frac{-18y}{20000} + \lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = x + y - 400 = 0$$

$$\frac{-2x}{20000} + \frac{18y}{20000} = 0$$

$$-2x + 18y = 0$$

$$2x(x + y = 400)$$

$$\begin{array}{rcl} -2x + 18y & = & 0 \\ 20 & & 800 \end{array}$$

$$\begin{array}{rcl} + 2x + 2y & = & 800 \\ 20 & & 20 \end{array}$$

$$\underline{y = 40}$$

$$\underline{x + y = 400}$$

$$x = 400 - 40$$

$$x = \underline{\underline{360}}$$

b) S.O.C

$$\left. \begin{array}{cc} \frac{\partial^2 L}{\partial x^2} & -2 \\ \frac{\partial^2 L}{\partial^2 x^2} & -18 \end{array} \right\} < 0 \text{ stationary point is max provided } |H| > 0$$

$$\frac{\partial^2 L}{\partial x \partial y} = \frac{\partial^2 L}{\partial x \partial y} = 0$$

$$|H| = \begin{bmatrix} -2 & 0 \\ 0 & -18 \end{bmatrix} \quad 0.9 - 0.7 > 0 > 0 \quad \text{Critical point is max 20000}$$

Hence target x group/ those who have lived in the town long range.

9.5 THE TOTAL DIFFERENTIAL

Introduction

If z is a function of x and y, the total differential of z is given as:

$$Dz = \frac{\partial z}{\partial x} \Delta x + \frac{\partial z}{\partial y} \Delta y$$

Example

At a certain factory, the daily output is given by the function;

$Q = 60K^{1/2} L^{1/3}$ units where K = capital investment measured in thousands of £s and L is the size of the labour force measured in worker- hours. The current capital investment is £ 900,000 and 1000 worker-hours of labour used each day.

Required:

Estimate the change in output that will result if K increases by £ 1000 and L increases by 2 worker-hours.

Solution:

Apply the approximation formula with K = 900, L = 1000, $\Delta K = 1$ and $\Delta L = 2$ to obtain:

$$\Delta Q = \frac{\partial Q}{\partial K} \Delta K + \frac{\partial Q}{\partial L} \Delta L$$

$$\Delta Q = 30K^{-1/2} L^{1/3} \Delta K + 20K^{1/2} L^{-2/3} \Delta L$$

$$\Delta Q = 30(30(900))^{-1/2}(1000)^{1/3} * 1 + 20(900)^{1/2}(1000)^{-2/3} * 2$$

$$\Delta Q = 30(\frac{1}{30})(10) * 1 + 20(30)(\frac{1}{100}) * 2$$

$$\Delta Q = 22 \text{ units}$$

Hence, output is expected to increase by approximately 122 units..

9.6 Summary

In this lecture, we have studied how to perform differentiation when we have two or more predictor variables. We dealt with the case when a single predictor is varying while the others are constant and applications. We also looked at the case when all predictors are varying at the same time and their business applications. Finally, we looked at optimisation in the case where there is a single constraint of equality type.

Activity

The return on two types of investments is given by the following function:

$$R = -0.00008x_1^2 - 0.00012x_2^2 + 0.00016x_1x_2 + 0.4x_1 + 0.8x_2$$

Where x_1 and x_2 are the sums allocated to investment 1 and 2 respectively.

Required:

- a) Determine the amounts to allocate to investments 1 and 2 in order to maximize return on investment if the amount of money available to invest is unlimited.
- b) What amount would be allocated to investments 1 and 2 if the investible money available is limited to \$ 10,000 and all of it is unlimited?
- c) What is the difference in return on investment between parts (a) and (b) above?
- d) In part (b), suppose an additional \$1 is availed for investment, what would be the increase or decrease of the return on investment?

LECTURE TEN

FURTHER DIFFERENTIATION: IMPLICIT, EXPONENTIAL AND LOGARITHMIC FUNCTIONS

Lecture Outlines

- 10.1 Introduction
- 10.2 Objectives
- 10.3 Differentiation of Implicit Functions
- 10.4 Differentiation of Exponential Functions
- 10.5 Differentiation of Logarithmic Functions
- 10.6 Applications
- 10.7 Summary

10.1 Introduction

Welcome to this lecture. In lecture 9 we dealt with differentiation of multivariate functions. In this lecture, we extend the use of differentiation to functions which can not be explicitly stated in terms of the independent variable, exponential and logarithmic functions.

10.2 Objectives

At the end of this lecture, you should be able to

1. Differentiate implicit functions
2. Differentiate exponential functions
3. Differentiate logarithmic functions
4. Apply the above differentiations in solving business problems

10.3 Differentiation of Implicit Functions

Introduction

(Insert introduction to differentiation of implicit functions here)

Evaluate the derivatives for the following functions:

Example 1:

$$X^2 + y^2 = 24$$

$$2x + 2y \frac{\partial y}{\partial x} = 0$$

$$\frac{\partial y}{\partial x} = \frac{-2x}{2y} =$$

$$= \frac{-x}{y}$$

Example 2:

$$X^3 + 3y^2 = 2x + y$$

$$3x^2 + 6y \frac{\partial y}{\partial x} = 2 + \frac{\partial y}{\partial x}$$

$$6y \frac{\partial y}{\partial x} - \frac{\partial y}{\partial x} = 2 - 3x^2$$

$$\frac{\partial y}{\partial x} (6y - 1) = 2 - 3x^2$$

$$\frac{\partial y}{\partial x} = \frac{2 - 3x^2}{6y - 1}$$

Example 3:

$$2xy^2 - y = 50$$

$$2x \cdot 2y \frac{\partial y}{\partial x} + y^2 \cdot 2 - \frac{\partial y}{\partial x} = 0$$

$$\frac{\partial y}{\partial x} (4xy - 1) = -2y^2$$

$$\frac{\partial y}{\partial x} = \frac{-2y^2}{4xy - 1}$$

NOTE: According to the product rule, if $y = UV$ (where u and v are functions of x);

$$\text{Then: } \frac{\partial y}{\partial x} = U \frac{\partial v}{\partial x} + V \frac{\partial u}{\partial x}$$

Example 4:

$$12x^3y + 2x = y^2 - 1$$

$$12x^3 \cdot \frac{\partial y}{\partial x} + y \cdot 36x^2 + 2 = 2y \frac{\partial y}{\partial x}$$

$$12x^3 \frac{\partial y}{\partial x} - 2y \frac{\partial y}{\partial x} = -36x^2y - 2$$

$$\frac{\partial y}{\partial x} (12x^3 - 2y) = -36x^2y - 2$$

$$\frac{\partial y}{\partial x} = \frac{-36x^2y - 2}{12x^3 - 2y}$$

$$\frac{\partial y}{\partial x} = \frac{-18x^2y - 1}{6x^3 - 2y}$$

Exercise

Without solving the equation, show that $2x^2 + 5xy + y^2 = 19$ defines an implicit function $y = f(x)$ for which when $x = 2$, $y = 1$. Find dy/dx when $x = 2$.

Give the geometrical interpretation of the result.

Solution

When $x = 2$ and $y = 1$:

Substituting,

$$2(2)^2 + 5(2)(1) + (1)^2 = 8 + 10 + 1$$

= 19... proved!

The Derivative

$$4x + 5x \cdot \frac{\partial y}{\partial x} + y \cdot 5 + 2y \frac{\partial y}{\partial x} = 0$$

$$4x + \frac{\partial y}{\partial x} (5x + 2y) + 5y = 0$$

$$\frac{\partial y}{\partial x} (5x + 2y) = -4x - 5y$$

$$\text{therefore: } \frac{\partial y}{\partial x} = \frac{-4x - 5y}{5x + 2y}$$

When $x=2$, $y=1$ and hence substituting $\frac{\partial y}{\partial x} = \frac{-4(2)-5(1)}{5(2)+2(1)}$

$$= (-8-5)/(10+2)$$

$$\frac{\partial y}{\partial x} = -13/12$$

Interpretation

In geometric terms, this means that the slope of the contour of the function

$$2x^2 + 5xy + y^2 = 19 \text{ is } \frac{-13}{12} \text{ at the point } (x,y) = (2,1)$$

Partial Derivatives: The Chain Rule

In many practical situations, a particular quantity is given as a function of two or more variables, each of which can be thought of as a function of yet another variable and the objective is to find the rate of change of the quantity with respect to this other variable. E.g. the demand for a certain commodity may depend on the price of the commodity itself and on the price of a competing commodity, both of which are increasing with time, and the objective may be to find the rate of change of the demand with respect to time.

This problem can be solved by using a generalization of the chain rule as follows:

In the case of a single predictor variable, we have a rule for ‘a function of a function’(chain rule).

For multivariate function, this rule is extended as follows:

$$\frac{\partial y}{\partial t} = \underbrace{\frac{\partial f}{\partial x} * \frac{\partial x}{\partial t}}_{(y \text{ constant})} + \underbrace{\frac{\partial y}{\partial y} * \frac{\partial y}{\partial t}}_{(x \text{ constant})} \dots \dots \dots \frac{\partial F}{\partial x} \text{ is the sum of the two terms}$$

Example

Let $F=xy$, $x=3+4t$; $y=6-2t$

$$\frac{\partial F}{\partial X} = Y \quad \frac{\partial F}{\partial Y} = X \quad \frac{\partial x}{\partial y} = 4; \quad \frac{\partial y}{\partial t} = -2$$

Hence, $\frac{Df}{dt} = y*4 + x*-2$

$$\frac{Df}{dt} = (6-2t)*4 + (3+4t)*-2$$

$$\frac{Df}{dt} = 24 - 8t - 6 - 8t$$

$$\frac{Df}{dt} = 18 - 16t$$

Example

Suppose that $F = x^2y + y^2$

$$X = 3t^3 + 3; \quad y = t^3 - 7$$

Use the chain rule to find $F'(2)$, i.e. $\frac{\partial f}{\partial t}$ when $t=2$

Solution

$$\frac{\partial f}{\partial x} = 2xy \quad \text{and} \quad \frac{\partial f}{\partial y} = x^2 + 2y$$

$$\frac{Dx}{dt} = 6t; \quad \frac{Dy}{dt} = 3t^2$$

Therefore: $\frac{\partial f}{\partial t} = 2xy*6t + (x^2 + 2y)*3t^2$

$$\begin{aligned} \text{When } t=2, \quad \frac{\partial f}{\partial t} &= 2xy*6*2 + (x^2 + 2y)*3*2^2 \\ &= 24xy + 12x^2 + 24y \end{aligned}$$

When $t = 2$ $x = 3(2)^3 + 3 = 15$

$y = 2^3 - 7 = 1$

So $\frac{\partial f}{\partial t} = 24(15)(1) + 12(15)^2 + 24(1) = 3084$

Application

A health store carries 2 kinds of multivitamins, Brand A and Brand B. Sales figures indicate that if Brand A is sold for Sh x per bottle and Brand B for Sh y per bottle, the demand for Brand A will be:

$Q(x,y) = 300 - 20x^2 + 30y$ bottles per month. It is estimated that t months from now the price of Brand A will be: $x = 2 + 0.05t$ shillings per bottle and the price of Brand B will be $y = 2 + 0.1t^{1/2}$ shillings per bottle.

Required:

Determine the rate at which Brand A sales demand will be changing with respect to time 4 months from now.

Solution:

Objective: $\frac{DQ}{dt}$ for $t = 4$

$$\text{Now, } \frac{DQ}{dt} = \frac{DQ}{dx} * \frac{\partial x}{\partial t} + \frac{DQ}{dy} * \frac{\partial y}{\partial t}$$

$$= -40x * 0.05 + 30 * 0.05t - 0.5$$

$$\text{For } t=4, x = 2 + 0.05(4) = 2.2$$

$$\begin{aligned} \text{Hence, } \frac{DQ}{dt} &= -40(2.2)(0.05) + 30(0.05)(4) - 0.5 \\ &= -3.65 \end{aligned}$$

Thus, 4 months from now, the monthly demand for Brand A will be decreasing at the rate of 3.65 bottles per month.

10.4 Differentiation of Exponential Functions

$$f(x)$$

If $y = ae^{f(x)}$ where $a = \text{constant}$

the $\frac{\partial y}{\partial x} = f'(x) a e^{f(x)}$

Note

$f'(x) = \text{first derivative of } x = \frac{\partial y}{\partial x}$

Example

i) $y = 18 e^{x^2}$

$$\frac{\partial y}{\partial x} = 2x \times 18 e^{x^2}$$

derivative of the function over times function $x^2 = 2x$

e.g. $\frac{\partial y}{\partial x} = 36 x e^{x^2}$

ii) $y = e^x = \frac{\partial y}{\partial x} = 1 \times e^x = e^x$

iii) $y = \frac{1}{2} x^{2x}$

$$\begin{aligned} \frac{\partial y}{\partial x} &= 2 \times \frac{1}{2} e^{2x} \\ &= e^{2x} \end{aligned}$$

iv) $y = (5e^2)^4 = 5^4 x^{4x}$

$$4 \times 5^4 e^4$$

$$-x - 5^4 - 25 \times e^4$$

v) $y = e^{-x}$

$$\frac{\partial y}{\partial x} = 3x^2 - x (e^{x^3-x^2})$$

$$\text{vi) } y = 10x e^{x^2}$$

$$\frac{\partial y}{\partial x} = \frac{u \partial v}{\partial x} + \frac{v \partial u}{\partial x}$$

$$u = 10x \quad \frac{\partial y}{\partial x} = 10$$

$$v = e^{x^2} \quad \frac{\partial v}{\partial x} = 2x e^{x^2}$$

$$\therefore \frac{\partial y}{\partial x} = 10x (2x e^{x^2}) + e^{x^2} \cdot 10$$

$$(20x^2 + 10 e^{x^2}) = 10 e^{x^2} (2x^2 + 1)$$

$$\frac{\partial y}{\partial x} = 10e^{x^2} (2x^2 + 1)$$

$$\text{vii) } y = x^3 - (12 - e^{2x})^3$$

$$u = x^3 \quad \frac{\partial y}{\partial x} = 3x^2$$

$$v = (12 - e^{2x})^3 \quad \frac{\partial v}{\partial x} = 3 (12 - e^{2x})^2 \times 2e^{2x}$$

$$= -6e^{2x} (12 - e^{2x})^2$$

$$\frac{\partial y}{\partial x} = x^3 - 6e^{2x} (12 - e^{2x})^2 + (12 - e^{2x})^3 \times 3x^2$$

10.5 Differentiation of Logarithmic Functions

(Base e logarithmic $\ln$)

Rule If $y = \ln f(x)$

then $\frac{\partial y}{\partial x} = \frac{f(x)}{f(x)}$

e.g. i) $y = \ln x \quad \frac{\partial y}{\partial x} = \frac{1}{x}$

ii) $y = \ln 2x \quad \frac{\partial y}{\partial x} = \frac{2}{2x} = \frac{1}{x}$

iii) $y = \ln (10x^2 + 15x - 3)$

$$\frac{\partial y}{\partial x} = \frac{20x + 1}{10x^2 + 15x - 3}$$

iv) $y = \ln x^3 = \frac{\partial y}{\partial x} = \frac{3x^2}{x^3} = \frac{3}{x}$

v) $y = x \ln x \quad \frac{\partial y}{\partial x} = \frac{\partial v}{\partial x} \frac{\partial y}{\partial v}$

$$u = x \frac{\partial y}{\partial x} - 1$$

$$v = \ln x \quad \frac{\partial v}{\partial x} = \frac{1}{x}$$

$$x \frac{1}{x} + \ln x - 1$$

$$1 + \ln x$$

vi) $y = [\ln (20 - x^2)]$

$$\frac{\partial y}{\partial x} = 4 (\ln (20 - x^2))^4 \frac{(-2x)}{20x^2}$$

10.6 Applications (To Add More Material)

10.7



Summary

This lecture extended the concept of differentiation further by considering functions which cannot be easily expressed in terms of the dependent variable (implicit functions). Also dealt with is the differentiation of exponential and logarithmic functions and their uses.

The next lecture will introduce the concept of integration, which is the inverse of differentiation.

Activity

1. (a) Kiplimo Njoroge Certified Public Accountants have recently started to offer business advice to their clients. Acting as consultants, they have estimated the demand curve of a client's firm to be:

$$AR = 200 - 8q$$

Where AR is average revenue in millions of shillings and q is the output in units. Investigations of the client firm's profile shows that marginal cost (MC) is given by:

$$MC = q^2 - 28q + 173 \text{ (in millions of shillings)}$$

Further investigations have shown that the firm's fixed cost is Sh. 10 million.

Required:

Determine:

- i) Total revenue function,
 - ii) Total cost function and hence,
 - iii) The profit function.
- (b) What advice would you give the client in terms of production level and price to charge if its objective is to maximize profit? What is the maximum profit?
- (c) Compare the marginal revenue and average cost values when $q = 5$ units.

2. (a) A national record distribution company sells records and tapes by mail only. Advertising is done on local TV stations. A promotion program is planned for a major metropolitan area for a new country western album. The target audience-those who might be interested in this type of album- is estimated at 800,000. Past experience indicates that for this city and this type of album the percentage of the target market R actually purchasing an album or tape is a function of the length of the advertising campaign, t . Specifically, this sales response function is:

$$R = 1 - e^{-0.025t}$$

The profit margin on each album is \$ 2.50. Advertising costs include a fixed cost of \$ 20,000 and a variable cost of \$ 2,500 per day.

Required:

- i) Determine how long the campaign should be conducted if the goal is to maximize net profit (i.e. gross profit minus advertising costs).
- ii) What is the expected maximum net profit?
- iii) What percentage of the target market is expected to purchase the album?

(b) A company is hiring persons to work in its plant. For the job the persons will perform, efficiency experts estimate that the average cost C of performing the task is a function of the number of persons hired, x . Specifically,

$$C = f(x) = 0.005x^2 - 0.49 \ln x + 5$$

Required:

- i) Determine the number of persons who should be hired to minimize the average cost.
- ii) Given the demand relationship $p = 800 - 75 \ln q$, determine the elasticity of demand when demand equals 200.
- (c) The demand for a particular commodity is given by:

$$P = 15e^{-x/3} \text{ for } 0 \leq x \leq 8$$

Where p = price and x = quantity.

Determine the price and quantity for which revenue is maximum and price elasticity of demand at that point and interpret it.

LECTURE ELEVEN

INTEGRATION

Lecture Outline

- 11.1 Introduction
- 11.2 Objectives
- 11.3 Integration
 - 11.3.1 Indefinite Integration
 - 11.3.2 Definite Integration
- 11.4 Summary

10.1 Introduction

Welcome to Lecture 11. In lectures 6 up to 10, we have been studying differentiation. In this lecture, we begin the study and applications of integration, which is the inverse of differentiation.

11.2	Objectives At the end of this lecture, you should be able to:4 1. Recognize the notation for integration. 2. Write down the integrals of functions. 3. Evaluate definite integrals. 4. Apply integration in solving business problems.
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11.3 Integration

Integration has two interpretations.

1. As a procedure which is the inverse of differentiation. [known as indefinite integration]
2. As a means of obtaining the area under a curve [known as definite integration]

11.3.1 Indefinite Integration:

This involves finding the anti-derivative of a function if we know its derivative.

Usual Question "What function which when differentiated yields to this derivative?"

Examples:

$$1. \frac{\partial y}{\partial x} = 3x^2 \text{ then } y = x^3$$

$$2. \frac{\partial y}{\partial x} = 2x \text{ then } y = x^2$$



Symbols of integration: $\int$ (elongated 'S')

$$1. \int 3x^2 \partial x = x^3$$

$$2. \int 2x \partial x = x^2$$

Constant of integration:

For any integral, an arbitrary constant has to be added since the derivative of a constant is zero

e.g. i) $y = x^3 \quad \frac{\partial y}{\partial x} = 3x^2$

ii) $y = x^3 + 24 \quad \frac{\partial y}{\partial x} = 3x^2$

iii) $y = x^3 - 154 \quad \frac{\partial y}{\partial x} = 3x^2$

Since its not possible to know which function had been differentiated in order to obtain $\frac{\partial y}{\partial x} = 3x^2$

It is necessary to add a constant to the integral as follows:

If $\frac{\partial y}{\partial x} = 3x^2$ Then $y = \int 3x^2 \partial x = x^3 \quad \int 3x^2 \partial x \quad \therefore y = x^3 + c$ where c is a constant.

Rules of Integration:

1. The integral $\int K \partial x = KX + C$ where KC are Constants.

$$2. \int ax^n dx = \frac{ax^{n+1}}{n+1} + C \text{ where } a, n, C = \text{Constant}$$

3. The integral of a sum or difference is the sum or difference of the integrals of the terms forming the integral function, i.e. $\int (u + v) dx = \int u dx + \int v dx$ where $u, v = \text{functions of } x$

4. Exponential functions:

Recall: If $y = ae^{f(x)}$

$$\text{Then: } \frac{\partial y}{\partial x} = f(x)ae^{f(x)}$$

$$\text{Therefore if } \frac{\partial y}{\partial x} = f(x)ae^{f(x)}$$

$$\text{Then: } y = \int f'(x)ae^{f(x)} dx$$

$$Y = \frac{ae^{f(x)}}{f'(x)} + K$$

5. Logarithmic functions:

If: $y = \ln f(x)$

$$\text{Then: } \frac{\partial y}{\partial x} = f'(x) / f(x)$$

$$\text{Therefore: if } \frac{\partial y}{\partial x} = f'(x) / f(x)$$

$$\text{Then: } y = \int f'(x) / f(x) dx$$

$$Y = \ln f(x) + K$$

Examples:

$$1. \frac{\partial y}{\partial x} = 12e^{-3x} + 5x + 2$$

$$y = \int (12e^{-3x} + 5x + 2) dx$$

$$y = \frac{12}{3} e^{-3x} + \frac{5}{2} x^2 + 2x + K$$

$$\frac{\partial y}{\partial x} = 2 \cdot 4/x = 2/x$$

$$y = \int 2/x dx$$

$$y = 2 \ln x + K$$

Exercises:

Integrate the following functions:

$$1. \frac{\partial y}{\partial x} = 3 + 4x^2 - 3x \text{ then } y = 3x + \frac{4x^3}{3} - \frac{3x^2}{2} + c_1$$

$$2. \frac{\partial y}{\partial x} = 2 \quad y = 2x + c_2$$

$$3. \frac{\partial y}{\partial x} = 2x - 3 \quad y = \frac{2x^2}{2} - 3x + c_3$$

$$= x^2 - 3x + c_3$$

$$4. \frac{\partial y}{\partial x} = 5x^{3/2} - 1 \quad y = \frac{5x^{5/2}}{5/2} - x + c_4$$

$$5. \frac{\partial y}{\partial x} = \frac{2}{x^3} = 2x^{-3} = \frac{2x^{-2}}{-2} + c_5$$

$$y = -x^{-2} + c_5$$

$$y = \frac{-1}{x^2} + c_5$$

$$6. \frac{\partial y}{\partial x} = \frac{1}{x} \quad \therefore \frac{\partial y}{\partial x} = x^{-1} = x^0 y = \ln X + K_0 \quad \ln = \text{natural logarithm to base } e$$

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

n

Constants of Integration: Evaluation:

Question 20

a) $\frac{\partial y}{\partial x} = 2x^2 \quad \therefore y = \underline{2x^3} + k_1$

3 when $x = 0$, $y = 7$

Substituting:

$$7 = 2/3 \times 0 + k_1 \therefore k_1 = 7$$

$$y = 2/3 x^3 + 7$$

b) $\frac{\partial y}{\partial x} = 4x$

when $x = 3$ $y = 3$

$$y = \frac{4x^2}{2} + k_2 \quad y = 2x^2 + k_2$$

But when $x = 3$, $y = 3 \therefore 3 = 2(3)^2 + k_2$

$$k_2 = 3 - 2(3)^2$$

$$k_2 = \underline{-15}$$

Equation $\rightarrow y = 2x^2 - 15$ when $x = 4$ $y = 1$

c) $\frac{\partial y}{\partial x} = \frac{x^2 - 4}{x^2} = \frac{\partial y}{\partial x} = \frac{x^2}{x^2} = \frac{4x^2}{x^2} = 1 - \frac{4}{x^2}$

$$y = \int \frac{1 - 4}{x^2} dx \quad y = x + \frac{4}{x} + k_3$$

Substituting:

$$1 = 4 + \frac{4}{4} + k \quad 1 = 4 + 1 + k, \quad 1 = 5 + k \quad k = \underline{\underline{-4}}$$

Equation: $y = x + \frac{4}{x} - 4$

Question 21

A firm finds that the marginal revenue is given by the expression $20 - 2q$ while the marginal cost is given by the expression $4q - 10$. Its fixed costs are Sh.30. q represents quantities of output produced and sold.

Required:

Determine the following:

- Total revenue, total cost and hence profit function
- Profit maximizing
 - Output
 - Price
 - The maximum profit
- Average total cost at profit maximising output
- Breakeven output.

Solution:

$$MR = 20 - 2q \quad MC = 4q - 10$$

$$MR = \frac{\partial R}{\partial q} \quad MC = \frac{\partial c}{\partial q}$$

$$f = 30$$

$$a) \quad (i) \quad R = \int (20 - 2q) = 20q - \frac{2q^2}{2} + k$$

$$R = 20q - q^2 + k \text{ when } q = 0, R = 0 \quad \therefore k = 0$$

$$R = 20q - q^2$$

$$ii) \quad MC = \frac{\partial c}{\partial q} = 4q - 10 \quad C = \int (4q - 10) \partial q$$

$$C = \frac{4q^2}{2} - 10q + k \text{ where } k = \text{fixed cost}$$

$$\therefore C = 2q^2 - 10q + 30$$

$$iii) \quad \text{Profit } \pi = R - C \quad \pi = (20q - q^2) - (2q^2 - 10q + 30)$$

$$\pi = \underline{30q - 3q^2 - 30}$$

a) Optimisation:

$$i) \quad \text{FOC} \quad \frac{\partial \pi}{\partial q} = 30 - 6q = 0 \quad q = \frac{30}{6} = \underline{5 \text{ units}}$$

$$\text{SOC: } \frac{\partial^2 \pi}{\partial q^2} = -6 < 0 \rightarrow \text{Maximum}$$

$$\pi \text{ Maximum} = 30(5) - 3(5)^2 - 30 = \underline{\text{Sh.45}}$$

$$ii) \quad \text{Price, } P = \frac{R}{q} = \frac{20q - q^2}{q}$$

$$P = 20 - q \text{ (demand curve) when } q = 5, p = 20 - 5 = \underline{\text{Sh.15}}$$

$$b) \quad \text{Average total cost, ATC} = \frac{c}{q} = \frac{2q^2 - 10q + 30}{q}$$

$$\therefore \text{ when } q = 5, \text{ ATC} = 2 \times 5 - 10 + \frac{30}{5} = \underline{\text{Sh.6}}$$

Sh.6 -----Also same as average fixed cost.

c) At B.E.P, $R = c$ so that $\pi = 0$

$$\therefore 30q - 3q^2 - 30 = 0 \quad ax^2 + bx + c = 0$$

$$-3q^2 + 30q - 30 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

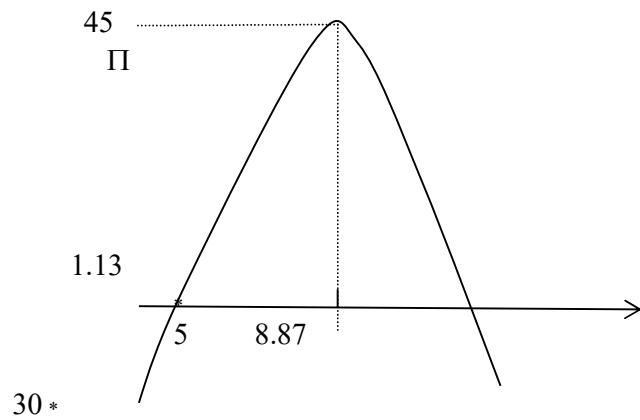
$$q = \frac{-30 \pm \sqrt{30^2 - 4(-3)(-30)}}{2(-3)}$$

$$q = \frac{-30 \pm \sqrt{540}}{-6}$$

$$q_1 = \frac{-30 + \sqrt{540}}{-6} = \underline{1.13 \text{ units}}$$

$$q = \frac{-30 \pm \sqrt{540}}{-6} = \underline{8.87 \text{ units}}$$

y ↑

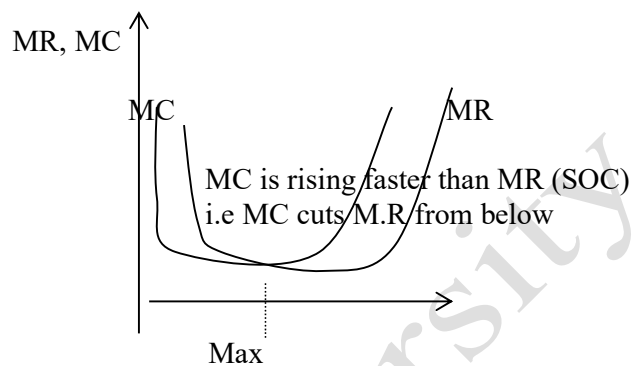


Hence, the correct (effective) B E P is $q = 1.13$ units, since after this the product becomes profitable.

Profit Maximisation: Economic Theory

Theorem:

Profit is maximum where $MR = MC$ (FOC)



Proving:

FOC: $MR = MC$ $20x - 2q = 4q - 10$

$$30 = 6q \quad q = \frac{30}{6} = \underline{5 \text{ units}}$$

SOC: $20 - 2q$ and $MC = 4q - 10$

$$\frac{\partial MR}{\partial q} = \frac{\partial^2 R}{\partial q^2} = -2 \quad \frac{\partial MC}{\partial q} = \frac{\partial^2 C}{\partial q^2} = 4$$

$$\therefore \text{Rate of change of MC} - \frac{\partial MC}{\partial q} > \frac{\partial MC}{\partial q} \rightarrow \text{Max. profit}$$

Question 22

$\pi = R - C$ (π , R, cost - function of quantity)

$$\text{FOC: } \frac{\partial \pi}{\partial q} = \frac{\partial R}{\partial q} - \frac{\partial C}{\partial q} = \frac{\partial R}{\partial q} = MR$$

$$\therefore MR = MC$$

SOC: For maximum

$$= \frac{\partial^2 \pi}{\partial q^2} = \frac{\partial^2 R}{\partial q^2} - \frac{\partial^2 C}{\partial q^2} < 0 (-ve)$$

$$\frac{\partial^2 R}{\partial q^2} < \frac{\partial^2 C}{\partial q^2} \quad \text{or} \quad \frac{\partial^2 C}{\partial q^2} > \frac{\partial^2 R}{\partial q^2}$$

$$\frac{\partial MC}{\partial q} > \frac{\partial MR}{\partial q} > \text{----- proved}$$

Question 23

$$X = c + s1 = \frac{\partial c}{\partial x} + \frac{\partial c}{\partial x} \text{ Note (Mpc + Mps) = 1}$$

$$\text{But } \frac{\partial c}{\partial x} = a \frac{-a}{x^{1/2}}$$

Substituting:

$$\frac{\partial c}{\partial x} - a \frac{-a}{x^{1/2}} = \frac{1 \partial c}{\partial x} = 1 - a \frac{+a}{x^{1/2}}$$

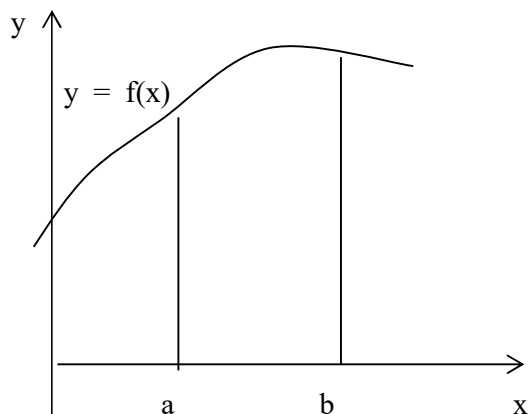
$$C = \int \left(1 - a + \frac{a}{x^{1/2}} \right) dx$$

$$C = x - ax + 2ax^{1/2} + k \text{ when } x = 0, C = 20a \rightarrow K = 20a$$

$$\therefore C = x - ax + 2ax^{1/2} + 20a$$

11.3.2 Definite Integration

Concerned with finding the area under a curve.



Problem:

To find the area bounded by $x = a$, $x = b$, and $y = f(x)$

Approximations:

- i) The area, $A = f(x) \Delta x$
- ii) $A = \sum f(x) \Delta x$ where $x = b$

$\sum_{x=a} f(x) \Delta x$

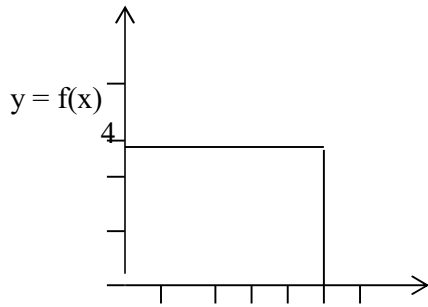
Division into smaller areas results in continuously better approximations. Eventually, we shall be summing up infinite small areas so that in the limit as $\Delta x \rightarrow 0$ i.e. (instantaneous rate of change)

$$x = b$$

$$\sum \rightarrow \int_a^b \text{ and [integral sign] and } \rightarrow \Delta x \text{ dx } \quad x = a$$

Question 24

a) Area bounded by



$$A = 5 \times 4 = 20 \quad \begin{pmatrix} 5 & 5 \\ 0 & 0 \end{pmatrix}$$

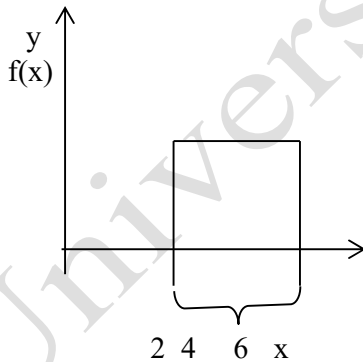
$$A = \int 4 dx \quad A = 4x$$

$$5 \quad A = 4 \times 5 - 4 \times 0 = \underline{20}$$

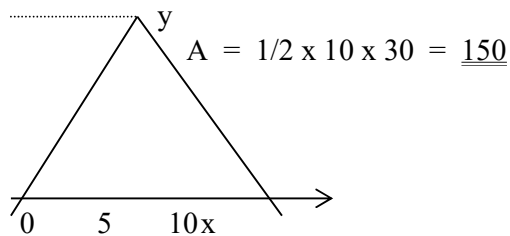
Take Note

Constant of integration for definite integration is zero since it is present in the upper limit as a positive and also in the lower limit as a negative; hence it cancels out.

b



$$A = \int_2^6 (7.5 dx) =$$
$$= 7.5 (6-2) = \underline{30}$$



$$A = \int_2^0 3x \, dx$$

$$= \frac{3x^2}{2} \Big|_2^0 = \frac{3(10)^2}{2} - 0 = \frac{300}{2} = \underline{150}$$

$$d) \quad A = \int_2^{10} 3x^2 \, dx$$

$$A = \frac{3x^3}{3} \Big|_2^{12} = \frac{x^3}{3} \Big|_2^{12} = \frac{(12)^3}{3} - \frac{(2)^3}{3} = \frac{1728}{3} - \frac{8}{3} = \underline{574}$$

$$e) \quad A = \int_5^9 (x^2 - x + 1) \, dx$$

$$\left(\frac{x^3}{3} - \frac{x^2}{2} + x \right) \Big|_5^9 = \left(\frac{9^3}{3} - \frac{9^2}{2} + 9 \right) - \left(\frac{5^3}{3} - \frac{5^2}{2} + 5 \right) = 177.33$$

December 2003 Question 25

beginning a promotional campaign. Advertising expenditures will cost the firm £5440 per day.

Marketing specialists estimate that the rate at which profit (exclusive of advertising expenses) will be generated from the promotion campaign will be estimated by the function.

$R^1 = 8000 - 40t^2$ where t represents the day of the campaign.

A company specializing in a mail order approach of sales is

R^1 is measured in pound per day.

Required:

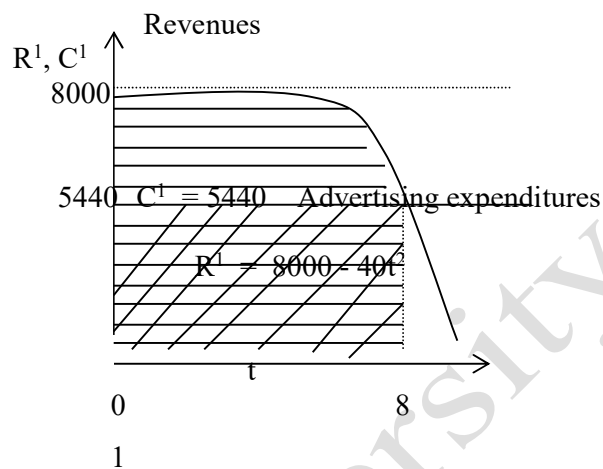
- Sketch the functions R^1 and C^1 on the same graph.
- How long should the campaign be conducted?
- What are total advertising expenditures expected to equal during the campaign/
- What net profit will be expected?

Solution:

$$R^1 = \frac{\partial R}{\partial t} = 8000 - 40t^2$$

$$C^1 = \frac{\partial c}{\partial t} = 5440$$

Sketches



- b) Conduct the campaign as long as $R^1 > C^1$ stop it when $R^1 = C^1$

$$\begin{aligned} R^1 &> C^1 \\ 8000 - 40t^2 &> 5440 \\ -40t^2 &> 5440 - 8000 - 40t^2 > -2560 \\ 40t^2 &< 2560 \quad t^2 < \frac{2560}{40} \end{aligned}$$

$\therefore t < \underline{8}$ stop campaign on day 8.

- c) Total expenditures during the campaign.

$$C = 5440 \times 8 = \underline{\underline{\pounds 43520}}$$

d) Profit = R - C

$$\int_0^8 R^1 - 43520 = \int_0^8 (8000 - 40t^2) dt$$

$$= \left(\frac{8000t - \frac{40t^3}{3}}{3} \right)^8$$

$$0 = 8000(8) - \frac{40}{3} (8)^3 = \underline{\underline{£57,173}}$$

$$\therefore \text{Profit } \pi = 57173 - 43520 = \underline{\underline{£13653}}$$

OR

$$\int_0^8 R^1 - C^1 = \int_0^8 (8000 - 40t^2 - 54440) dt$$

$$\int_0^8 (2560 - 40t^2) dt = 2560 \frac{40t^3}{3} \left(2560 \times 8 - \frac{40 \times 8^3}{3} \right) = \underline{\underline{£1363}}$$

11.4

Summary

This lecture introduced you to the concept of integration, which is the inverse of differentiation. The distinction between indefinite and definite integration was made. Rules and their applications were presented. The last lecture of this course unit will extend the methods of integration to certain special types of derivatives.

LECTURE ONE

MATRICES

1.1 Introduction

Welcome to lecture one on matrix algebra. Matrices have wide application areas in business and economics. Specifically, you can apply matrices in the following areas among others: Solving simultaneous linear equations; markov processes; input output analysis; linear programming; game theory; solutions of economic models e.g. National Economy Models and National Income Models; determining maxima and minima of multivariate functions; convenient way of storing and organizing a large mass data;

To be able to apply matrices, it is important to understand what they are, and how to manipulate them, i.e. you should understand matrix algebra.

1.2 Objectives

At the end of this lecture, you should be able to:

- Define a matrix
- Describe the notation and terminology of matrix algebra
- Describe the various types of matrices
- Add and subtract matrices
- Multiply a matrix by a scalar
- Multiply matrices together
- Calculate the determinant of a square matrix
- Find the inverse of a square matrix

1.3 Definition

A matrix is any rectangular array of numbers, written in the form:

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

$$A = a_{21} \ a_{22} \ \dots \ a_{2n} \text{ or } A = (a_{ij})$$

: : :

$$a_{m1} \ a_{m2} \ \dots \ a_{ij}$$

a_{ij} 's stand for real numbers and are the elements of the matrix.

In particular, a_{ij} is the element in the i^{th} row and the j^{th} column of the matrix A .

e.g $A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 8 \end{pmatrix}$ $C = \begin{pmatrix} 3 & 4 \\ 4 & 5 \end{pmatrix}$ are all matrices. $\left(\begin{array}{c} \\ \\ \end{array} \right)$



- A matrix is denoted using capital letters
- Each number in the matrix is an element a_{ij} of the matrix where i represents the row position and j represents the column position.

1.4 Order of a matrix

This is defined by the number of rows and columns in a matrix. Thus, if a matrix has m rows and n columns it is said to be an $m \times n$ (read 'm by n') matrix or a matrix of order $m \times n$. The order of matrices is frequently denoted by subscripts e.g. $A_{m \times n}$.

For example, the matrix,

$$A = \begin{pmatrix} 1 & -2 & 6 \\ 4 & 3 & -9 \end{pmatrix} \text{ is of order } 2 \times 3 \text{ since it has two rows and three columns.}$$

Similarly the matrix,

$$B = \begin{pmatrix} 6 & 6 & 3 & 1 \\ 3 & 8 & -2 & 7 \\ 1 & 4 & 5 & 0 \end{pmatrix} \quad \text{is of order } 3 \times 4 \text{ since it has three rows and 4 columns.}$$

1.5 Types of Matrices

1.5.1 Vectors

Row vector is a matrix with only one row. E.g. $V_1 = (10 \ -3 \ 4 \ 6)$ is a 1×4 row vector

Column vector is a matrix with only one column e.g. $V_2 = \begin{pmatrix} 7 \\ 3 \\ -5 \end{pmatrix}$ is a 3×1 column vector.

1.5.2 Square Matrix

If the number of rows and the number of columns of a matrix are equal, the matrix is said to be a square matrix. i.e. if $m = n$, the matrix is a square matrix.

e.g. (i) $S_1 = \begin{pmatrix} 2 & -5 & 3 \\ 7 & 4 & 6 \\ 10 & 8 & 9 \end{pmatrix}$ is a 3×3 square matrix

(ii) $S_2 = \begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix}$ is a 2×2 square matrix

1.5.3 Null Matrix

This is a matrix whose all elements are zero

Examples of null matrices

$$(i) N_1 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \text{ is a } 3 \times 2 \text{ null matrix}$$

$$(ii) N_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \text{ is a } 2 \times 2 \text{ null matrix which is also a square matrix.}$$

1.5.4 Sub matrix

A sub matrix is a matrix which is formed from any matrix by striking out one (or more) rows or columns from the original matrix.

Example

$$\text{Let the original matrix be P. If } P = \begin{pmatrix} 5 & 4 & 9 & 2 \\ 10 & 4 & 3 & -1 \\ -3 & 0 & -7 & -1 \end{pmatrix}$$

Sub-matrices:

$$B_1 = \begin{pmatrix} 4 & 9 \\ 3 & -1 \\ 0 & -7 \end{pmatrix} \text{ 1st column omitted}$$

$$B_2 = \begin{pmatrix} 5 \\ 10 \\ -3 \end{pmatrix} \text{ 2nd, 3rd, and 4th columns omitted}$$

$$B_3 = \begin{pmatrix} 4 & 3 & -1 \\ 0 & -7 & -1 \end{pmatrix} \text{ 1st row omitted}$$

1.5.5 Diagonal Matrix

A diagonal matrix is a square matrix, whose every element other than at least one element in the leading diagonal is equal to zero.

Examples

$$(i) D_1 = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

$$(ii) D_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$(iii) D_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

1.5.6 Scalar matrix

A scalar matrix is a diagonal matrix whose leading diagonal elements are equal.

Examples

$$S_1 = \begin{pmatrix} -5 & 0 \\ 0 & -5 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{pmatrix}$$

1.5.7 Identity Matrix

A square matrix for which all the leading diagonal elements are equal to 1 and all other elements are equal to 0 is called an identity matrix.

EXAMPLES

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ 2x2 identity matrix}$$

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ 3x3 identity matrix}$$

In matrix algebra, the identity matrix behaves like 1 in real number algebra

1.5.8 Transpose of a Matrix

Let A be a matrix. If the rows and columns of the matrix A are interchanged – so that the 1st row becomes the 1st column, the 2nd row becomes the 2nd column and so on, then the new matrix formed is known as the transpose of A, denoted A' or A^T.

For example, if $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ then the transpose of A is

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

1.5.9 Symmetric Matrix

If a matrix and its transpose are equal, then the matrix is said to be symmetric (about its main/leading diagonal)

$$M_1 = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$$

$$M_2 = \begin{pmatrix} 5 & 7 & 1 \\ 7 & 3 & -2 \\ 1 & -2 & 0 \end{pmatrix}$$

1.6 Matrix Operations

Matrix operations consist of ways of combining matrices in order to obtain other matrices and values of interest. This gives rise to matrix algebra.

1.6.1 Equality of Matrices

Two matrices $A = (a_{ij})$ and $B = (b_{ij})$ are equal if and only if A and B are of the same order and $a_{ij} = b_{ij}$ for all i and j .

1.6.2 Addition and Subtraction of Matrices

The matrices to be added or subtracted must be of the same order. The operations are carried out by adding or subtracting corresponding elements.

Examples

$$\text{Let } P = \begin{pmatrix} 4 & -3 & 1 \\ 3 & 6 & 2 \end{pmatrix} \text{ and } Q = \begin{pmatrix} 20 & 40 & -2 \\ -2 & & \end{pmatrix}$$

Find (i) $P+Q$ (ii) $P-Q$ (iii) $Q-P$

$$(i) P+Q = \begin{pmatrix} 4 & -3 & 1 \\ 3 & 6 & 2 \end{pmatrix} + \begin{pmatrix} 20 & 40 & -2 \\ -2 & & \end{pmatrix} = \begin{pmatrix} 24 & 37 & -1 \\ 1 & 6 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 6 & 2 & -1 & 2 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 4+20 & -3+40 & 1 & 2 \\ 3-1 & 6+2 & 2 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 24 & 37 & -1 \\ 2 & 8 & 0 \end{pmatrix}$$

$$(ii) P-Q = \begin{pmatrix} 4 & -3 & 1 \\ 3 & 6 & 2 & -1 \end{pmatrix} - \begin{pmatrix} 20 & 40 & -1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} & & & \\ & & & \end{pmatrix}$$

$$= \begin{pmatrix} 4-20 & -3-40 & 1-(-2) \\ 3--1 & 6-2 & 2--2 \end{pmatrix}$$

$$= \begin{pmatrix} -16 & -43 & 3 \\ 4 & 4 & 4 \end{pmatrix}$$

$$i) P = \begin{pmatrix} 4 & -3 & 1 \\ 3 & 6 & 2 & -1 \end{pmatrix} \quad Q = \begin{pmatrix} 20 & 40 & -1 \\ 2 & -1 \end{pmatrix}$$

$$P + Q = \begin{pmatrix} 4 & -3 & 1 \\ 3 & 6 & 2 & -1 \end{pmatrix} + \begin{pmatrix} 20 & 40 & -1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 24 & 37 & 0 \\ 5 & 5 & 1 & -2 \end{pmatrix}$$

$$P - Q = \begin{pmatrix} 4 & -3 & 1 \\ 3 & 6 & 2 & -1 \end{pmatrix} - \begin{pmatrix} 20 & 40 & -1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} -16 & -43 & 2 \\ 1 & 7 & 3 & 0 \end{pmatrix}$$

$$Q - P = \begin{pmatrix} 20 & 40 & -1 \\ 2 & -1 \end{pmatrix} - \begin{pmatrix} 4 & -3 & 1 \\ 3 & 6 & 2 & -1 \end{pmatrix} = \begin{pmatrix} 16 & 43 & -2 \\ -1 & 7 & -3 & 0 \end{pmatrix}$$

-1 2 -1 3 6 2 -4 -4 -3

Properties of matrix addition

- (i) Matrix addition is commutative i.e. $A+B=B+A$
- (ii) Matrix addition is associative i.e. $(A+B)+C=A+(B+C)$
- (iii) If ϕ denotes the null or zero matrix of the same order as that of matrix B, then

$$B + \phi = \phi + B = B$$

1.6.3 Multiplication of Matrices

Matrix multiplication can occur in one of two ways; by a scalar i.e. a real number or by another matrix.

(i) Multiplication by a scalar

If a matrix is multiplied by a scalar, every element in the matrix will be multiplied by that scalar.

Example

Let $R = \begin{pmatrix} 3 & -2 & 5 \\ -5 & 7 & 1 \end{pmatrix}$ and scalar, $k = -0.7$

$$\text{Then } k \times R = -0.7 \begin{pmatrix} 3 & -2 & 5 \\ -5 & 7 & 1 \end{pmatrix} = \begin{pmatrix} -2.1 & 1.4 & -3.5 \\ 3.5 & -4.9 & -0.7 \end{pmatrix}$$

(ii) Multiplication of two matrices

Two matrices can be multiplied if and only if the number of columns in one matrix is equal to the number of rows in the other matrix. In particular, the matrix product AB is defined if and only if the number of columns in A is the same as the number of rows in B , in which case the matrices A and B are said to be conformable for multiplication.

Further, if matrix A is of the order $m \times n$ and B is of the order $n \times p$, then the product AB will be of the order $m \times p$ i.e. it will have the same rows as A and columns as B.

Example

$$\text{If } A = \begin{pmatrix} -3 & 5 \\ -2 & -2 \\ 0 & -1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$AB = \begin{pmatrix} -3 & 5 \\ -2 & -2 \\ 0 & -1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$\begin{aligned} &= \begin{pmatrix} -3 \times 1 + 5 \times 0 & -3 \times 0 + 5 \times 1 \\ -2 \times 1 + -2 \times 0 & -2 \times 0 + -2 \times 1 \\ 0 \times 1 + -1 \times 0 & 0 \times 0 + -1 \times 1 \end{pmatrix} = \begin{pmatrix} -3 & 5 \\ -2 & -2 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

Properties of Matrix multiplication

(i) Matrix multiplication is not necessarily commutative

$AB \neq BA$. In fact BA may not be defined. Thus the sequence of matrix multiplication must always be preserved.

(ii) Matrix multiplication is associative i.e. $AB(C) = (AB)C$

(iii) Matrix multiplication is distributive i.e. $A(B + C) = AB + AC$

(iv) $\phi A = A\phi = \phi$

(v) If I is an identity matrix and matrix multiplication is defined, then $AI = IA = A$

1.6.4 Determinants

To each square matrix A , there corresponds a single value known as its determinant denoted by $|A|$.

In this lecture we will learn how to calculate the determinants of 2x2 and 3x3 matrices.

(i) Determinant of a 2x2 matrix

Consider the general 2x2 matrix $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

Determinant of the matrix A is given as the product of the leading diagonal elements minus the product of the secondary diagonal elements i.e.

$$\det A = a_{11}a_{22} - a_{21}a_{12}$$



- If the determinant of a matrix is equal to 0, it is said to be a singular matrix; otherwise it is non-singular.

Determinants of 3x3 matrices

Before we can discuss the determinant of a 3x3 matrix, we need to introduce a concept known as a cofactor.

Consider the general 3x3 matrix $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

Corresponding to each element a_{ij} of matrix A , there is a cofactor, C_{ij} . A 3x3 matrix has nine elements, so there are nine cofactors to be computed. The cofactor C_{ij} is defined to be the

determinant of the 2x2 matrix obtained by deleting row i and column j of the matrix A , multiplied by '+1' or '-1' according to the following pattern:

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

For example, suppose we wish to calculate C_{23} , which is the cofactor of element a_{23} of the matrix A .

The element a_{23} lies in row 2 and column 3. Consequently we delete 2nd row and 3rd column as indicated below:

$$\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array} \begin{pmatrix} & & \\ & & | \\ & a_{21} & a_{22} & a_{23} \\ & & & \end{pmatrix}$$

The 2x2 matrix whose elements have not been deleted is $\begin{pmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{pmatrix}$

We obtain the determinant of this matrix and multiply by -1 since it is located at a position with a negative sign in the pattern above.

Thus the cofactor of a_{23} is: $C_{23} = -1(a_{11}a_{32} - a_{31}a_{12})$

The same procedure is repeated for all the elements of the matrix.

Find the cofactors of the following matrix

$$A = \begin{pmatrix} 2 & 4 & 1 \\ 4 & 3 & 7 \\ 2 & 1 & 3 \end{pmatrix}$$

1.6.5 Inversion

$$AB = BA = I_n \text{ (identity matrix of order } n \times n \text{)}$$

For a given $n \times n$ matrix A , the $n \times n$ matrix B is the inverse of A if

Obtaining the inverse of a matrix is referred to as matrix inversion.

Matrix inversion

The following steps may be followed in obtaining the inverse by adjoint method.

Step 1

From the matrix A , form a new matrix by replacing every element a_{ij} by its cofactor, which we may call C .

Step 2

Transpose the matrix C , by writing rows as columns and columns as rows. This new matrix is called the adjoint of A .

Step 3

Calculate the value of the determinant of A i.e. $|A|$

Step 4

The inverse is given by

$$A^{-1} = \frac{1}{|A|} C^T$$

Inverse of a 3 x 3 matrix:

Let A be 3×3 matrix.

Steps.

- Obtain the determinant.
- Determine the matrix of minor elements, M (minor matrix)
- From M, above, obtain the matrix of Cofactor elements, C (cofactor matrix)
- Transpose the cofactor matrix C to obtain C transpose, C^T also called the adjoint/adjunct matrix of A.
- The inverse of A will be: $A^{-1} = \frac{1}{|A|} \times C^T$

Examples

3 1 2
Consider the matrix A = $\begin{pmatrix} 5 & 1 \\ 4 & -2 & -1 \end{pmatrix}$

- Minor element matrix M = ignore the row and column of e.g. element 3 i.e.

$\begin{pmatrix} 0 & 1 & 2 \\ 2 \\ 4 \end{pmatrix}$ and remain with $= -5 - 2 = -3$

M = $\begin{pmatrix} -3 & -6 & -24 \\ 1 & -11 & -10 \\ 9 & -1 & 13 \end{pmatrix}$

- Cofactor matrix C (multiply M by $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$)

$$C = \begin{bmatrix} -3 & 6 & -24 \\ -9 & 1 & 13 \end{bmatrix}$$

- Adjoint matrix $C^T = \begin{bmatrix} -3 & -9 \\ 6 & 1 \\ -24 & 10 & 13 \end{bmatrix}$

- Inverse matrix $N_1^{-1} = \frac{1}{|N_1|} \times C^T$

$$= \frac{1}{-3} \begin{bmatrix} -3 & -9 \\ 6 & 1 \\ -24 & 10 & 13 \end{bmatrix}$$

ii) $N_3 = \begin{bmatrix} 7 & -3 & 6 \\ 4 & 2 & 8 \\ -1 & 0 & 2 \end{bmatrix}$

Determinant $= |N_3| = 88$

a) Minor element matrix $M = \begin{bmatrix} 4 & 16 & 2 \\ -6 & 20 & -3 \\ -36 & 32 & 26 \end{bmatrix}$

b) Co factor matrix $C = \begin{bmatrix} 4 & -16 & 2 \\ -6 & 20 & 3 \\ -36 & -32 & 26 \end{bmatrix}$

c) Adjust (Transpose) matrix $C^T = \begin{bmatrix} 4 & -6 & -36 \\ -16 & 20 & -32 \\ 2 & 3 & 26 \end{bmatrix}$

d) Inverse matrix $N_3^{-1} = \underline{1}$

$$/N_3 / \times C^T$$

$$= \underline{1} \quad \begin{matrix} 4 & 6 \\ 88-16 & 20 & -32 \end{matrix} \begin{pmatrix} 36 \\ 2 & 3 & 26 \end{pmatrix}$$

Find the inverse of the following matrices

$$(a) A = \begin{pmatrix} 2 & 0 & -1 \\ 3 & 1 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

$$(b) B = \begin{pmatrix} 5 & 2 \\ -3 & 4 \end{pmatrix}$$

LECTURE TWO

MATRIX APPLICATIONS

Welcome to lecture two. Having gone through the first lecture, introducing you to matrix algebra, we now have a good basis for this lecture on matrix applications. In this lecture, we will see how matrix algebra can be used in solving simultaneous linear equations and also in allocating overhead costs.

2.1 Objectives

At the end of this lecture, you should be able to:

- Represent a system of linear equations using matrix notation
- Use matrix inverse method to solve systems of linear equations
- Use crammers rule to solve systems of linear equations
- Apply systems of linear equations in solving business problems
- Allocate overhead costs using matrix operations

2.1.1 Linear Equations and the Matrix notation

Using matrices can greatly simplify the statement and solution of a system of linear equations. Consider the following general set of linear equations:

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\
 a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\
 &\vdots \\
 a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n &= b_n
 \end{aligned}$$

This system of linear equations can be written in matrix notation as follows:

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

Now, if we let

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}, \quad X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \text{ and } B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

Then $AX = B$

2.1.2 Solving a system of linear equations using Inverse Method

If $AX = B$ is any system of linear equations, and if A^{-1} exists, then

$$\begin{aligned}
 A^{-1}AX &= A^{-1}B \Rightarrow IX = A^{-1}B \\
 \Rightarrow X &= A^{-1}B \text{ since } IX = X
 \end{aligned}$$

Example 1

$$\begin{aligned}
 x + 2y &= 3 \\
 x - 2y &= 11
 \end{aligned}$$

$$A X = B$$

$$A^{-1} = \begin{bmatrix} -1 & -2 \\ 4 & -1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -1-28 \\ 4-8 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -29 \\ -4 \end{bmatrix}$$

Example 2

Use the matrix inverse method to solve for the values of the unknowns in the following system of linear equations:

$$\begin{bmatrix} 3x - 2y - z = 2 \\ -4x + y - z = 3 \\ 2x + z = 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & -2 & -1 \\ -4 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

$A X = B$

$$A^{-1} = \frac{1}{\begin{vmatrix} 3 & -2 & -1 \\ -4 & 1 & -1 \\ 2 & 0 & 1 \end{vmatrix}} \begin{bmatrix} 3 & -2 & -1 \\ -4 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix}$$

(Determinant)

$$A^{-1} = \frac{1}{\begin{vmatrix} 3 & -2 & -1 \\ -4 & 1 & -1 \\ 2 & 0 & 1 \end{vmatrix}} \begin{bmatrix} 3 & -2 & -1 \\ -4 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} - & \text{Minor element matrix} & M = \begin{bmatrix} 12 & -2 \\ -25 & 4 \\ 37 & -5 \end{bmatrix} \end{bmatrix}$$

- Co factor matrix C (signed matrix)

$$\begin{pmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{pmatrix}$$

- $CT =$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{pmatrix}$$

- Inverse matrix $A^{-1} = \frac{1}{|A|} CT$

$|A| =$

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{vmatrix}$$

$$y = 2 \begin{vmatrix} x & 1 \\ 5 & 7 \end{vmatrix} - 2 \begin{vmatrix} 2 & 3 \\ 4 & 15 \end{vmatrix} + 3 \begin{vmatrix} 2 & 2 + 6 + 3 \\ 1 & 15 + 7 \end{vmatrix} = 11 \begin{vmatrix} x & 1 \\ y & 26 \end{vmatrix}$$

$$z = 2 \begin{vmatrix} x & 1 \\ 5 & 7 \end{vmatrix} - 2 \begin{vmatrix} 2 & 3 \\ 4 & 15 \end{vmatrix} + 3 \begin{vmatrix} 2 & 2 + 6 + 3 \\ 1 & 15 + 7 \end{vmatrix} = 21 \begin{vmatrix} x & 1 \\ y & 26 \end{vmatrix}$$

Self Assessment tests

- What is a cofactor of an matrix
- Solve the following system of linear equations

$$3x + 10y = 15$$

$$x + 3y = 4$$

Solve the following system of linear equations using inverse method

$$x_1 + 3x_2 + x_3 = 4$$

$$x_1 - 2x_2 - 2x_3 = 3$$

$$2x_1 + x_2 - x_3 = 9$$

2.1.3 Solving a system of linear equations using Cramer's Rule

If $AX = B$ is any system of linear equations, then the value of the i th variable can be found using the

following formula: $X_i = \frac{\Delta_i}{\Delta}$

Where Δ = determinant of the matrix A

Δ_i = determinant of the matrix A with column i replaced by vector B.

Example 1

$a_{11}x_1 + a_{12}x_2 = b_1$ where: $a_{ij}, b_i = \text{constants}$.

$a_{21}x_1 + a_{22}x_2 = b_2$ x_j 's = unknowns/variables

Matrix equation:

$$= \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$\begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \end{vmatrix} \begin{vmatrix} a_{12} & b_1 \\ a_{22} & b_2 \end{vmatrix} \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}$$

Example 2

Use Cramer's Rule to evaluate the following equations

Equation: Matrix Notation:

$$\begin{aligned} x + 2y &= 3 \\ 2x + 3y &= 1 \end{aligned} \quad \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\begin{aligned} x &= \frac{\begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}} = \frac{3(3) - 2(1)}{1(3) - 2(2)} = \frac{9 - 2}{3 - 4} = \frac{7}{-1} = -7 \\ y &= \frac{\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}} = \frac{1(1) - 3(2)}{1(3) - 2(2)} = \frac{1 - 6}{3 - 4} = \frac{-5}{-1} = 5 \end{aligned}$$

The equilibrium condition for three related markets is given by:

$$11P_1 - P_2 - P_3 = 31$$

$$-P_1 + 6P_2 - 2P_3 = 26$$

$$-P_1 - 2P_2 + 7P_3 = 24$$

Required:

Using matrix inverse method, find the equilibrium price for each market.

A company owns three types of trucks numbered 1, 2, and 3 which are equipped to transport three different products per load in tons according to the following chart:

PRODUCT	TRUCK		
	1	2	3
A	2	2	2
B	0	2	4
C	4	2	2

Required:

Using matrix inverse method, determine how many trucks of each type should be made available to transport 24 tons of product A, 20 tons of product B and 32 tons of product C at one load. Assume that each truck will be fully loaded.

2.1.4 Allocation of Overhead Costs

Example

Consider the following information.

	<u>Service Departments</u>		<u>Production Departments</u>	
	<u>Maintenance</u>	<u>Electricity</u>	<u>Machining</u>	<u>Assembly</u>
Hours of maintenance time	-	3,000	16,000	1,000
Units of electricity consumed	20,000	-	130,000	50,000

Department: Cost before any

allocation of service departments £50,000 £4,000 140,000 206,000

Required:

- Determine total costs to be allocated to production departments using matrix algebra.
- Explain the advantages of this method over the traditional allocation procedures.

Solution:

	<u>Service Departments</u>		<u>Production Departments</u>	
	<u>Maintenance</u>	<u>Electricity</u>	<u>Machining</u>	<u>Assembly</u>
Hours of maintenance time	-	$\frac{3,000}{20,000} = 0.15$	$\frac{16,000}{20,000} = 0.8$	$\frac{1,000}{20,000} = 0.05$
Units of electricity consumed	$\frac{20,000}{200,000} = 0.10$	-	$\frac{130,000}{200,000} = 0.65$	$\frac{50,000}{200,000} = 0.25$

Let x = total overhead cost of maintenance department.

y = total overhead cost of electricity department.

$$x = 50,000 + 0.1y - 0.1x - 0.1y = 50,000$$

$$y = 4,000 + 0.15x - 0.15x - 0.15x = 4,000$$

Matrix equation:

$$\underbrace{\begin{pmatrix} 1 & -0.1 \\ -0.15 & 1 \end{pmatrix}}_A \underbrace{x}_X = \underbrace{\begin{pmatrix} 50000 \\ 4000 \end{pmatrix}}_B$$

$$X = A^{-1} B$$

$$A^{-1} = \begin{pmatrix} 10.1 & 1 \\ 0.15 & 10.95 \end{pmatrix}$$

Therefore:

$$\begin{matrix} x = \underline{1} & 10.1 & 50,000 & (=51,168) \\ y & 0.985 & 0.15 & 1 & 4,000 & (=1,675) \end{matrix} \begin{pmatrix} \\ \end{pmatrix} \begin{pmatrix} \\ \end{pmatrix}$$

Step 2:

Service Dept. cost allocation to production departments:

$$\begin{matrix} \text{Machining} & = & 0.80 & 0.65 & 51,168 & (=48,523) \\ \text{Assembly} & & 0.05 & 0.25 & 1,675 & (=5,477) \end{matrix} \begin{pmatrix} \\ \end{pmatrix} \begin{pmatrix} \\ \end{pmatrix}$$

A manufacturing company has two service departments (maintenance and administration), and two production departments, (assembly and finishing). The direct overheads of each department are as follows:

Maintenance Administration Assembly Finishing

£2500 £3250 £22500 £11400

The labour hours of each service department's time used by the other departments is as follows:

Maintenance Administration Assembly Finishing

Maintenance - 200 1000 800

Administration 150 400 350

Using matrix algebra, calculate the total cost of each production department including the allocation of service department costs.

LECTURE THREE

INPUT - OUTPUT ANALYSIS

4.1 Introduction

In the last lecture, we dealt with the application of matrix algebra in solving simultaneous linear equations and also in allocating overhead costs. In this lecture, we will see how matrix algebra can further be used to study the interdependence of organizational units.

As you may be aware, industries or economic sectors do not produce in isolation. They rely on each other. The output from one sector may be the input to the same sector or to another productive sector, or may be consumed as final demand. Thus industries are related in the sense that each must use each other's products in order to operate.

4.2 Objectives

At the end of this lecture, you should be able to:

- Recognize interdependence between organizational units
- Describe the components of an input/output table
- Explain the assumptions of Input/Output Analysis
- Calculate and interpret the matrix of technical coefficients
- Calculate the total output given the final demand
- Apply input/output models in analyzing interdependence

4.3 Components

Input-output table has five components

Inter industry transaction

This consists of inputs sold and purchased within the sectors, which must exclude final demand. They represent intermediate consumption of goods and services.

The final demand

This accounts for final consumption of goods and services. It is usually made up of the following components (a) consumption (b) investment (c) Government services and (d) export (e) accumulation of inventory. Basic objective is to justify of existence of firms.

Total output

This is the total of inter industry demand and final demand

Primary inputs

These are inputs not produced by the industries themselves. They are non-industry inputs. Examples include labour (wages/salaries), land (rents), machinery (depreciation), entrepreneurship (profits).

They are meant to transform inter - industry inputs. They are also known as value added

Total inputs

This is the amount of goods and services used in production. It is equal to the sum of inter industry inputs and primary inputs (profit is included)

As profit is included, total value of output must be accounted for by the total cost of products.

4.4 Analysis

We begin by dividing an economy into n industries each of which is thought to be producing only one type of output.

In addition, an economy must usually produce some finished products for final demand. The data for input output analysis can be tabulated as in the table below:

PRODUCER	<u>USER</u>					Total output
	1	2		n	Final demand	
1	b_{12}	b_{12}	...	b_{1n}	d_1	X_1
2	b_{21}	b_{22}		b_{2n}	d_2	X_2
...	...	...	...	...	...	...
n	b_{n1}	b_{n2}		b_{nn}	d_n	X_n

Where

b_{ij} = shilling amount of the products of industry i used by industry j

d_i = final demand for the products of industry i

X_i = total output of industry i .

Input – output analysis determines the production level of each of the industries if final demand changes, assuming the structure of the economy do not change.

Technology matrix (unit cost structure)

The structure of the economy can be described by the technology matrix

$$A = (a_{ij}), \text{ where } a_{ij} = \frac{b_{ij}}{X_j}$$

a_{ij} is defined as the shilling value of the output of industry i that industry j must purchase to produce one shillings worth of its own products.

Therefore, to satisfy the needs of all other industries, industry i must produce outputs

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \text{ for } i = 1, 2, 3, \dots, n$$

Inter industry demand vector can thus be written as AX where,

$$A = \begin{matrix} & \begin{matrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & \dots & \dots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{matrix} \end{matrix} \text{ and } X = \begin{matrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{matrix}$$

The production of the economy must be adjusted to meet both inter industry needs and final demand.

The final demand vector is

$$D = \begin{matrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{matrix} \quad d_i \geq 0 \quad i = 1, 2, \dots, n$$

This requirement can be written

$$X = AX + D$$

$$X - AX = D$$

$$(I - A)X = D$$

$$X = (I - A)^{-1}D$$

The matrix $(I - A)$ is known as the Leontief matrix, $[I - A]^{-1}$ the Leontief inverse matrix, and I is identity matrix. Therefore, given a set of positive final demands, production levels consistent with the set of final demands can be determined.

Example

Consider a very simple hypothetical economy of two industries A and B given below:

PURCHASER (USER) OF INPUTS

PRODUCER	Agri	Manuf	Service	Final demand	Total production
Agricultural	144	156.5	25.5	250	576
Manufacturer	144	78	77	300	599
Service	46	62.5	25.5	250	384
Primary inputs	242	302	384	800	
Total value of inputs	576	599	384		1559

Assumptions of Input Output Model

- Each sector produces only one homogenous product or service. However, the conditions still holds for a multi product sector product ratio of the product mix remains constant.
- Each industry uses a fixed input- output ratio. This ensures that the technical coefficient matrix is stable.
- Production in every industry is subject to constant returns to scale i.e. there are no economies or diseconomies of scale.

	USER		FINAL DEMAND	TOTAL OUTPUT
PRODUCER	I	II		

I	500	350	150	1,000
II	320	360	120	800

- Determine the matrix of technical Coefficients
- Determine the total output vector of the economy if final demand changes to 200 for I and 100 for II.
- Determine the total primary input

A simple hypothetical economy of three industries A, B and C is represented in the following table (data are in millions of Dollars of products)

	USER			FINAL DEMAND	TOTAL DEMAND
PRODUCER	A	B	C		
A	50	40	30	30	150
B	30	100	60	10	200
C	30	40	40	10	120

LECTURE FOUR

MARKOV CHAIN MODELS

4.1 Introduction

In the last lecture, we saw how matrix algebra can be used to forecast production levels for various organizational units. In this lecture, we continue with matrix based forecasting models.

5.2 Objectives

At the end of this lecture, you should be able to:

- Define a Markov process
- Explain the assumptions of a Markov process
- Determine the transition matrix
- Forecast future states given initial states
- Determine the steady states distribution
- Discuss the concept of sink or absorbing states
- Apply Markov chains in solving business problems

Definition

A Markov process is a stochastic/probabilistic system for which the occurrence of a future state depends on the immediate preceding state.

The condition of the system at any point in time is called the state of the system.

A Markov chain model is specified by:

- Defining its possible states;
- Specifying the initial probability distribution for these states i.e. determining the initial states of the system which describe the condition (or system state) at the point in time at which the analysis begin;
- Specifying transition probabilities, which measure the probability of the system moving from one state to another state, during a specified time period.

4.2 Markov Analysis

4.2.1 Transition Matrix

Let the states, or outcomes, of the process be denoted by $S_1, S_2, \dots, S_n$. We can form a matrix of transition probabilities, T as follows:

To	
$S_1 \ S_2 \dots S_n$	
$S_1 P_{11} \ P_{12} \dots P_{1n}$	
$T =$	From $S_2 P_{21} \ P_{22} \dots P_{2n} :$
$\vdots$	
$S_n P_{n1} \ P_{n2} \dots P_{nn}$	

Where,

P_{ij} = probability of the phenomenon being in state S_j in the next period if the current state is S_i .

P_{ii} = probability of repeating state S_i

4.2.2 Properties of the transition matrix

- The sum of probabilities across a row is 1
- Transition matrix T , is a square matrix
- Probabilities, P_{ij} 's are obtained empirically i.e. by observation and data collection.

4.2.3 Analysis

If T is the transition matrix and the probability distribution of the states on trial t

is $P_1, P_2, \dots, P_n$, then the probability distribution of the states on trial $t + 1$ is determined by;

$$(P_1, P_2, \dots, P_n) \times T$$

4.2.4 Steady state / Equilibrium

If Markov process conditions hold and the transitions go on sufficiently long, there shall be attained a dynamic equilibrium which is independent of any initial state. It only depends on the transition matrix such that the following equation will hold true;

$$P \times T = P$$

P = long term position



- A Markov process approaches a steady state or equilibrium as the number of observations increase (i.e. as n approaches infinity)
- When a Markov process is in equilibrium, the probability of each possible state or outcome is constant from observation to observation
- The equilibrium probability distribution is unique and depends only on the transition matrix and not on the initial probability distribution of the state

If $(P_1, P_2, P_3, \dots, P_n)$ is the steady state distribution, then

$$(P_1, P_2, P_3, \dots, P_n) T = (P_1, P_2, P_3, \dots, P_n)$$

$$\text{and } P_1 + P_2 + P_3 + \dots + P_n = 1$$

You will then have $n+1$ equations and n unknowns which you can solve simultaneously to obtain P_i 's.

4.2.4 Absorbing State

A system is in an absorbing or trapping state if it is impossible to exit that state and move to some other state. Once that absorbing state is reached, it cannot be left i.e. there is zero probability of moving from that state to any other state.

When a system is in an absorbing or trapping state, the normal methods of Markov analysis that have so far been considered cannot be used. For example, it is no longer a question of what proportion of the total will reach a given state, all of the total will actually be in the absorbing states. So that with the presence of absorbing states, it is possible to determine:

- The average number of periods the various state categories will spend in each non-absorbing state before finally becoming absorbed;
- The average number of periods it will take for the state categories (the system) to move from any beginning non-absorbing state to an absorbing state;

- The probability of moving from any given non-absorbing state to each of the absorbing states;
- The proportion of the system that will end up in each of the absorbing states.

Fundamental matrix

When a system has an absorbing state, a special matrix called a fundamental matrix is designed as follows:

- Arrange or rearrange the transition matrix so that the absorbing states are first;
- Partition the rearranged transition matrix into four parts:

I	O
A	N

Where,

I=the identity matrix;

O=a matrix of zeros;

A =a matrix of the transition probabilities of the remaining states of the system being absorbed in the next period;

N =a transition matrix for the non-absorbing states showing movement among these states

- Fundamental matrix $F = (I - N)^{-1}$. It indicates the expected number of times the system will be in any non-absorbing state before absorption i.e. it shows the average number of periods the system will be in each non-absorbing state before absorption occurs. N and I are of the same dimensions.

After computing F , it is possible to determine the probabilities of moving from any non-absorbing state to each absorbing state. This is found by multiplying the fundamental matrix F by A i.e. determining $F \times A$

A Training Problem

East Rochester Training College is a proprietary, two-year college offering a specialized technical training program. In order to plan for the future, the administration is interested in knowing how many students will graduate each year, continue in the program, or drop out. The administration has developed the following transition probabilities. Notice that, at the conclusion of the first year (row 1, first year), a student may either continue (taking second-year courses or first-year courses) or leave school (transferring as well as dropping out). It is not possible to graduate from the first year. At the conclusion of the second year, a student can graduate, drop out, or repeat second-year courses. With this program, it is not possible to take first-year courses after second year. A student's status in the next school year depends only upon his or her status this year.

	1 st	2 nd	Grad	Leave
1 st	0.25	0.57	0.18	
2 nd	0	0.13	0.68	0.19
Grad	0	0	1	0
Leave	0	0	0	1

The school's administration needs to know what proportion of students in the first and second year will eventually graduate and what proportion will drop out.

The solution to this problem reveals that 59 percent of current first-year students will graduate and 41 percent will drop out. Of the current second-year students, 78 percent will graduate and 22 percent will drop out. Thus if there are currently 2,000 first-year students and 1,800 second-year students, 2,584 will graduate.

4.3 Assumptions Markov Analysis

- There is a finite number of possible states;
- The system is stable i.e. the number and composition of both the states and subjects of analysis remain intact. E.g. in market share analysis, this means that the competing firms/brands remain

those and only those ones for the period of analysis. Similarly customer number and composition remain unaltered;

- The probability of changing states remains the same (i.e. transition matrix is stable);
- The various states are both mutually exclusive and collectively exhaustive;
- Time period is constant for any particular analysis;
- We can predict any state by using the initial (current) state and the matrix of transition probabilities.

4.4 Applications of Markov Analysis

Generally, Markov analysis is used to study the evolution of systems in which there is probabilistic movements from one condition to another condition, over repeated periods of time. It analyses the current behaviour of a system in order to predict the future behaviour of that system.

It can be applied to any situation that seeks to determine in a sequential manner, the probability of certain events happening or not happening.

More specifically, it can be used in the following areas:

- Consumer brand switching;
- The aging of accounts receivable;
- Consumer response to advertising;
- Forecasting share prices in the stock exchange;
- Manpower planning;
- Education;
- Inventory models;
- Maintenance of capital equipment; forecasting rates of breakdown in machinery so as to determine appropriate replacement programs etc.

Self Assessment test

Exercises

University of Nairobi